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APPLIED REGRESSION ANALYSIS (STA602)

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ANALYSIS OF FACTORS THAT INFLUENCE MONTHLY RESTAURANTS REVENUE:

Number of Customers, Menu Pricing, and Average Customer Spending

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CHAPTER 1: INTRODUCTION

1.1 Background of Study

Managing a restaurant is highly challenging since one must consider the target customers, the desired menu and suitable prices for the food. There are many ways to increase the value of a restaurant such as location, marketing efforts and famous trends online. This research is focused on the various factors affecting a restaurant's monthly revenue. The factors included are the number of customers, menu price, number of reviews received by the restaurant, average customer spend, and marketing spending. It is very crucial for a restaurant to have more than enough customers per day as customers are the main source of income. However, the thing that keeps the customers coming is the menu price and number of reviews received by the restaurant. Therefore, one can maintain customers' average spending by increasing their marketing strategy.

Every independent variable in this research can result in various outcomes in terms of the restaurant's monthly revenue, it can either be low, moderate or high income. This research explores the effectiveness and relationship of different strategies on the restaurant revenue prediction. It is also aimed to develop more predictive models in forecasting restaurants' revenue. Having these models can help restaurant owners in discovering their weak spots and improvising to achieve the optimum revenue.

1.2 Data Description

The data of restaurant performance, an interesting opportunity for analysts to monitor and track the revenue of the restaurant over time.

Variable	Description
Restaurant Revenue (Monthly), Y	The total revenue generated by the restaurant each month in monetary units (e.g., dollars).
Number of Customers, X_1	The total number of customers who visited the restaurant in a given month.
Menu Price, X_2	The price of items on the restaurant's menu, typically expressed in monetary units.
Average Customer Spending, X_3	The average amount of money spent by a customer per visit to the restaurant.

Table 1:Data Description

1.3 Research Objectives

The researcher has a few objectives in this research. The objectives are as follows:

1. To examine the relationship between restaurant revenues (dependent variable) and factors such as number of customers, menu price and average customer spend (independent variables).
2. To find the best regression model.
3. To select the best predictor variables for the best fit model $\left(R_p^2, R_{a,p}^2 \; and \; C_p\right)$.

1.4 Research Questions

To complete this research, there are a few questions that the researcher wants to answer. The questions are:

1. Is there any relationship between restaurant revenues and factors such as number of customers, menu price, number of reviews received, average customer spend, and marketing spend?
2. What is the best regression model?
3. Which predictor variables are the best for the best fit model?

1.5 Theoretical Framework

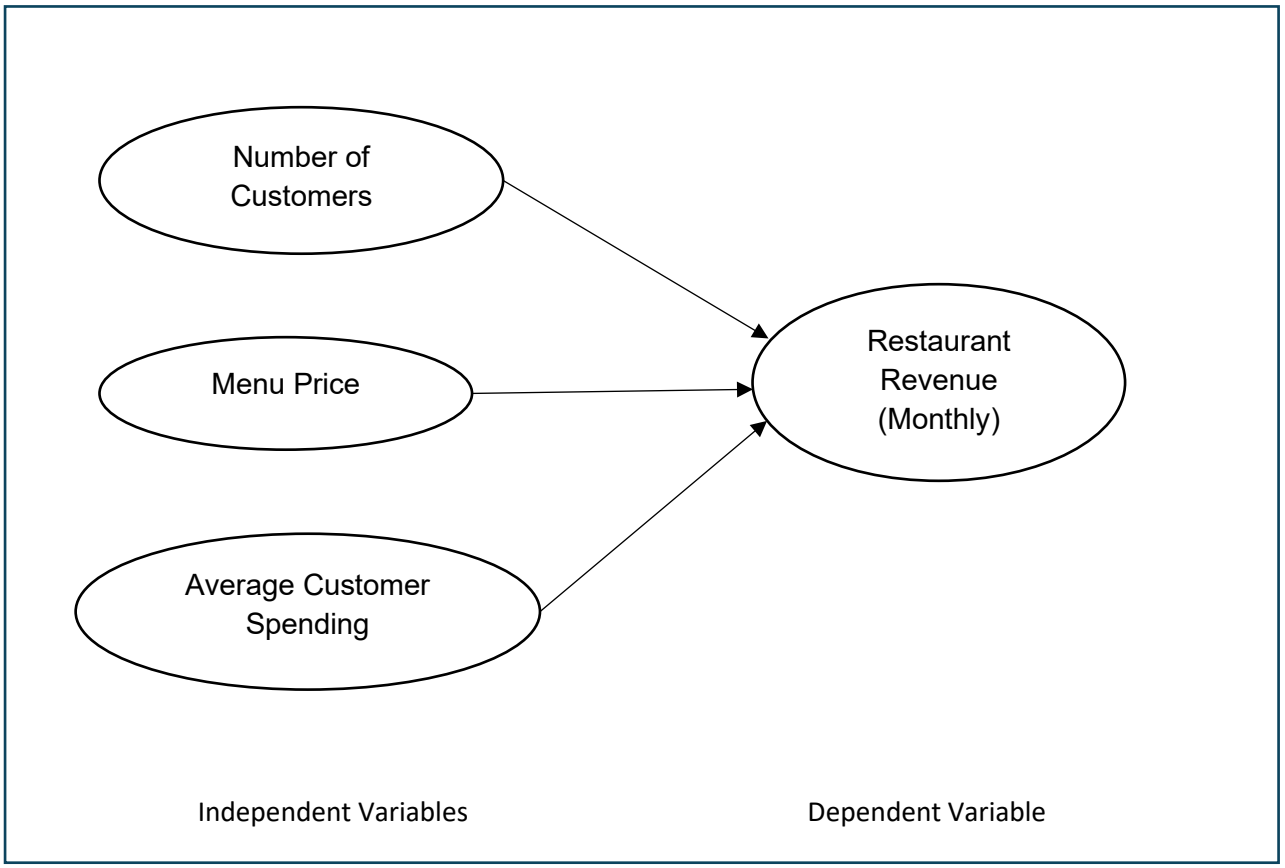


Figure 1:Theoretical Framework

CHAPTER 2: LITERATURE REVIEW

2.1 Restaurants Revenue

Revenue management is a strategic practice to maximize revenue by balancing supply and demand using pricing and inventory controls. Initially developed in the airline industry, it has since been applied across various sectors such as hospitality, transportation and retail (Talluri et al., 2004). In this research, the dependent variable of sales revenue was measured to evaluate the impact of revenue management on small businesses. Monthly statistics were utilized instead of annual to encourage respondents to reveal what could be interpreted as sensitive financial information on sales income (Shields, 2006)

2.2 Number of Customer

Customers are long-term investments for companies that must be given the best service for the continuity of their business (Donny et al., 2018). The number of customers directly influences business performance, especially for revenue. Research shows that higher customer volume enhances profitability as business capitalize on economies of scale and increased product utilization (Gómez, 2014). For example, customer engagement strategies and customer-centric approaches are closely tied to the number of repeat customers, loyalty, and revenue growth (Abrokwah-Larbi, 2023).

2.3 Menu Price

Menu prices have a direct influence on consumer demand and profitability, making them a crucial component of restaurant revenue management. Costs of ingredients, competition, and consumer opinions are all taken into account when setting prices (Kimes,2010). Techniques like dynamic pricing, which maximizes income during peak hours and increases off-peak traffic, modify rates in response to demand patterns (Chiang et al., 2007).

Charm pricing (e.g., $9.99 instead of $10) is an example of psychological pricing that affects consumer behavior and perceived value (Baker et al., 2015). Furthermore, different client demographics and cuisines have different price elasticity, fine-dining patrons are less price-sensitive than fast-casual diners (Jang et al., 2011). Advancements in technology now allow restaurants to optimize prices dynamically using data analytics, ensuring competitiveness and profitability.

2.4 Average Customer Spending

Revenue management in restaurants involves predicting customer demand to optimize inventory and pricing. Key strategies include capacity management, price management and duration management. These strategies help restaurants maximize revenue by efficiently utilizing space and adjusting prices based on demand (Tyagi & Bolia, 2021). Studies on average customer spending highlight various influencing factors, including pricing strategies, marketing efforts, customer satisfaction and socioeconomic conditions. For example, the COVID-19 pandemic significantly altered consumer spending patterns, shifting preferences and financial priorities, as observed globally and within specific markets like Slovakia (Valaskova et al., 2021).

CHAPTER 3: METHODOLOGY

3.1 Introduction

Methodology refers to the systematic design that is used to ensure the results are valid and reliable. Besides that, methodology is also used to address the research objectives and questions. Accordingly, research methodology shows the steps and the list of action undertaken to analyze the data within the study.

3.2 Multiple Linear Regression

Multiple regression analysis is among the most famously used statistical methods, particularly valuable in experimental situations where the predictor variables can be controlled by the experimenter. The multiple regression model involves one dependent variable and multiple independent variables, extending the basic linear regression framework to include two or more explanatory variables in the prediction equation for the response variable. Typically, the variables X1, X2, ..., Xp-1 in the model do not necessarily need to represent distinct predictor variables. The general form of the Multiple Linear Regression is as follows:

$$Y_i = \beta_0 + \beta_1 \mathrm{X}_1 + \beta_2 \mathrm{X}_2 + \cdots + \beta_{p-1} \mathrm{X}_{p-1} + \varepsilon_i$$

Where:

$Y_i =$ Dependent Variable

$\mathrm{X}_{1,} \mathrm{X}_{2,} \mathrm{X}_{p-1} =$ Independent Variable

$\beta_0, \beta_1, \beta_2, \beta_{p-1} =$ Parameters

$\varepsilon_i =$ Error Term

The aim of the usage of Multiple Linear Regression is to test whether the estimators $\beta_0, \beta_1, \beta_2, \beta_{p-1}$ are significant by using t-test.

$$t - test = \frac{\hat{\beta}_{\ i} - \beta_i}{s(\widehat{\beta_\iota})}$$

Where:

$\widehat{\beta_\iota} =$ Estimate Coefficient

$\beta_i =$ Hypothesis Value

$s(\widehat{\beta_\iota}) =$ Estimated Standard Error

With six assumptions as follows:

1. There is a linear relationship between each predictor variable and independent variable.
2. The predictors are independent of each other, or in other words, no multicollinearity.
3. There is no outlier in the data.
4. The error terms are normally distributed with a mean equal to zero.
5. The error variance is constant.
6. The error terms are independent.

3.2.1 Model Adequacy Checking for Multiple Linear Regression

Model adequacy checking is related to assumptions of regression models. If there are one of assumptions that do not fulfill, we need to apply transformation of variables to make sure the model is adequate to make analysis. There are six conditions that must be examined when we want to check for adequacy for multiple linear regression which are:

1. There is linear relationship between each predictor & dependent variable

- Scatter-plot matrix or partial plot is used to determine whether there is a relationship between dependent and independent variables.
- There is linear relationship if the points form a pattern that approximates a straight line with a positive slope, this suggests a positive linear relationship between independent variables (x_i) and dependent variable (y).

2. The error variable must be normally distributed

- P-P plot, histogram, and Kolmogorov Smirnov Test are used to examine whether error variable is normally distributed or not.
- Errors are normally distributed if:

Statistical Tool	**Conditions of errors are normally distributed**
P-P Plot	The plot of residuals close or on 45° line
Histogram	The histogram shows a bell-shaped curve
Kolmogorov Smirnov Test	p-value is greater than alpha (a) value

Table 2:Table of Conditions of Errors are Normally Distributed

3. The error variance must be constant

- Plot of residuals against predicted values is used to check whether the distribution of the error term (residuals) has constant variance or not. It means, we want to check the constancy of the error variance (Homoscedasticity).
- The distribution of the error terms has a constant if the error terms are randomly scattered and there is no obvious pattern.

Homoscedasticity in regression analysis refers to the condition where the residuals (errors) have constant variance across all levels of the independent variables, ensuring the reliability of the regression model's estimates and statistical inferences. It is a key assumption for the efficiency of ordinary least squares (OLS) estimators, valid hypothesis testing, and accurate confidence intervals.

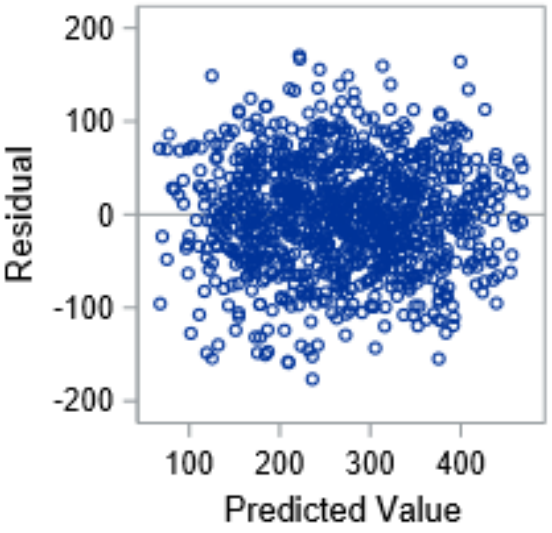


Figure 2: Scatter Plot Residuals against The Predicted Value

- Based on the graph residual versus predicted value for Restaurant Revenue in Figure 2, the pattern of the graph shows that it is randomly scattered. Therefore, it indicates that variance is constant (Homoscedasticity).

4. The predictors are independent of each other (No multicollinearity)

- Effect of Multicollinearity

a) Effects on the Regression Coefficient

The estimated regression coefficients tend to have large sampling variability and they vary depending on which predictor variables are included in the model. Many of the estimated regression coefficients individually may be statistically insignificant even though a definite statistical relationship may exist between the dependent and independent variables. Sometimes the sign of the regression coefficient may be wrong. We cannot interpret the regression coefficients as the change in the mean of the response variable when the given predictor variable is increased by one unit assuming the other variables are held constant.

b) Effects on Extra of Squares

The marginal contribution of any one predictor variable varies depending on which other variables are already in the regression model. Hence, there is no unique sum of squares that can be ascribed to any one predictor variable reflecting its effect in reducing the total variation in Y.

c) Effects on the standard error of Regression Coefficients (S_{β_k})

High correlation among the predictor variables causes the standard errors of the regression coefficients to become larger.

d) Effects on fitted Values and Predictions

High multicollinearity among predictor variables does not prevent the mean square error, measuring the variability of the error terms, from being steadily reduced as additional variables are added into the regression model. The precision of fitted values within the range of observations on the predictor is not affected with the addition of correlated variables into the regression model.

e) Effects on Simultaneous Tests of β_k

When testing the significance of each of the independent variables used in a multiple regression model, we normally use the *t* statistic. When predictor variables are highly correlated this might cause problems. The *t* values of each of the predictor variables may be so small that the test may indicate that the predictors are not significant in explaining Y (Since the standard error of the regression coefficients are large, this will cause that the t

values to be small). Even if the *F* statistic is very large indicating the significance of the model, the individual *t* tests may indicate many of the independent variables to be not significant.

- Method to detect Multicollinearity

a) Check the correlation matrix

The correlation matrix shows the correlation coefficients of each pair of data. If the correlation coefficient of any pair of predictor variables is more than 0.8, we can assume that there is multicollinearity.

b) The Variance Inflation factor (VIF)

A formal method of detecting the presence of multicollinearity is the variance inflation factor. This measures how much the variance of an estimated regression coefficients is inflated as compared to when predictor variables are not linearly related. Suppose X_1 has a VIF value of 6. This means that the variance of $\boldsymbol{\beta_1}$ has been increased six times more that what it would be if there is no multicollinearity.

The VIF is calculated for each of the predictor variable where $VIF_i = \frac{1}{1-R_i^2}$

R_i^2 is the value of R-square when each of the predictor variables is regressed on the remaining predictor variables. A VIF value greater than 10 indicates the presence of multicollinearity.

c) Tolerance Statistic

This is the reciprocal of the Variance Inflation Factor. (*TOL)* = $1 - R_i^2 = \frac{1}{(VIF)_t}$. A tolerance value of less than 0.2 indicates multicollinearity.

How to solve multicollinearity:

a) Drop one or more of the correlated independent variables. Stepwise regression may be used to select the best set of predictor variables.

b) If you decide to keep all the independent variables in the model.

- Avoid making inferences about individual β parameters.
- Restrict inferences about *E*(y) and future y values to within the range of values of the independent variables that were used in constructing the model.

c) Use a designed experiment.

d) To stabilize regression coefficients, use ridge regression to estimate the β parameters.

5. The error terms are independent

- Durbin-Watson is obtained to check whether the error terms are independent or not.
- Error terms are independent if the value of the Durbin Watson statistics is between 1.5 and 2.5. If the error terms are not independent, autocorrelation needs to be applied.

6. There is no outlier

Outliers are extreme values which can affect the coefficient of the regression line

6.1 Detection of outlying or extreme Y observation

- Studentized residuals (r_i) is used to detect outlying/extreme Y observations. Also can use studentized deleted residual.
- ($\pm$ 2se_i indicates outlier)

6.2 Detection of outlying X observation

- Leverage value is used to detect outlying x observations
- ($h_{ii} > \frac{2p}{n}$ indicates outlier)

6.3 Identifying Influential Cases

- **DFFITS**
  - To measure differences between the fitted value $\hat{Y}_i$, for i th case when all n cases are used in fitting the regression function and the predicted value $\hat{Y}_{i(i)}$ for the i th case obtained when the i th case omitted in fitting the regression function.
  - Influential case if |DFFITS| > $2\sqrt{\frac{p}{n}}$ (for large data)

- **DFBETAS**
  - To measure difference between the estimated regression coefficient b_k based on all n cases & the regression coefficient obtained when the i th case is omitted, denoted by $b_{k(i)}$
  - Influential case if |DFBETA| > $\frac{2}{\sqrt{n}}$ (for large data)
- **Cook's Distance**
  - To measure how much the residuals of all cases would change if a particular case were included from the calculation of the regression coefficients

- Influential case if D_i > $\frac{4}{n-k-1}$

3.2.2 Multiple Linear Regression Analysis

1. Finding standard error of the estimate (S_e)

The smaller the standard error of estimation the better the model.

Standard error of estimate (S_e) is used to know which model is the better. We can define the better model by looking at the smallest (S_e) because the smaller (S_e), the better the model. We also can consider the model is good if (S_e). is less than 10% of $\bar{y}$.

2. Testing the validity of the regression model

Before making the inference or using model for forecasting, the validity of the model should be tested. ANOVA test is being conducted to test the significance of the model. Unlike t-test, which are limited to comparing the means of two groups, ANOVA can handle comparisons across multiple groups simultaneously. This makes it a more efficient tool for experiments with more than two conditions, ANOVA also helps in identifying and partitioning the sources of variability within the data. It separates the total variance observed in the data into variance between groups and variance within groups, helping to understand if the group differences are greater than what would be expected by chance.

In multiple linear regression, the variability of dependent is called sum of squares total, SSTO. This variation in dependence is partly because of the independent variable. The part of the variation in dependence that is because of independence is called sum of squares regression, SSR. This is also known as the explained variation. The difference between the two values is unexplained part of the variation in dependent sum of squares error, SSE. Hence, the variation in dependent is composed of two components, SSR and SSE. The general ANOVA table for multiple linear regression analysis is shown in table 3.

Source of Variation	Sum of Squares (SS)	Degree of Freedom (df)	Mean Sum of Squares (MS)	F
Regression	$\sum(\hat{Y}-\bar{Y})^2$	$p-1$	$\frac{SSR}{p-1}$	$\frac{MSR}{MSE}$
Error	$\sum(Y-\bar{Y})^2$	$n-2$	$\frac{SSE}{n-2}$	
Total	$SSR+SSE$	$n-1$		

Table 3: ANOVA table

- Testing the validity of the model:

1) $H_0: \beta_0 = \beta_1 = \ \ldots\ldots = \beta_{p-1} = 0$

   $H_1: At\ least\ one\ \beta \neq 0$

2) Test Statistic

   $F = \frac{MSR}{MSE}$

3) Critical value at level of significance α

   $F_c = F_{a,p-1,n-p}$

4) Decision

   If $F > F_c$, Reject H_0

   Otherwise, Reject H_1

5) Conclusion

   The model is not valid as the test statistic is greater than critical value for a given α level and we reject H_0

   OR

   The model is valid as the test statistic is lower than critical value for a given α and we fail to reject H_0

- Testing the Significance of Each of the Independent Variable:

Test for β_k (two tailed test)

1) Hypothesis

   $H_0: \beta_k = 0$

   $H_1: \beta_k \neq 0$

2) Test Statistic

$$t - test = \frac{b_k - \beta_k}{s(b_k)}$$

3) Critical value at significance level α

$$t_c = t_{\frac{\alpha}{2}, n-p}$$

4) Decision

$|t| \leq t$, conclude H_0

Otherwise, conclude H_1

5) Conclusion

The model is not valid as the test statistic is greater than critical value for a given α level and we rejected H_0.

OR

The model is valid as the test statistic is lower than critical value for a given α level and we fail to reject H_0.

3. Testing the significance of each of the independent variables

- Significance if p-value < 0.05

4. Coefficient of Multiple Determination and The Adjusted Coefficient of Multiple Determination

The coefficient of multiple determination, denoted by R^2, is defined as follows

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

It indicates the proportion of the variation in the response variable, Y that is explained by the (p-1) predictor variable. R^2 can also be interpreted as the proportion reduction of the total variation in Y associated with the use of the set of variables $X_1, X_2, \ldots . X_{p-1}$.

Adding more X variables to the regression model can only increase R^2 and never reduce it, because SSE can never become larger with more X variables and SSTO is always the same for given set of responses. Since R^2 usually can be made larger by including a larger number of predictor variables, it is sometimes suggested that a modified measure be used that adjusts the number of X variables in the model. Hence, the adjusted coefficient of multiple determination, denoted by R_a^2 is used.

$$R_a^2 = 1 - \frac{SSE/n-p}{SSTO/n-1} = 1 - (\frac{n-1}{n-p})\frac{SSE}{SSTO}$$

The adjusted coefficient of multiple determination may become smaller when another X variable is added into the model, because any decrease in SSE may be offset by the loss of a degree of freedom in the denominator n-p. it measures the proportion of the variation in the response variable, Y that is explained by the (p-1) predictor variables after considering the samples size, n and the number of independent variables.

The differences between R^2 and R_a^2:

R^2	R_a^2
The value of R^2 generally increases as more explanatory variables are added to the regression model. R^2 can never be negative.	The value of R_a^2 may increase, decrease or stay the same as more explanatory variable are added to the regression model. If a new variable added to the regression model contributes significantly to explain the variation in Y, the R_a^2 increases, otherwise it decreases. R_a^2 c. an be negative

Table 4: Table of Comparison between R^2 and R_a^2

Interpretation of R^2

About R^2 % of the total variation in Y is explained by $X_1, X_2, \ldots . X_{p-1}$ while the balance $1 - R^2\%$ is explained by other factors.

Interpretation of R_a^2

After considering the sample size about $R^2\%$ of the total variation in Y is explained by $X_1, X_2, \ldots . X_{p-1}$ while the balance $1 - R^2\%$ is explained by other factors.

3.2.3 Selecting of Predictor Variables

In choosing the best predictor variable for model selection, there are four criterions can be used that are ${R^2}_p$, ${R^2}_a$, and C_p.

3.2.4 $R^2{}_p$ or SSE_p Criterion

This criterion uses the coefficient of multiple determination R^2, to identify several "good" subsets of X variables for which R^2 is high. $R^2{}_p$ indicates that there are p parameters (with p-1 predictor variables) in the regression function on which $R^2{}_p$ is based.

The R, criterion is equivalent to using the error sum of squares SSE_p criterion where subsets of regression models for which SSE_p is small is considered good.

$$R^2{}_p = 1 - \frac{SSE_p}{SSTO}$$

Since *SSTO* is constant for all possible regression models, $R^2{}_p$ varies inversely with SSE_p. The $R^2{}_p$ criterion is used to find the point where adding more X variables is not worthwhile because it leads to a very small increase in $R^2{}_p$.

3.2.5 $R^2{}_{a,p}$ or MSE_P Criterion

Since R, does not consider the number of parameters in the regression model and since max $R^2{}_p$ can never decrease as p increases, the adjusted coefficient of multiple determination is used as an alternative criterion.

$$R^2{}_{a,p} = 1 - \left(\frac{n-1}{n-p}\right)\frac{SSE_P}{SSTO} = 1 - \frac{MSE_P}{\frac{SSTO}{n-1}}$$

$R^2{}_{a,p}$ ta kes the number of parameters in the regression model through the degrees of freedom.

$R^2{}_{a,p}$ only increases if MSE_P decreases. The largest $R^2{}_p$ for a given number of parameters can decrease as *p* increases. This is because the increase in max ($R^2{}_p$) becomes so small that it is not sufficient to offset the loss of an additional degree of freedom. Based on the above table max ($R^2{}_{a,p}$) occurs when X_1, X_2, and X_3 are selected.

3.2.6 Mallow's C_p Criterion

This criterion is concerned with the total mean squared error of the *n* fitted values for each subset regression model.

$$C_p = \frac{SSE_p}{MSE(X_{1,\ldots,}X_{P-1})} - (n - 2p)$$

In using the C_p criterion, we seek to identify subsets of X variables for which the C_p value is small and the C_p is near p. Subsets with small C_p values have a small total mean squared error,

and when the C_p value is also near p, the bias of the regression model is small. Based on this criterion, we see that X_1, X_2, and X_3 are the variables that should be selected for the model.

3.2.7 Press P Criterion

The $PRESS_p$ (prediction sum of squares) criterion is a measure of how well the fitted values for a subset model can predict the observed responses Y_i. Each fitted value $\widehat{Y}_i$ for the *PRESS* criterion is obtained by deleting the *i*th case from the data set, estimating the regression function for the subset model from the remaining n-1 cases, and then using the fitted regression function to obtain the predicted value $\widehat{Y_{i(i)}}$ for the *ith* case. $\widehat{Y_{i(i)}}$ is the predicted value for the *i*th case where the *i*th case was omitted when the regression function was fitted. Hence, the PRESS prediction error for the *i*th case is

$$Y_i - \widehat{Y_{i(i)}}$$

And the PRESS_p, criterion is the sum of the squared prediction errors over all n cases.

$$\mathrm{PRESS}_p = = \sum_{i=1}^{n} (Y_i - \widehat{Y_{i(i)}})^2$$

Models with small PRESS_p values are considered good candidate models. The reason is that when the prediction errors are small, so are the squared prediction errors and the sum of the squared prediction errors. Again, using the table above we find that the model with X_1, X_2, and X_3 has the smallest PRESS_p value.

CHAPTER 4: ANALYSIS AND FINDINGS

4.1 FIRST TIME OF CHECKING ADEQUACY

4.1.1 Model Adequacy Checking for Original Data

4.1.2 Linearity of The Model

Pearson Correlation Coefficients, N = 1000 Prob > |r| under H0: Rho=0				
	Number_of_Customers	Menu_Price	Average_Customer_Spending	Monthly_Revenue
Number_of_Customers	1.00000	0.03347 0.2903	-0.01390 0.6607	0.74524 <.0001
Menu_Price	0.03347 0.2903	1.00000	0.02081 0.5110	0.25958 <.0001
Average_Customer_Spending	-0.01390 0.6607	0.02081 0.5110	1.00000	-0.03566 0.2598
Monthly_Revenue	0.74524 <.0001	0.25958 <.0001	-0.03566 0.2598	1.00000

Table 5: Pearson Correlation Coefficient for Second Output Model

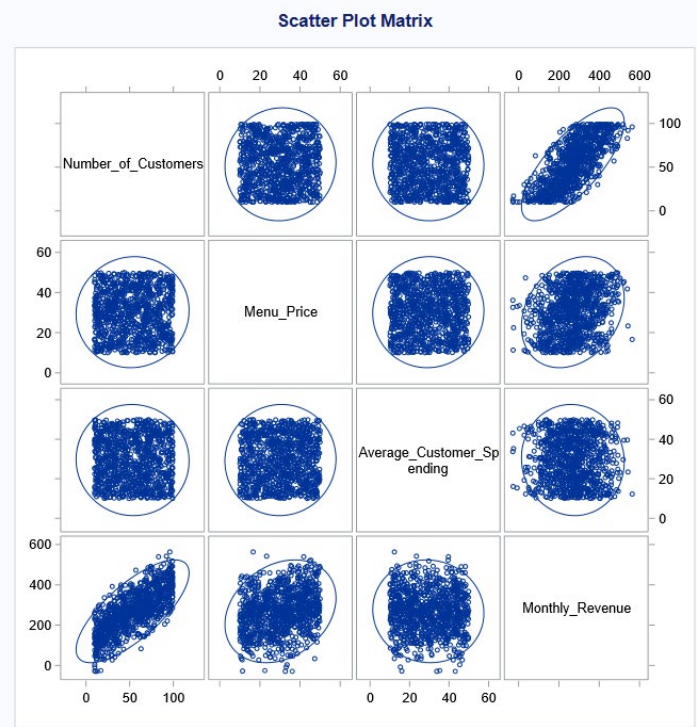


Figure 3: Scatter Plot Matrix for Second Output

Based on the figure above, for the relationship between the number of customers and monthly revenue, the value indicates there is a strong positive linear relationship. Next, for the relationship between the number of customers and menu price, the value indicates there is a very weak positive linear relationship. Regarding the relationship between the number of customers and average customer spending, the value indicates there is a very weak negative

linear relationship. Meanwhile, for the relationship between the monthly revenue and menu price, the value indicates there is a weak positive linear relationship. Lastly, for the relationship between the monthly revenue and average customer spending, the value indicates there is a very weak negative linear relationship.

4.1.3 Normality of Error Variable

P-P Plot

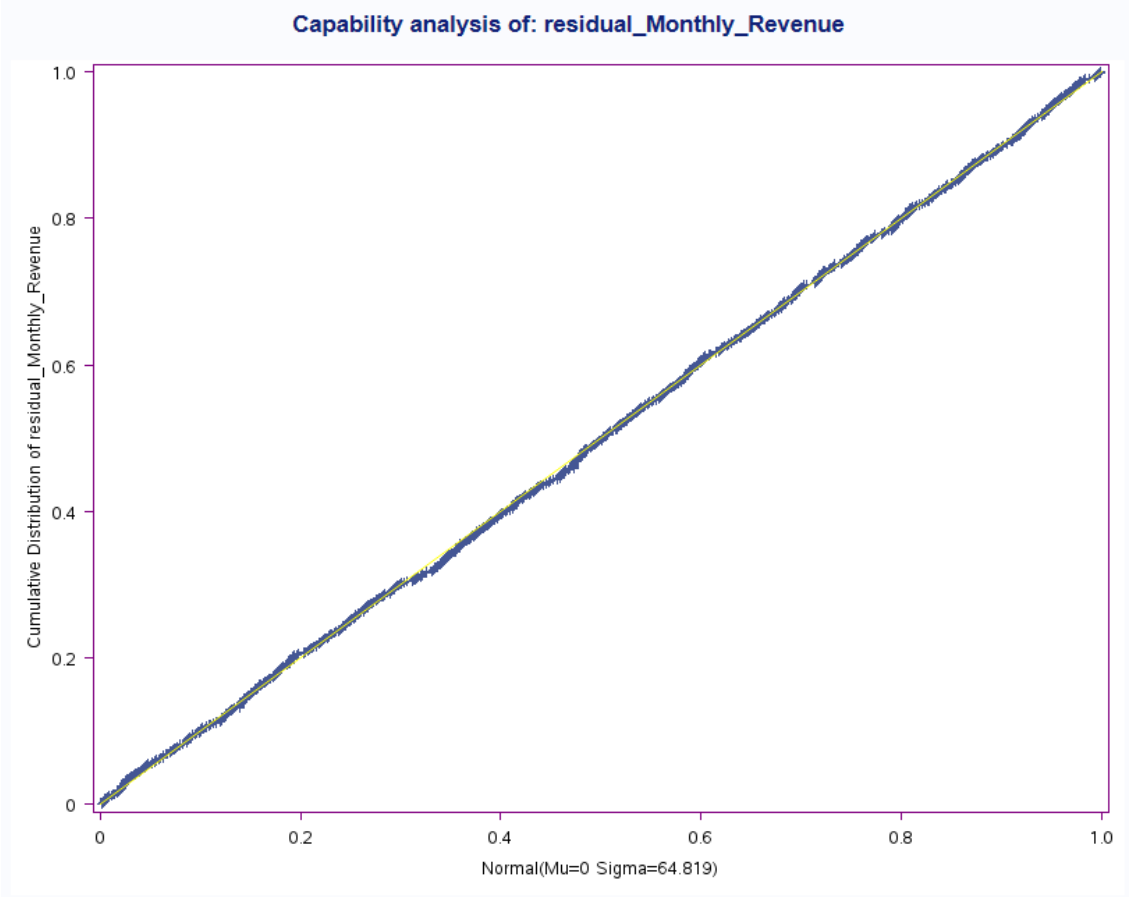


Figure 4:P-P Plot of Residual

The normal probability plot of residuals produces a plot of the residuals against their probability on a 45-degree line. Since the plots are close to or on the 45-degree line, we can assume that the errors are normally distributed.

Histogram

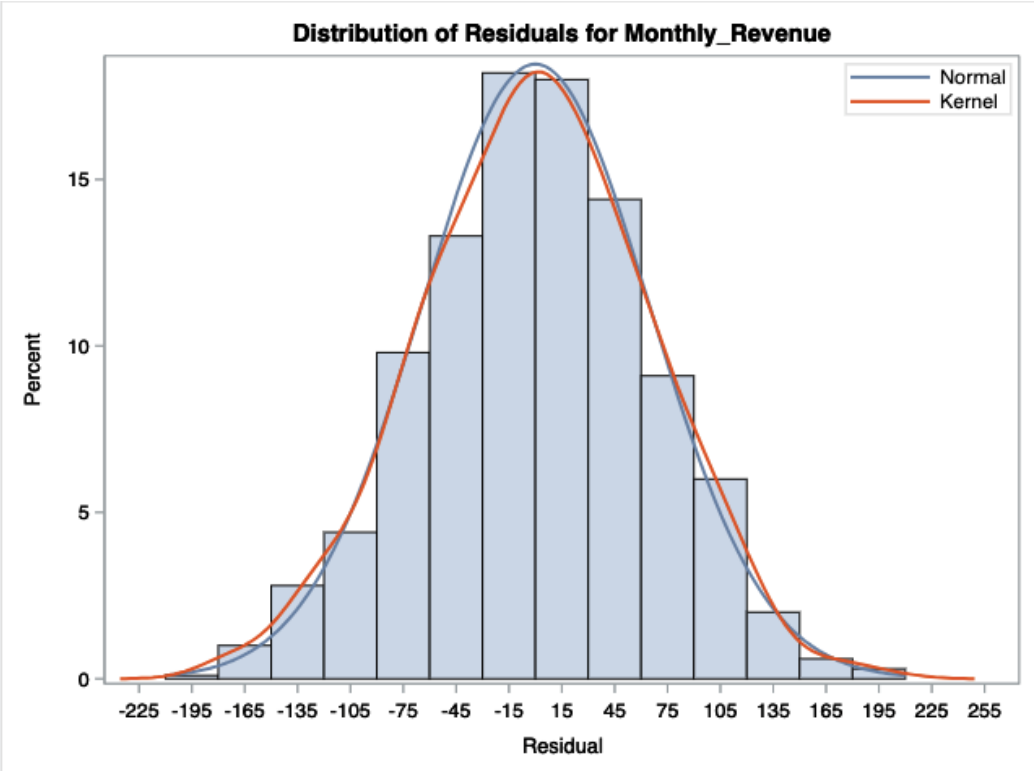


Figure 5: Histogram of Residual

Based on Figure 5, it shows that the histogram shows a bell-shaped curve. So, we can assume the residual is normally distributed.

Kolmogorov Smirnov

Goodness-of-Fit Tests for Normal Distribution				
Test	**Statistic**		**p Value**	
Kolmogorov-Smirnov	**D**	0.01458460	**Pr > D**	>0.150
Cramer-von Mises	**W-Sq**	0.01871405	**Pr > W-Sq**	>0.250
Anderson-Darling	**A-Sq**	0.15744849	**Pr > A-Sq**	>0.250

Table 6: Goodness of Fit Tests for MLR

Based on the P-P plot in Figure 4, it shows that the points lie close 45-degree line. Hence, it does follow a normal distribution. Referring to Histogram of residual in Figure 5, the histogram did show a bell-shaped curve. So, it is normally distributed. Furthermore, Kolmogorov-Smirnov test in Table 6 is also used to support the result from P-P plot and Histogram of residual. Since p-value is > 0.150 is more than the significance level of $\alpha = 0.05$, therefore the residuals do follow a normal distribution.

4.1.4 Checking the constancy of error variance

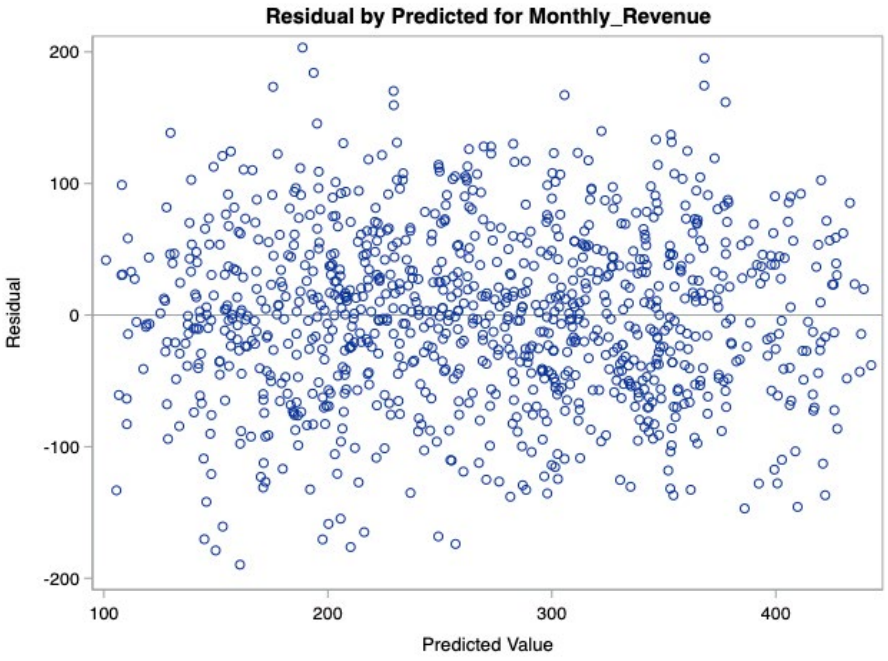


Figure 6: Plot of Residual against Predicted Value

Based on the graph residual versus predicted value for monthly revenue in Figure 6, the pattern of the graph shows that it is randomly scattered. It also does not have any obvious pattern. Therefore, it indicates that the error variance is constant (Homoscedasticity).

4.1.5 Detecting the multicollinearity

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	56.37349	8.80441	6.40	<.0001	.	0
Average_Customer_Spending	1	-0.27487	0.17910	-1.53	0.1252	0.99935	1.00065
Menu_Price	1	2.17157	0.18224	11.92	<.0001	0.99843	1.00158
Number_of_Customers	1	2.90646	0.07795	37.28	<.0001	0.99867	1.00134

Table 7: Parameter estimates for MLR

Multicollinearity occurs when independent variables used in a multiple regression model are highly correlated. To detect multicollinearity, we used Variance Inflation Factor (VIF) and Tolerance Statistics (TOL). A VIF value greater than 5 indicates the presence of multicollinearity meanwhile for a tolerance value less than 0.2 indicates multicollinearity. Based on the VIF and TOL in Table 7, there is no presence of multicollinearity for any independent variables.

4.1.6 Checking independence of error term

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: Monthly_Revenue

Durbin-Watson D	2.141
Number of Observations	1000
1st Order Autocorrelation	-0.071

Table 8: Autocorrelation for MLR

From Table 8, the Durbin-Watson value is 2.141 which indicates that there is no need for an autocorrelation in the model because it is between the interval of 1.5 and 2.5. Therefore, the error terms are independent.

4.1.7 Detection of outlier

Studentized residual

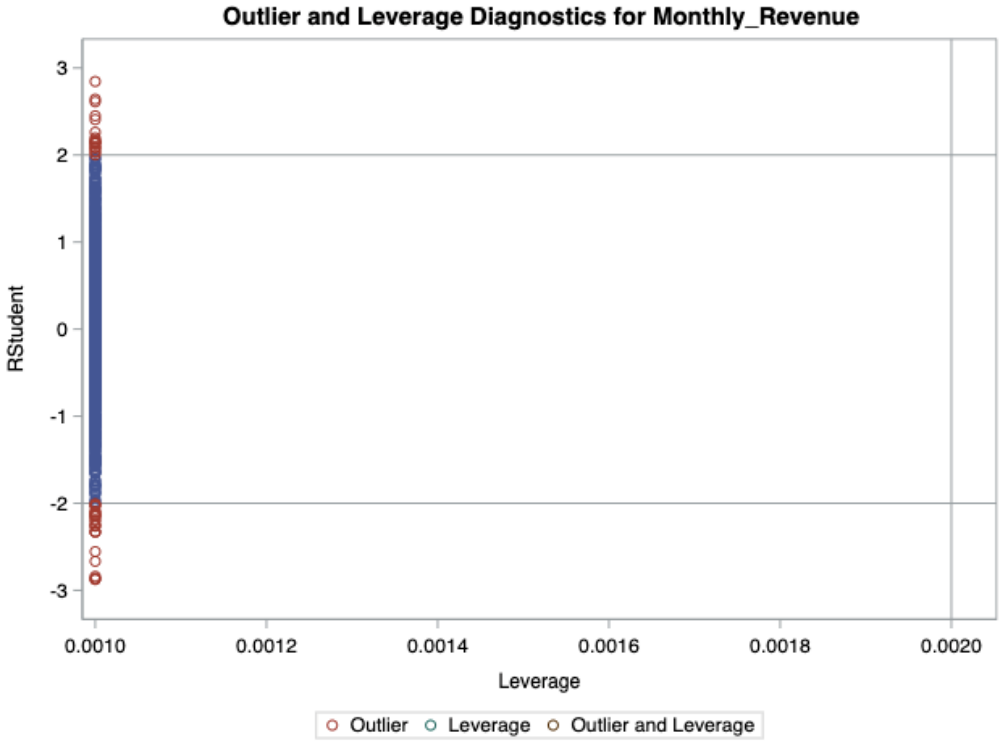


Figure 7: Studentized Residuals

From Figure 7, the residuals are randomly scattered around zero without discernible patterns, indicating that the model satisfies the assumptions of linearity and homoscedasticity. No residuals exceed the threshold of ± 3 , suggesting there are outliers influencing the model.

Leverage Value

$$h_{ii} > \frac{2p}{n}$$

$$h_{ii} > \frac{2(4)}{1000}$$

$$h_{ii} > 0.008$$

DFFITS

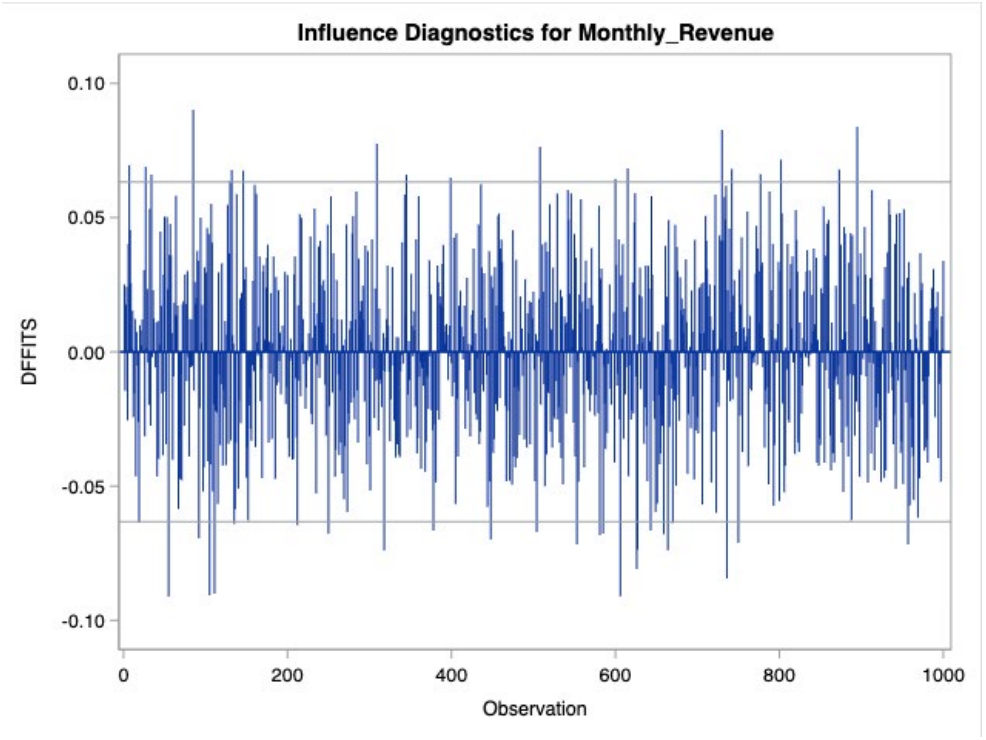


Figure 8: DFFITS

This measures the difference between the fitted value $\hat{Y}_i$ for the i th case when all n cases are used in fitting the regression function and the predicted value $\widehat{Y_{i(t)}}$ for the ith case obtained when the ith case is omitted in fitting the regression function. DFFITS for each case can be obtained as follows.

$$DFFITS_i = t_i \left(\frac{h_{ii}}{1 - h_{ii}}\right)^{\frac{1}{2}}$$

As a guideline for identifying influential cases:

$$|DFFITS| > 2\sqrt{\frac{p}{n}} \quad \text{for large data sets}$$

$$|DFFITS| > 2\sqrt{\frac{4}{1000}}$$

$$|DFFITS| > 0.1265$$

DFBETAS

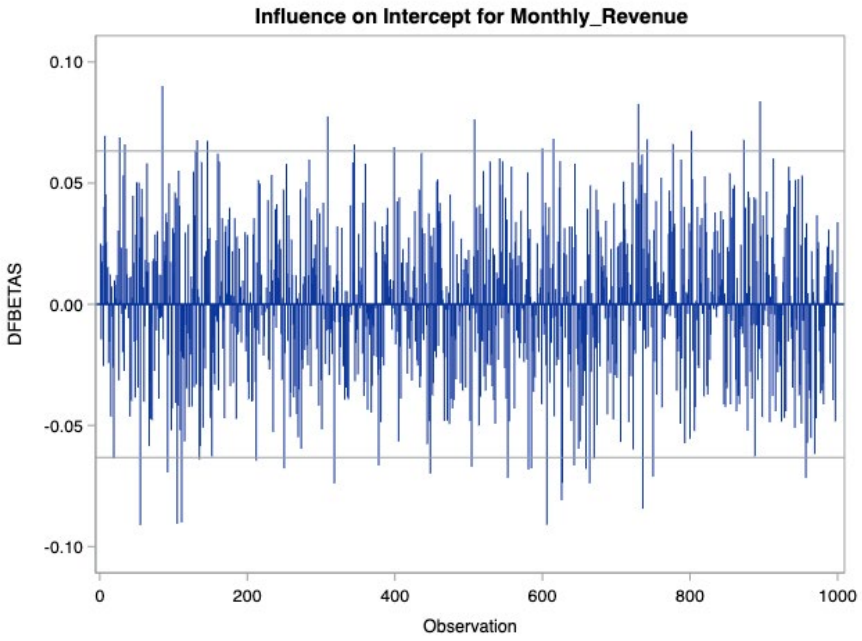


Figure 9: DFBETAS

This measures the difference between the estimated regression coefficient b_k based on all n cases and the regression coefficient obtained when the i th case is omitted, denoted by $b_{k(i)}$.

$$(DFBETAS)_{k(i)} = \frac{b_k - b_{k(i)}}{\sqrt{MSE_{(i)}c_{kk}}}$$

Where C_{kk} is the k th diagonal element $(X'X)^{-1}$

As a guideline for identifying influential cases:

$|DFBETA| > \frac{2}{\sqrt{n}}$ for large data sets

$$|DFBETA| > \frac{2}{\sqrt{1000}}$$

$$|DFBETA| > 0.06325$$

Cook's Distance

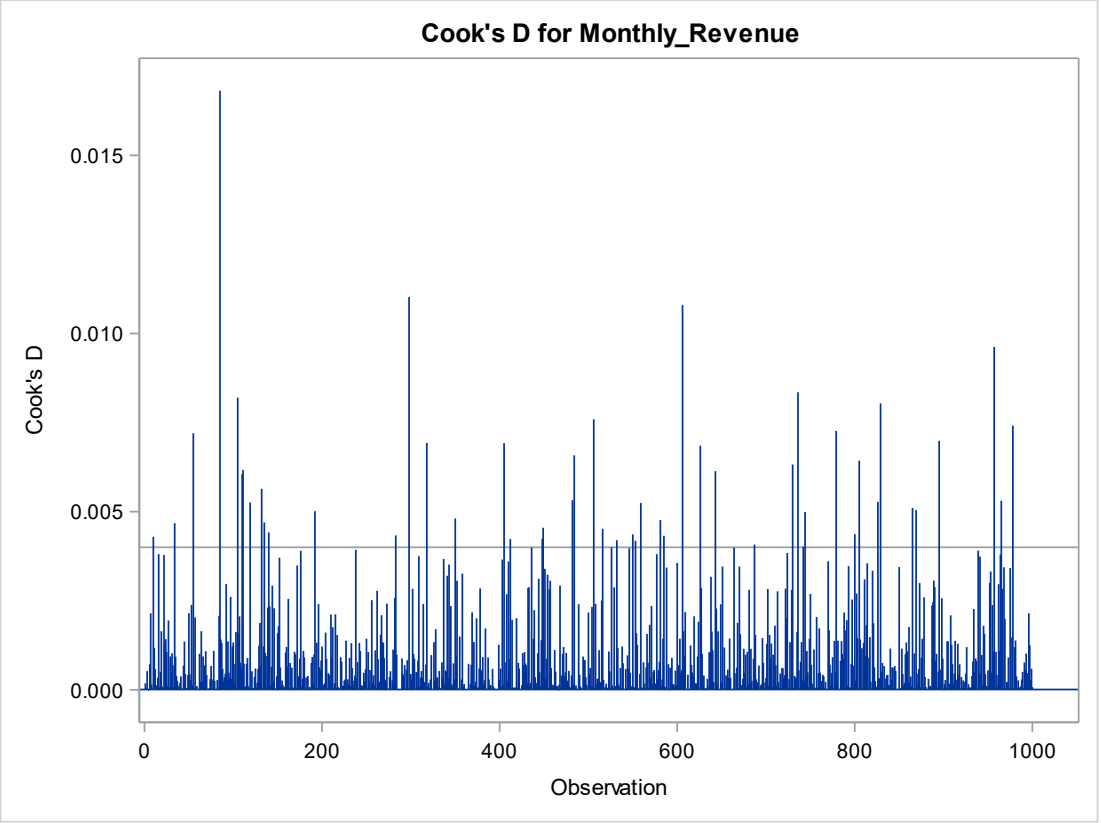


Figure 10: Cook's Distance

Cook's distance D_i measures how much the residuals of all cases would change if a particular case were excluded from the calculation of the calculation of the regression coefficients. A large Cook's D indicates that excluding a case form computation of the regression statistics, changes the coefficients substantially.

$$D_i = \frac{e_i^2}{pMSE}\left[\frac{h_{ii}}{1-h_{ii}}\right]$$

For interpreting Cook's measure D_i, we relate this measure to F (p,n-p) distribution and ascertain the corresponding percentile value. If the percentile value is 20 percent, the ith case has little influence on the fitted value. But if it is more than 50 percent, it means that the observation is influential. The rule of thumb is

$$D_i > \frac{4}{n-k-1} \text{ the case is influential}$$

$$D_i > \frac{4}{1000-3-1}$$

$$D_i > 0.00402$$

Process of remove outliers (68 outliers)

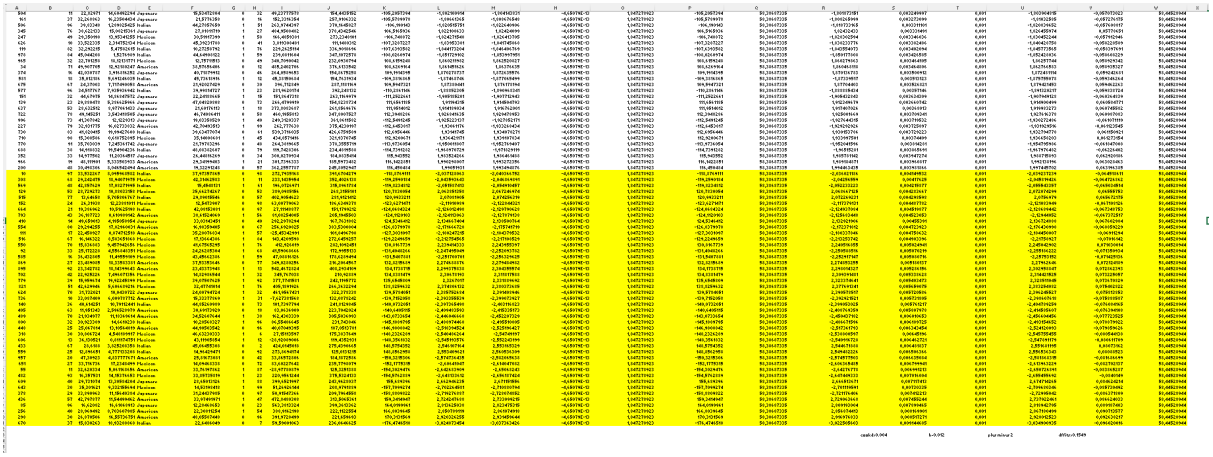


Table 9: Derived First Outlier

4.2 SECOND TIME OF CHECKING ADEQUACY

4.2.1 Model Adequacy Checking for Data After Remove Outlier (First Time)

4.2.2 Linearity of The Model

Pearson Correlation Coefficients, N = 932 Prob > |r| under H0: Rho=0				
	Number_of_Customers	Menu_Price	Average_Customer_Spending	Monthly_Revenue
Number_of_Customers	1.00000	-0.00181 0.9561	-0.01596 0.6265	0.75647 <.0001
Menu_Price	-0.00181 0.9561	1.00000	0.00388 0.9058	0.24952 <.0001
Average_Customer_Spending	-0.01596 0.6265	0.00388 0.9058	1.00000	-0.03498 0.2861
Monthly_Revenue	0.75647 <.0001	0.24952 <.0001	-0.03498 0.2861	1.00000

Table 10: Pearson Correlation Coefficient for Second Output Model

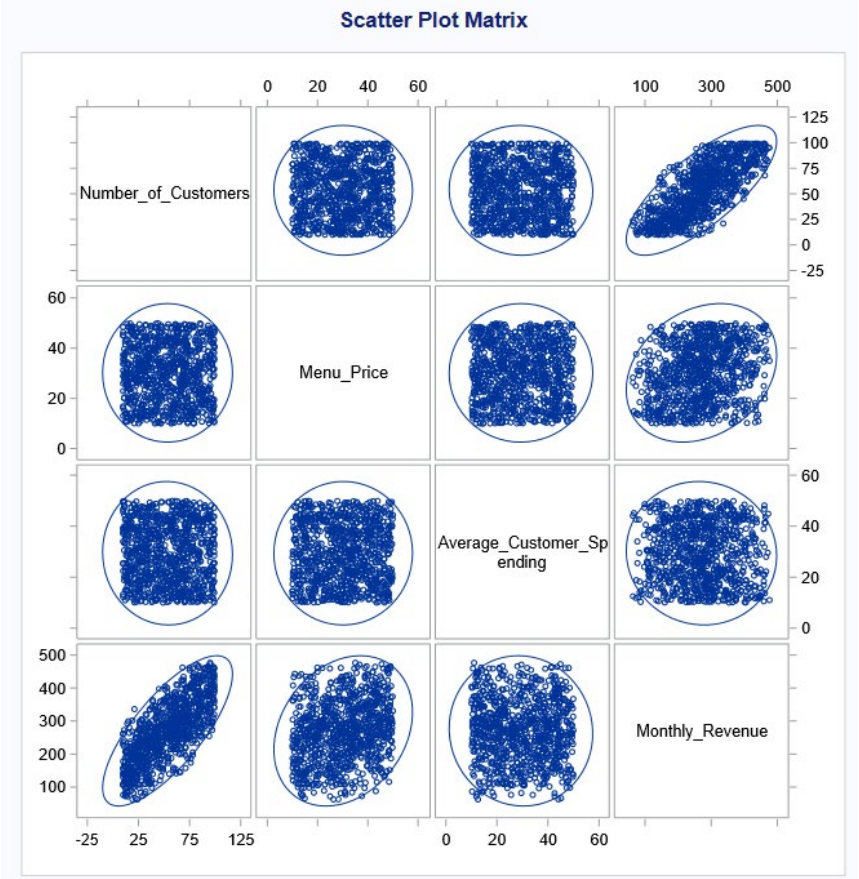


Figure 11: Scatter Plot Matrix for Second Output

Based Figure 11, for the relationship between the number of customers and monthly revenue, the value indicates there is a strong positive linear relationship. Next, for the relationship between the number of customers and menu price, the value indicates there is a very weak negative linear relationship. Lastly, regarding the relationship between the number of

customers and average customer spending, the value indicates there is a very weak negative linear relationship. Meanwhile, for the relationship between monthly revenue and menu price, the value indicates there is a weak positive linear relationship, and for the relationship between monthly revenue and average customer spending, the value indicates there is a very weak negative linear relationship.

4.2.3 Normality of Error Variable

P-P Plot

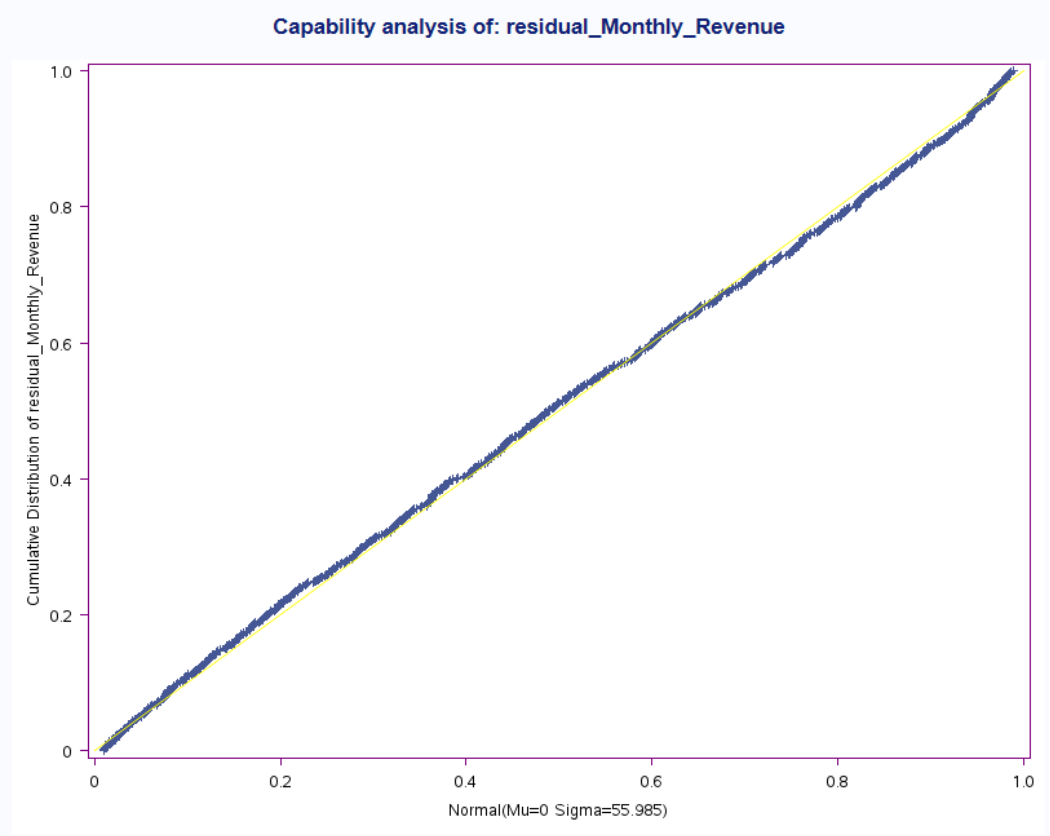


Figure 12: P-P Plot of Residual

Based on Figure 12, the normal probability plot of residuals produces a plot of the residuals against their probability on a 45-degree line. Since the plots are close to or on the 45-degree line, we can assume that the errors are normally distributed.

Histogram

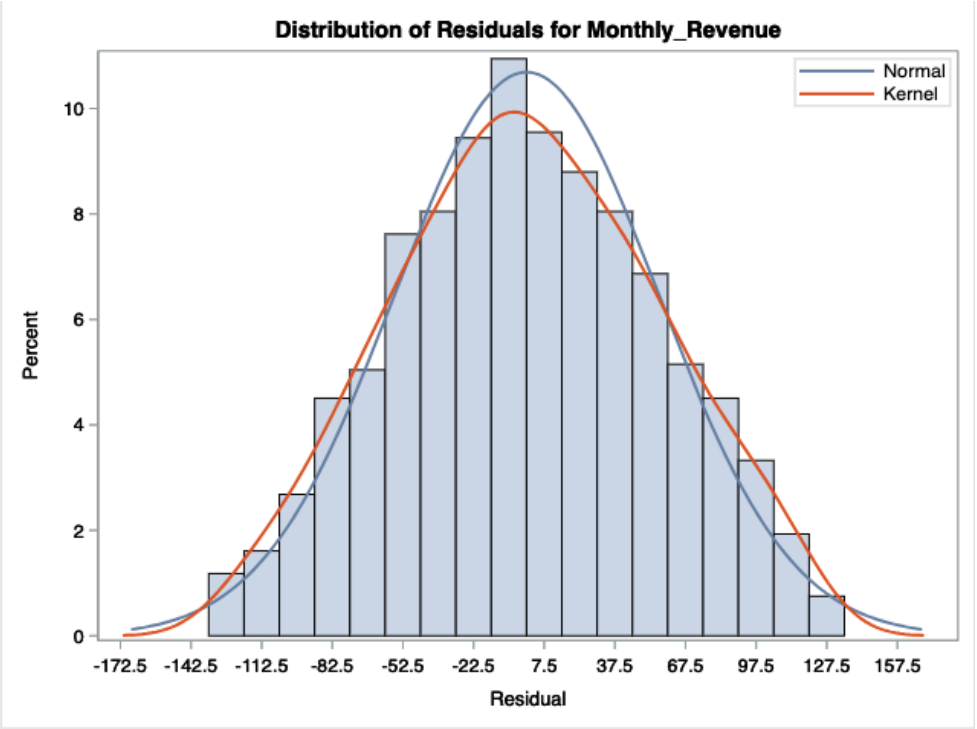


Figure 13: Histogram of Residual

Based on Figure 13, it shows that the histogram shows a bell-shaped curve. So, we can assume the residual is normally distributed.

Kolmogorov Smirnov

Goodness-of-Fit Tests for Normal Distribution				
Test	Statistic		p Value	
Kolmogorov-Smirnov	D	0.02181855	Pr > D	>0.150
Cramer-von Mises	W-Sq	0.09455623	Pr > W-Sq	0.136
Anderson-Darling	A-Sq	0.85357345	Pr > A-Sq	0.029

Table 11: Goodness of Fit Tests for MLR

Based on the P-P plot in Figure 12, it shows that the points lie close 45-degree line. Hence, it does follow a normal distribution. Referring to Histogram of residual in Figure 13, the histogram did show a bell-shaped curve. So, it is normally distributed. Furthermore, Kolmogorov-Smirnov test in Table 11 is also used to support the result from P-P plot and Histogram of residual. Since p-value $is > 0.150$ is more than the significance level of $\alpha = 0.05$, therefore the residuals do follow a normal distribution.

4.2.4 Checking the constancy of error variance

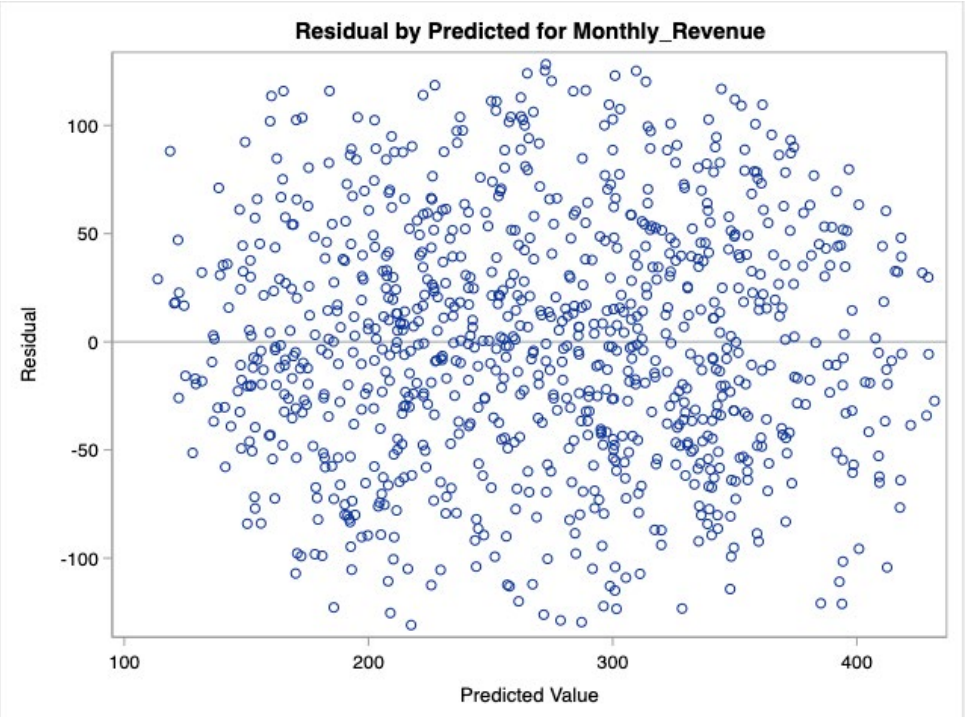


Figure 13: Plot of Residual Against Predicted Value

Based on the graph residual versus predicted value for monthly revenue in Figure 14, the pattern of the graph shows that it is randomly scattered. It also does not have any obvious pattern. Therefore, it indicates that the error variance is constant (Homoscedasticity).

4.2.5 Detecting the multicollinearity

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	68.61265	8.05015	8.52	<.0001	.	0
Average_Customer_Spending	1	-0.19328	0.16039	-1.21	0.2285	0.99973	1.00027
Menu_Price	1	2.07019	0.16342	12.67	<.0001	0.99998	1.00002
Number_of_Customers	1	2.71379	0.07108	38.18	<.0001	0.99974	1.00026

Table 12: Parameter estimates for MLR

Multicollinearity occurs when independent variables used in a multiple regression model are highly correlated. To detect multicollinearity, we used Variance Inflation Factor (VIF) and Tolerance Statistics (TOL). A VIF value greater than 5 indicates the presence of multicollinearity meanwhile for a tolerance value less than 0.2 indicates multicollinearity. Based on the VIF and TOL in Table 12, there is no presence of multicollinearity for any independent variables.

4.2.6 Checking independence of error term

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: Monthly_Revenue

Durbin-Watson D	0.017
Number of Observations	932
1st Order Autocorrelation	0.986

Table 13: Autocorrelation for MLR

From Table 13, the Durbin-Watson value is 0.017 which indicates that there is need for an autocorrelation in the model because it is not between the interval of 1.5 and 2.5. Therefore, the error terms are not independent.

4.2.6.1 Autocorrelation

Estimates of Autocorrelations

Lag	Covariance	Correlation	-1.0 -0.5 0.0 0.5 1.0
0	3130.9115	1.0000	
1	3086.6001	0.9858	
2	3070.3259	0.9806	

Preliminary MSE	79.4576

Estimates of Autoregressive Parameters

Lag	Coefficient	Standard Error	t Value
1	-0.678763	0.031177	-21.77
2	-0.311493	0.031177	-9.99

Figure 14: Process of autocorrelation

Yule-Walker Estimates			
SSE	39486.9986	DFE	929
MSE	42.50484	Root MSE	6.51957
SBC	6160.82849	AIC	6146.31649
MAE	5.32998022	AICC	6146.34235
MAPE	53.1344577	HQC	6151.85088
Durbin-Watson	1.9622	Transformed Regression R-Square	0.0000
		Total R-Square	0.9865

Table 14: Durbin-Watson after autocorrelation

After doing autocorrelation second time (second order), the Durbin-Watson value is 1.9622 which indicates that there is no need for an autocorrelation anymore in the

model because it is between the interval of 1.5 and 2.5. Therefore, the error terms are independent.

4.2.7 Detection of outlier

Studentized Residual

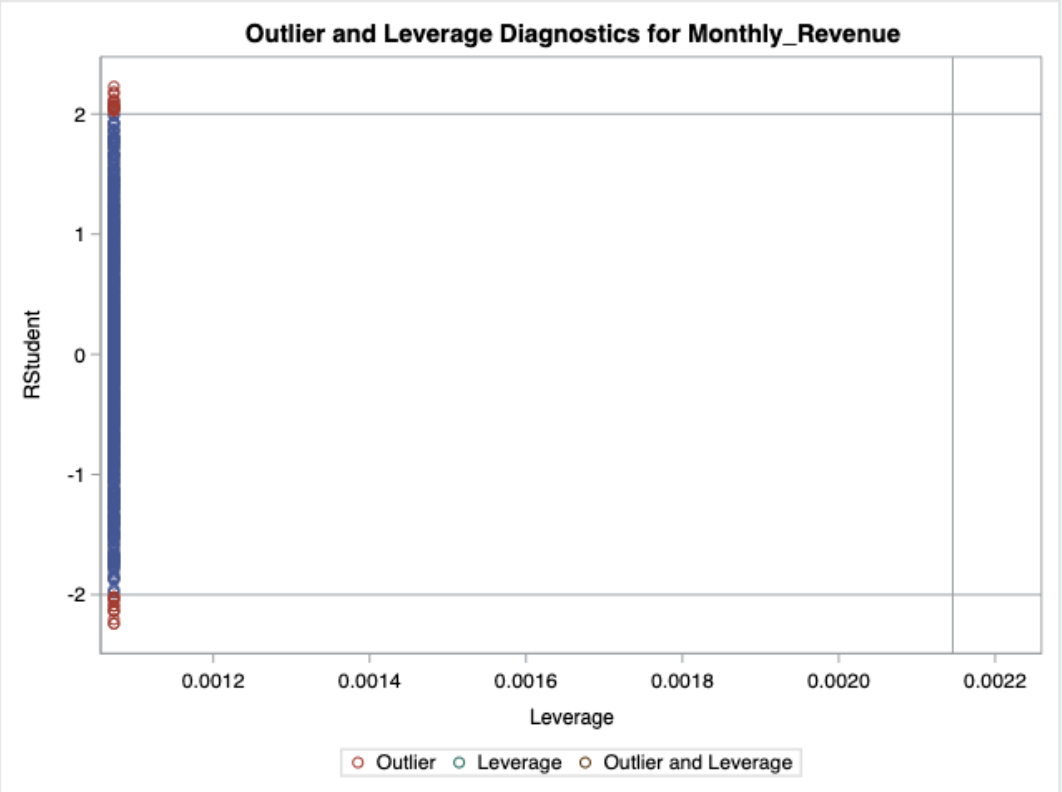


Figure 15: Studentized Residual

From Figure 15, the residuals are randomly scattered around zero without discernible patterns, indicating that the model satisfies the assumptions of linearity and homoscedasticity. No residuals exceed the threshold of ± 3 , suggesting there are no significant outliers influencing the model.

Leverage Value

$$h > \frac{2p}{n}$$

$$h > \frac{2(4)}{932}$$

$$h > 0.008584$$

DFFITS

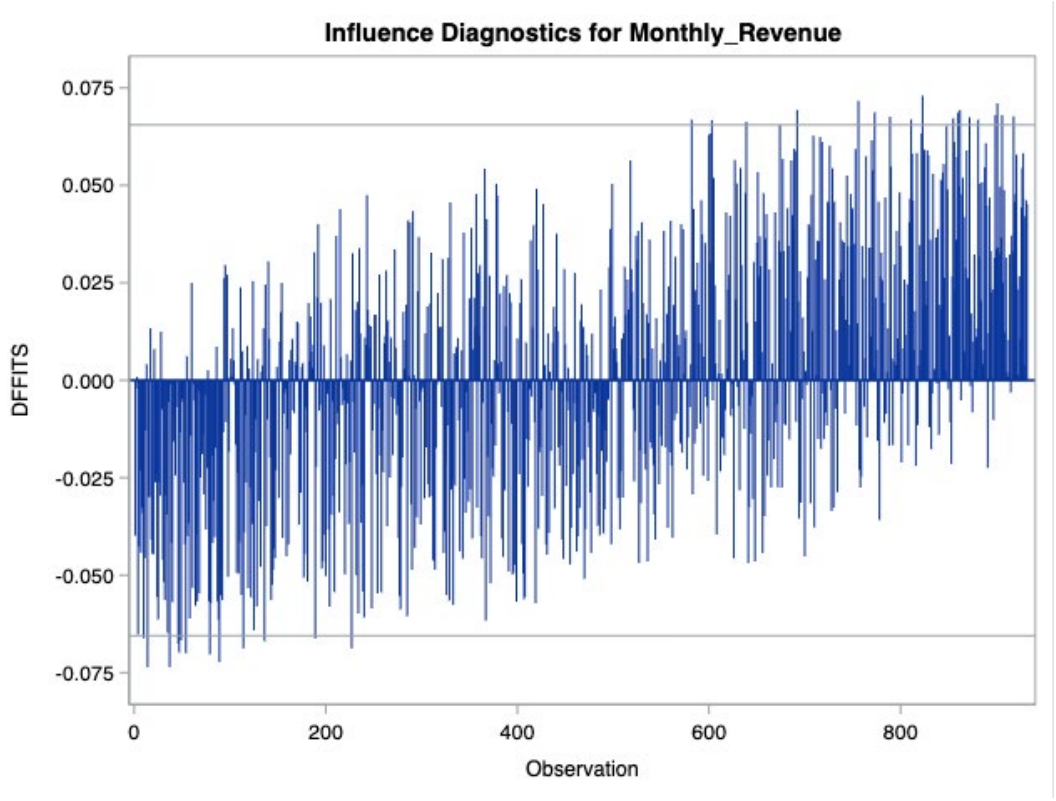


Figure 16: DFFITS

This measures the difference between the fitted value for the ith case when all n cases are used in fitting the regression function and the predicted value for the case obtained when the ith case is omitted in fitting the regression function. DFFITS for each case can be obtained as follows.

$$DFFITS_i = t_i \left(\frac{h_{ii}}{1 - h_{ii}}\right)^{\frac{1}{2}}$$

As a guideline for identifying influential cases:

$$|DFFITS| > 2\sqrt{\frac{p}{n}} \text{ for large datasets}$$

$$|DFFITS| > 2\sqrt{\frac{4}{932}}$$

$$|DFFITS| > 0.1310$$

DFBETAS

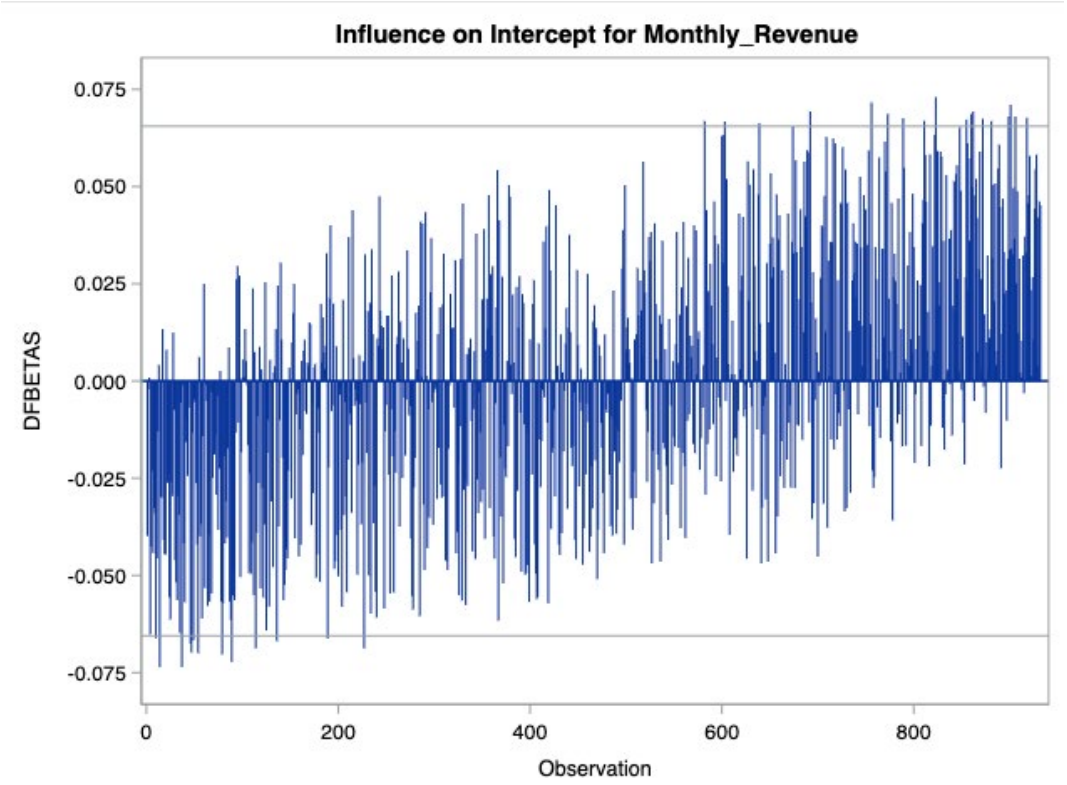


Figure 17: DFBETAS

This measures the difference between the estimated regression coefficient based on all n cases and the regression coefficient obtained when the i th case is omitted, denoted by $b_{k(i)}$.

$$(DFBETAS)_{k(i)} = \frac{b_k - b_{k(i)}}{\sqrt{MSE_{(i)} c_{kk}}}$$

Where is the k th diagonal element $(X'X)^{-1}$

As a guideline for identifying influential cases:

$$|DFBETA| > \frac{2}{\sqrt{n}} \text{ for large datasets}$$

$$|DFBETA| > \frac{2}{\sqrt{932}}$$

$$|DFBETA| > 0.06551$$

Cook's Distance

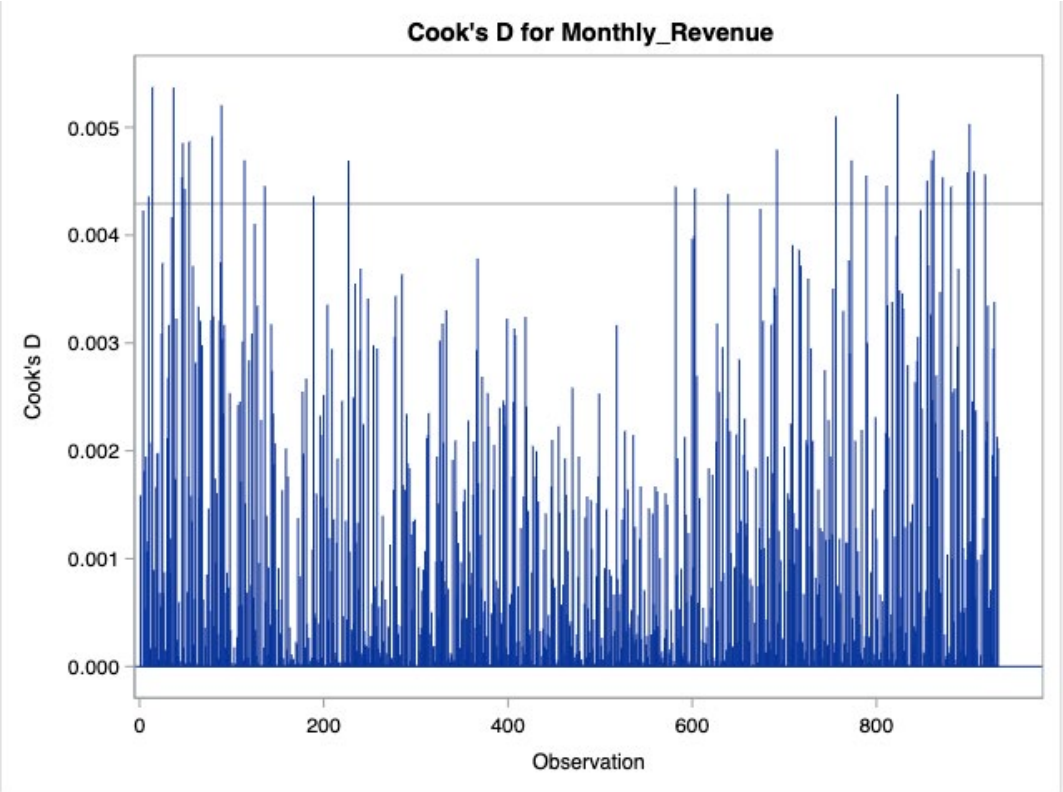


Figure 18: Cook's Distance

Cook's distance D_i measures how much the residuals of all cases would change if a particular case were excluded from the calculation of the calculation of the regression coefficients. A large Cook's D indicates that excluding a case form computation of the regression statistics, changes the coefficients substantially.

$$D_i = \frac{e_i^2}{pMSE}\left[\frac{h_{ii}}{1-h_{ii}}\right]$$

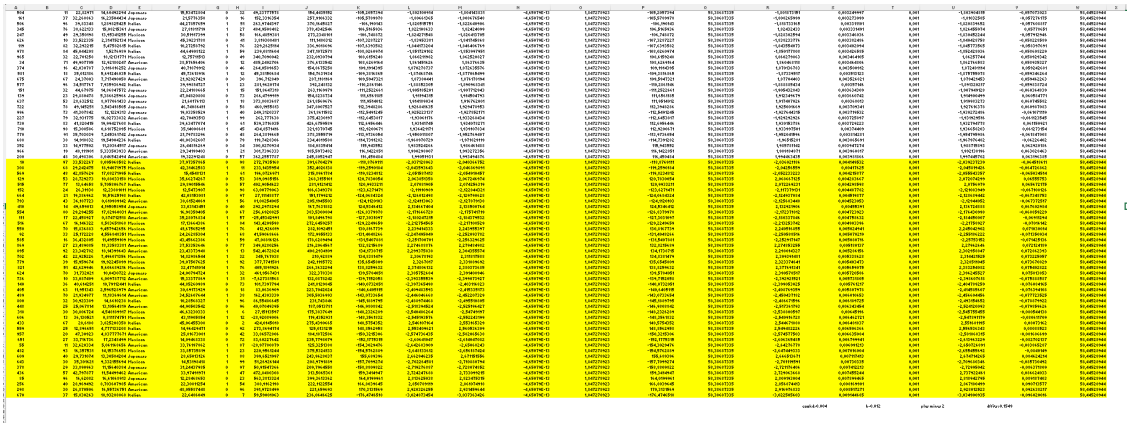
For interpreting Cook's measure D_i, we relate this measure to F (p,n-p) distribution and ascertain the corresponding percentile value. If the percentile value is 20 percent, the i th case has little influence on the fitted value. But if it is more than 50 percent, it means that the observation is influential. The rule of thumb is

$D_i > \frac{4}{n-k-1}$ the case is influential

$$D_i > \frac{4}{932-3-1}$$

$$D_i > 0.00431$$

Process of checking outliers



	AD	AE	AF	AG	AH	AI	AJ	AK
1	ual_Monthly_Revenue3	stdr_Monthly_Revenue	student_Monthly_Revenue3	cookd_Monthly_Revenue	h_Monthly_Revenue	rstudent_Monthly_Revenue3	dffits_Monthly_Revenue	stdi_Monthly_Revenue
2	-298,8306253	98,7933758	-3,024804274	0,009570545	0,001044932	-3,037793467	-0,098249262	98,89666216
3	-267,4010408	98,7933758	-2,706669741	0,007663244	0,001044932	-2,715679215	-0,087831343	98,89666216
4	-243,8681294	98,7933758	-2,468466407	0,006373772	0,001044932	-2,475075423	-0,08004966	98,89666216
5	-236,7261387	98,7933758	-2,396174205	0,006005911	0,001044932	-2,402145046	-0,077690923	98,89666216
6	-235,7302084	98,7933758	-2,386093263	0,005955482	0,001044932	-2,391978309	-0,077362107	98,89666216
7	-225,8388425	98,7933758	-2,28597151	0,005466178	0,001044932	-2,291045818	-0,074097717	98,89666216
8	-225,256954	98,7933758	-2,280081556	0,005438046	0,001044932	-2,28511047	-0,073905754	98,89666216
9	-224,5452135	98,7933758	-2,272877222	0,005403735	0,001044932	-2,277850956	-0,073670965	98,89666216
10	-221,9825735	98,7933758	-2,246937831	0,005281098	0,001044932	-2,251715984	-0,072825699	98,89666216
11	-213,9009296	98,7933758	-2,16513433	0,004903563	0,001044932	-2,169326894	-0,070161046	98,89666216
12	-212,3105176	98,7933758	-2,149035964	0,004830916	0,001044932	-2,153118761	-0,069636837	98,89666216
13	-210,8471286	98,7933758	-2,134223341	0,004764549	0,001044932	-2,138206701	-0,069154547	98,89666216
14	-208,0936399	98,7933758	-2,106352154	0,00464092	0,001044932	-2,110152435	-0,068247207	98,89666216
15	-204,9120467	98,7933758	-2,074147634	0,004500092	0,001044932	-2,077742834	-0,067199006	98,89666216
16	-199,1518439	98,7933758	-2,015842077	0,004250648	0,001044932	-2,019083266	-0,06530182	98,89666216
17	-198,1718123	98,7933758	-2,005922064	0,004208916	0,001044932	-2,009105203	-0,064979106	98,89666216
18	-197,8781035	98,7933758	-2,002949104	0,004196449	0,001044932	-2,006114967	-0,064882395	98,89666216
19	-194,6574366	98,7933758	-1,970349075	0,004060958	0,001044932	-1,973329168	-0,063822026	98,89666216
20	-194,5668473	98,7933758	-1,969432117	0,004057179	0,001044932	-1,972407081	-0,063792204	98,89666216
21	-191,354034	98,7933758	-1,936911584	0,003924295	0,001044932	-1,939708025	-0,062734641	98,89666216
22	-189,6010183	98,7933758	-1,91916732	0,003852723	0,001044932	-1,921869096	-0,062157689	98,89666216
23	-189,0282356	98,7933758	-1,913369536	0,00382948	0,001044932	-1,916040797	-0,061969188	98,89666216
24	-187,661385	98,7933758	-1,899534088	0,003774299	0,001044932	-1,902133351	-0,061519389	98,89666216
25	-183,4457607	98,7933758	-1,856862965	0,003606632	0,001044932	-1,859247388	-0,060132358	98,89666216
26	-182,0032102	98,7933758	-1,842261273	0,003550132	0,001044932	-1,844574648	-0,059657808	98,89666216
27	-174,8405285	98,7933758	-1,769759633	0,003276202	0,001044932	-1,771738451	-0,057302117	98,89666216
28	-174,0324743	98,7933758	-1,761580398	0,003245989	0,001044932	-1,763523344	-0,057036421	98,89666216

933	98,7933758	1,925253745	0,003877199	0,001044932	1,927987776	0,062355581	98,89666216
934	98,7933758	1,929949279	0,003896134	0,001044932	1,932708347	0,062508255	98,89666216
935	98,7933758	1,932155658	0,003905048	0,001044932	1,934926538	0,062579997	98,89666216
936	98,7933758	1,953251493	0,003990786	0,001044932	1,956136815	0,063265986	98,89666216
937	98,7933758	1,953585904	0,003992153	0,001044932	1,956473062	0,063276861	98,89666216
938	98,7933758	1,97251775	0,004069902	0,001044932	1,975509993	0,063892559	98,89666216
939	98,7933758	1,973188499	0,00407267	0,001044932	1,976184507	0,063914375	98,89666216
940	98,7933758	2,021733136	0,004275528	0,001044932	2,025009096	0,065493475	98,89666216
941	98,7933758	2,036839477	0,00433966	0,001044932	2,040205619	0,065984965	98,89666216
942	98,7933758	2,077424318	0,004514322	0,001044932	2,081040071	0,067305646	98,89666216
943	98,7933758	2,111426443	0,004663307	0,001044932	2,115259668	0,068412387	98,89666216
944	98,7933758	2,12093334	0,004705396	0,001044932	2,12482875	0,068721873	98,89666216
945	98,7933758	2,163409208	0,004895753	0,001044932	2,167589924	0,070104868	98,89666216
946	98,7933758	2,166338781	0,004909021	0,001044932	2,170539628	0,070200268	98,89666216
947	98,7933758	2,169370283	0,004922769	0,001044932	2,173592024	0,07029899	98,89666216
948	98,7933758	2,214009933	0,005127448	0,001044932	2,218546725	0,071752929	98,89666216
949	98,7933758	2,216852856	0,005140624	0,001044932	2,221410185	0,07184554	98,89666216
950	98,7933758	2,226849823	0,005187092	0,001044932	2,231479818	0,072171215	98,89666216
951	98,7933758	2,232863231	0,005215145	0,001044932	2,237537277	0,072367127	98,89666216
952	98,7933758	2,237853627	0,005238482	0,001044932	2,242564424	0,072529717	98,89666216
953	98,7933758	2,257696325	0,005331792	0,001044932	2,26255501	0,073176259	98,89666216
954	98,7933758	2,281866933	0,005446566	0,001044932	2,28690958	0,073963942	98,89666216
955	98,7933758	2,349437043	0,005773906	0,001044932	2,355016601	0,07616668	98,89666216
956	98,7933758	2,501247193	0,006544182	0,001044932	2,508159062	0,081119661	98,89666216
957	98,7933758	2,546185763	0,006781446	0,001044932	2,553526758	0,082586957	98,89666216
958	98,7933758	2,714263502	0,007706304	0,001044932	2,723357341	0,088079671	98,89666216
959			COOKD>0.0042		plus minus 2	DFFITS>0.1584	

Table 15: Derived last outlier

4.3 Analysis of Multiple Linear Regression

4.3.1 Finding Standard Error of Estimate

Before Remove Outlier

Root MSE	64.91691	R-Square	0.6114
Dependent Mean	268.72417	Adj R-Sq	0.6102
Coeff Var	24.15745		

Table 16: R-Square and Adjusted R-Sq for SLR

After Remove Outlier

Before Autocorrelation

Root MSE	56.07501	R-Square	0.6358
Dependent Mean	270.38107	Adj R-Sq	0.6346
Coeff Var	20.73925		

Table 17: Standard error of estimates for last data (before autocorrelation)

For standard error of estimate, the smaller the value of root MSE, the better the model. In our study, Table 17 shows the standard error for last data (after removing outlier) $= 56.07501$ while Table 16 shows standard error for original data $= 64.91691$. In conclusion, we can conclude that the standard error of second data = 56.07501 is less than original data $= 64.91691$. Therefore, last data is the best model for this data set.

After Autocorrelation

Yule-Walker Estimates			
SSE	39486.9986	DFE	929
MSE	42.50484	Root MSE	6.51957
SBC	6160.82849	AIC	6146.31649
MAE	5.32998022	AICC	6146.34235
MAPE	53.1344577	HQC	6151.85088
Durbin-Watson	1.9622	Transformed Regression R-Square	0.0000
		Total R-Square	0.9865

Table 18: Standard error of estimates for last data (after autocorrelation)

For standard error of estimate, the smaller the value of root MSE, the better the model. In our study, Table 18 shows the standard error for last data (after autocorrelation) $= 6.51957$ while Table 18 shows standard error for last data (before autocorrelation) $= 56.07501$. In conclusion, we can conclude that the standard error of last data (after autocorrelation) = 6.51957 is less than last data (before autocorrelation) $= 56.07501$. Therefore, last data (after autocorrelation) is the best model for this data set.

4.3.2 Testing the Validity of The Regression Model

Before Remove Outlier

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	6604293	2201431	522.38	<.0001
Error	996	4197348	4214.20514		
Corrected Total	999	10801641			

Table 19: Analysis of Variance (ANOVA)

Based on the Table 19, the regression model has a large F-value of 522.38 and a highly significant p-value of less than 0.0001 , indicating that the independent variables explain a substantial portion of the variance in the dependent variable. However, the high explained variance from the model and large error variance suggest that the results might be influenced by extreme values or outliers.

After Remove Outlier

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	5093241	1697747	539.93	<.0001
Error	928	2918010	3144.40685		
Corrected Total	931	8011251			

Table 20: Analysis of Variance (ANOVA) after removing outlier

Based on Table 20, the F-value increase to 539.93 but the model remains statistically significant with a p-value of less than 0.0001. The explained variance from the model and residual variance are reduced, reflecting a more accurate and unbiased evaluation of the model after addressing the outliers.

After Autocorrelation

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	24087	8029.01504	487.17	<.0001
Error	928	15294	16.48080		
Corrected Total	931	39381			

Table 21: Analysis of Variance (ANOVA) after autocorrelation

Based on Table 21, the F-value decrease to 487.17 but the model remains statistically significant with a p-value of less than 0.0001.

4.3.3 Testing the Significance of Each of Independent Variables

Before removing outlier

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	56.37349	8.80441	6.40	<.0001	.	0
Number_of_Customers	1	2.90646	0.07795	37.28	<.0001	0.99867	1.00134
Menu_Price	1	2.17157	0.18224	11.92	<.0001	0.99843	1.00158
Average_Customer_Spending	1	-0.27487	0.17910	-1.53	0.1252	0.99935	1.00065

Table 22:: Parameter estimates for MLR

To test the significance of each independent variable, we need to refer column $Pr > |t|$ from Table 22. It shows that all variables are significant excluding the average customer spending variable, X_3 which is $p - value > 0.05$.

After removing outlier

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	68.61265	8.05015	8.52	<.0001	.	0
Number_of_Customers	1	2.71379	0.07108	38.18	<.0001	0.99974	1.00026
Menu_Price	1	2.07019	0.16342	12.67	<.0001	0.99998	1.00002
Average_Customer_Spending	1	-0.19328	0.16039	-1.21	0.2285	0.99973	1.00027

Table 23: Parameter estimates for MLR after removing outlier

To test the significance of each independent variable, we need to refer column $Pr > |t|$ from Table 23. It shows that all variables are significant excluding the average customer spending variable, X_3 which is $p - value > 0.05$.

We can conclude that, before and after removing outliers, average customer spending, X_3 is not significant.

4.3.4 Selection of Predictor Variable

Number in Model	R-Square	Adjusted R-Square	C(p)	MSE	Variables in Model
1	0.5722	0.5718	161.8204	3684.77292	Number_of_Customers
1	0.0623	0.0613	1461.153	8077.92344	Menu_Price
1	0.0012	0.0001	1616.661	8603.70834	Average_Customer_Spending
2	0.6352	0.6344	3.4521	3145.93719	Number_of_Customers Menu_Price
2	0.5728	0.5719	162.4835	3684.21433	Number_of_Customers Average_Customer_Spending
2	0.0636	0.0615	1459.861	8075.47494	Menu_Price Average_Customer_Spending
3	0.6358	0.6346	4.0000	3144.40685	Number_of_Customers Menu_Price Average_Customer_Spending

Table 24: Best subset

Based on Table 24, we can conclude that the variables that should be selected for the model are number of customers, X_1 and menu price, X_2. It is because $R^2 = 0.6352$ and $R_\alpha^2 = 0.6344$ is the second largest. As for the C_p value should be the second smallest is 3.4521. Meanwhile,

$X_1X_2X_3$ is not the best model even though it has the largest value in R^2 & R_α^2 and smallest value in C_p because X_3, average customer spending is not significant.

Therefore, the best subset is number of customers and menu price as it is considered good for the model as it follows the criteria.

4.3.5 Coefficient of Multiple Determination and The Adjusted Coefficient of Multiple Determination

Before remove outlier

Root MSE	64.91691	R-Square	0.6114
Dependent Mean	268.72417	Adj R-Sq	0.6102
Coeff Var	24.15745		

Table 25: R-Square and Adjusted R-Sq for SLR

From Table 25, we can see that about 61.14% of total variation in monthly revenue is explained by number of customers, menu price and average customer spending while the balance 38.86% is explained by other factors.

After considering the sample size, we can see that about 61.02% of total variation in monthly revenue by number of customers, menu price and average customer spending while the balanced 38.98% is explained by other factors.

After remove outlier

Before Correlation

Root MSE	56.07501	R-Square	0.6358
Dependent Mean	270.38107	Adj R-Sq	0.6346
Coeff Var	20.73925		

Table 26: R-Square and Adjusted R-Sq for SLR after removing outlier

From Table 26, we can see that about 63.58% of total variation in monthly revenue is explained by number of customers, menu price and average customer spending while the balance 36.42% is explained by other factors.

After considering the sample size, we can see that about 63.46% of total variation in monthly revenue by number of customers, menu price and average customer spending while other factors explain the balanced 36.54%.

After Autocorrelation

Yule-Walker Estimates			
SSE	39486.9986	DFE	929
MSE	42.50484	Root MSE	6.51957
SBC	6160.82849	AIC	6146.31649
MAE	5.32998022	AICC	6146.34235
MAPE	53.1344577	HQC	6151.85088
Durbin-Watson	1.9622	Transformed Regression R-Square	0.0000
		Total R-Square	0.9865

Table 27: Autocorrelation table

From Table 27, we can see that about 98.65% of total variation in monthly revenue is explained by number of customers, menu price and average customer spending while the balance 1.35% is explained by other factors.

The comparison of the Coefficient of Multiple Determination (R-Square) and the Adjusted Coefficient of Multiple Determination before and after removing the outlier highlights a clear improvement in the model's performance after the outlier's removal. Before removing the outlier, the R-Square was 0.6114, and the Adjusted R-Square was 0.6102, indicating that the predictors explained 61.14% of the variation in monthly revenue. After removing the outlier, these values increased to 0.6358 and 0.6346, respectively, showing that the predictors now explained 63.58% of the variation. Additionally, the Root Mean Squared Error (RMSE) decreased significantly from 64.91691 to 56.07501, further indicating an improved model fit with reduced error. This comparison suggests that the removal of the outlier enhanced the model's accuracy and reliability, making the analysis after outlier removal more robust and representative of the relationship between the predictors and monthly revenue.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

In conclusion, our analysis concludes the three factors affecting restaurant revenue. At first, we used Multiple Linear Regression (MLR) to analyze the model of the relationship between restaurant revenue and our three independent variables. To check the model adequacy of our data, there are six assumptions we need to fulfil. Our initial data is already normal based on Kolmogorov-Smirnov test. However, to ensure the best validity of the regression model, we must eliminate the outliers and assess the relationships of each independent variable with restaurant revenue. Even when our data is already normal, outliers or influential points can still affect the regression results. We removed 68 outliers from our dataset and observed the results from before and after. Based on the residual plot analysis, we can see that the residuals became more homoscedastic and evenly spread out randomly compared to before removing the outliers. From the standardized residual, we can see that the red dots (outliers) have lessened after outliers' elimination. Subsequently, we want to choose the best variables that affect restaurant revenue through the processes of Coefficient of Multiple Determination(R^2) selection, Analysis of Variance (ANOVA) and Adjusted Coefficient of Multiple Determination $\left(R^2_{a,p}\right)$. From the processes, we conclude that the best predictor variables that affect restaurants' monthly revenue are the number of customers, X_1 and menu price, X_2. After removing the outliers, the R^2 increased from 0.6114 to 0.6358, $R^2_{a,p}$ has increased from 0.6102 to 0.6346. The Root Mean Squared Error (RMSE) also decreased from 64.91691 to 56.07501, showing that our model has improved with reduced error.

Therefore, based on table 20, the relationship of our model can be described by the equation:

$$Y = 68.6127 + 2.7138\,X_1 + 2.0702\,X_2$$

Where:

$Y =$ Restaurants' monthly revenue

$X_1 =$ The number of customers

$X_2 =$ Menu price

5.2 Recommendations

Based on the analysis, several recommendations can be made for future researchers looking to enhance their understanding, particularly in the context of improving Multiple Linear Regression (MLR) models. Firstly, to improve the Multiple Linear Regression (MLR) model, researchers should consider incorporating a broader range of variables beyond type of cuisine, operating hours, active promotions and more to better understand external factors influencing revenue of the restaurant. Based on article Multiple Linear Regression: Definition, Example and Applications by Mayur Badole, studies have shown that incorporating such diverse variables can lead to more precise predictions and better understanding of the underlying factors affecting the dependent variable.

Moreover, increasing the sample size will enhance the accuracy and generalizability of the findings. Results derived from a larger and more diverse group of participants are more likely to be applicable to the broader population, thus enhancing the external validity of the research, reducing sampling error with more data points, and decreasing the margin of error, leading to more precise and reliable estimates of the population parameters. This increased precision allows for more accurate and confident conclusions and better capture of data variability within the data set. It also enables the researcher to observe a wider range of responses and behaviors, which can lead to the identification of patterns and trends that might not be visible in a smaller sample. This comprehensive view helps in understanding the underlying factors and relationships within the data.

Another than that, based on our best model that we gain from the analysis which is the number of customers, X^1 and menu price, X^2. In enhancing the restaurant's revenue, a few strategies can be made. Firstly, attract more customers with well-priced offers. Use competitive pricing for entry-level dishes such as appetizers to attract more customers, then upsell higher-priced items such as main courses, drinks or desserts. Secondly, targeted promotions. For example, offer discounts on menu prices to large groups of families to encourage group dining, which brings in more customers at once. Lastly, gather feedback on menu prices to identify items that customers find overpriced or undervalued and adjust accordingly to optimize both sales and satisfaction.

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APPENDIX

1. Original Data

Obs	Number_of_Customers	Menu_Price	Marketing_Spend	Cuisine_Type	Average_Customer_Spending	Promotions	Reviews	Monthly_Revenue
1	61	43.11763549	12.66379252	Japanese	36.23613252	0	45	350.91204
2	24	40.02007699	4.577892097	Italian	17.95256244	0	36	221.3190913
3	81	41.98148524	4.652910771	Japanese	22.60042029	1	91	326.5297629
4	70	43.00530707	4.416052596	Italian	18.9840976	1	59	348.1905731
5	30	17.45619947	3.475051913	Italian	12.76614271	1	30	185.0091206
6	92	19.42766984	13.11447345	American	43.09994967	1	10	399.8675067
7	96	35.35034255	16.9603177	American	20.18162172	0	86	496.6536867
8	84	46.31464522	14.48634939	Italian	28.92110013	1	1	417.1586067
9	84	22.64786329	15.84187268	American	11.73261093	0	26	352.1480652
10	97	33.53226703	8.095968582	Italian	37.97357865	0	98	272.7935168
11	33	37.31928254	13.08151591	Mexican	23.22244009	0	1	318.7732089
12	12	28.07794405	7.084724809	American	39.78946911	0	20	189.2388689
13	31	38.55129819	10.51009244	American	26.49612242	0	57	218.4735501
14	62	45.98669781	9.632696931	American	17.83022426	0	4	308.9711494
15	11	34.96406177	3.537742903	Mexican	30.52700329	0	51	117.1533293
16	97	31.59124248	7.741575207	Italian	15.22018239	1	44	292.7543309
17	39	27.54978702	19.27644551	Japanese	27.62156639	0	84	252.5846616
18	47	33.09945084	4.971605154	American	22.6922191	1	92	182.9190408
19	11	24.21449807	2.676074726	Italian	46.47838337	1	19	60.37322071
20	73	25.65928441	8.371612432	Mexican	12.43714887	0	60	301.1840968
21	69	31.27429919	15.80675472	Japanese	37.76069656	0	37	294.335597
22	30	12.6647645	11.6429646	American	17.31004664	1	19	276.4921286
23	42	19.16101564	13.61949654	Mexican	22.76180297	1	24	308.198028
24	85	31.71397242	3.793896963	Mexican	26.62056252	1	1	267.4949482
25	67	27.26112573	12.37896386	Japanese	46.32060357	1	31	368.3336437
26	31	23.31263144	6.470007898	Italian	19.70313064	0	25	165.8841612
27	98	39.22196547	16.68606732	American	13.6523521	0	69	494.2657908
28	58	37.7487043	12.58597612	American	22.74758416	0	40	273.3665072
29	68	16.6692304	3.977125556	Italian	14.18703882	1	66	345.2649163
30	51	45.14516461	1.292333597	Italian	15.14385944	0	96	255.775191
31	69	29.81694506	0.852664186	Italian	32.95402392	1	0	203.9593522
32	89	39.65843645	7.428171353	Japanese	12.43607732	0	50	442.8549585
33	24	32.92603397	5.317022733	Japanese	11.23723526	1	4	178.3033493
34	71	49.90770464	12.92180247	American	38.57656406	0	12	485.2402706
35	71	40.0961248	9.9866767	Japanese	24.46588379	0	18	262.1708567
36	56	38.27921845	5.32038556	Mexican	15.01501397	1	11	343.6727362
37	71	41.14287756	2.984755559	American	20.14451083	1	47	309.3590956
38	60	15.72511942	2.543995254	Japanese	35.68346029	1	24	268.8777339
39	64	18.18178373	17.29320784	Italian	21.02146269	1	56	249.757181
40	73	38.56256329	4.869559368	Mexican	29.780845	1	34	304.3846129
41	12	29.75925949	13.03421296	Italian	14.86781227	0	75	116.7904526
42	60	40.18541167	16.21371856	American	33.5237455	1	93	305.9655992
43	16	14.11664123	12.28162622	Italian	22.64310735	0	4	138.2032058
44	30	31.45925406	17.19130463	Japanese	35.07873738	0	97	277.2829196
45	82	25.15286262	7.847874431	Japanese	42.33579529	0	21	415.4999787
46	48	28.28000877	19.12444381	American	34.46585678	0	76	324.8181694
47	27	34.15831488	2.662900084	Italian	21.8299025	0	56	218.4098265
48	13	30.0915339	2.042990593	American	20.22974806	1	46	142.2740039
49	98	31.59442548	13.17384292	Italian	23.09683835	1	53	362.5184822
50	69	29.45429928	18.85662283	Mexican	16.58482053	0	2	433.9088069
51	23	26.35820708	16.17766923	Japanese	17.24687403	0	52	220.8440634
52	18	40.87527922	1.528337189	American	41.62764408	1	83	155.9520951
53	99	10.48812289	5.587354279	Mexican	25.38930306	1	30	433.1772878
54	62	33.93770805	18.6063777	Mexican	27.43980722	0	68	345.1718044
55	11	32.62033389	5.061168656	American	33.76197862	1	87	-28.97780879
56	93	38.64716299	15.62340848	American	33.58613981	1	32	387.0318044
57	69	33.96117459	16.50264392	Mexican	12.69507486	1	71	424.5034802
58	80	43.07195311	9.909042069	Italian	43.26163145	0	87	385.9386454
59	53	48.36299178	7.726978197	Mexican	17.09306213	1	49	291.8133288
60	17	23.70097029	9.002488648	American	38.92286544	0	83	136.685594
61	56	19.09403932	4.539293887	Mexican	40.90683953	0	15	239.0277439
62	44	26.9438745	17.78781832	Italian	38.59598136	0	75	328.2755963
63	87	21.51718466	9.012393135	Japanese	33.15910586	0	76	313.5640786
64	90	34.59801053	17.64918881	Japanese	10.58435988	0	38	459.3053087
65	45	46.47409635	2.113869598	Mexican	24.41501969	1	88	312.9120107
66	59	15.56464777	19.76550542	Italian	34.19357717	0	45	325.1018811
67	13	14.03178412	8.865461749	Japanese	25.49805184	0	95	76.65350206
68	11	20.24062127	0.956616408	Mexican	44.5391958	0	87	163.8073293
69	15	39.04382344	0.760436965	Mexican	22.35209101	1	46	113.2107886
70	63	33.71851621	14.92414515	Italian	21.7413917	1	43	242.1051453
71	13	14.08850657	13.18071068	American	20.63990772	1	8	112.0015302
72	63	46.75002089	16.04261116	Mexican	23.31123476	1	86	326.2939592
73	72	41.60338548	2.107350679	American	32.29619857	1	49	330.2542096
74	27	10.92090419	0.711417341	American	38.28061177	1	76	178.572959
75	99	36.05468844	10.17158579	Japanese	36.50271671	1	59	363.2381375
76	53	40.84986674	7.75190463	American	42.3087128	0	95	326.9734099
77	43	24.97741479	9.238938733	Italian	45.43655982	0	14	233.3247689

78	83	12.75686157	19.83176818	Japanese	42.64316169	1	35	367.6532909
79	71	13.09284262	19.41499915	Japanese	10.75781613	0	62	262.5682778
80	23	14.16987864	18.84630779	Japanese	39.43671814	0	68	140.8462541
81	57	43.61758528	15.67685354	Mexican	18.33598294	1	85	308.708379
82	24	46.42744793	7.096827205	Mexican	15.31712171	0	79	250.1765183
83	81	14.91243127	8.242618605	American	35.53955139	0	40	308.4835922
84	87	19.43624138	10.99232087	Japanese	22.78972782	1	73	251.2093448
85	96	16.62082041	16.61068913	Japanese	12.28468653	0	23	563.3813324
86	71	17.45283518	1.37425665	Italian	15.50518938	1	81	221.3899269
87	49	43.49963156	8.485937718	American	10.03717667	1	48	354.0511695
88	94	23.28585651	1.652353269	Italian	46.12894509	0	17	327.1516584
89	89	22.45777056	10.29315173	American	10.98471935	0	6	334.9565374
90	91	19.09582484	17.97801648	American	18.37823078	0	3	391.8035009
91	62	34.31577575	15.64969849	Mexican	26.65051123	1	89	380.1445807
92	33	25.17222794	4.556848351	Mexican	24.26285304	1	68	41.59068666
93	35	39.76998607	19.51981699	Italian	46.46903508	1	59	199.9911442
94	98	18.22319234	13.34700796	American	11.61155783	0	54	432.7392218
95	69	41.51159555	0.208553988	Japanese	30.48858831	0	15	279.7553194
96	50	34.14886127	5.108269694	Mexican	37.2191785	1	98	325.8872649
97	38	14.570692	5.214265998	Japanese	47.26791895	0	89	98.06199096
98	24	26.58018337	1.385086126	Italian	34.81485008	0	81	128.4403363
99	54	44.54072562	19.19653036	Mexican	42.15020944	1	28	371.9437925
100	74	46.91840927	9.565105892	Mexican	13.34899751	1	22	307.7723269
101	98	28.62800489	17.91629195	American	25.43758044	0	13	365.4749116
102	80	29.23347912	6.444310069	Italian	24.95788538	1	78	419.892437
103	18	46.738189	2.504671409	American	45.45152866	0	48	135.0952899

104	97	33.48272897	9.486722198	Mexican	16.05484685	1	32	412.5826517
105	10	11.31386681	2.279132694	Italian	16.07544689	1	24	-27.61027601
106	17	46.50997852	9.72031006	Mexican	23.142326	1	20	131.2350416
107	97	19.92905771	18.40594362	Italian	12.17864189	0	16	449.0210765
108	72	33.10527816	10.41536284	Mexican	17.29832163	0	22	401.9888859
109	20	16.6205327	12.32999454	Japanese	48.60838554	1	56	98.58966672
110	90	11.35516641	3.132768739	Japanese	27.12735759	1	80	204.7213793
111	17	22.45892685	0.874712518	American	35.28076834	1	57	-25.45342991
112	44	41.2198283	2.216159244	Japanese	39.04445037	0	13	196.3214949
113	44	21.10348691	11.97063151	Mexican	24.67845628	1	92	251.8456981
114	42	18.80388856	1.90023398	Mexican	15.88908298	1	19	194.4723994
115	14	18.50723137	7.519857799	Japanese	18.26324065	1	29	83.55896435
116	50	30.6069473	11.00183881	American	22.53065906	1	18	365.8905363
117	37	49.02165263	7.773055973	Japanese	35.44273681	0	52	241.2438154
118	16	28.35955952	3.273676029	Italian	25.93122196	0	55	154.8721373
119	82	32.29221476	5.47582615	Italian	10.27258792	1	76	229.2625114
120	81	44.42528072	18.56635066	Italian	30.9362874	0	81	376.7581595
121	21	31.40278876	13.51964551	Japanese	46.86053568	0	22	130.0347865
122	43	17.37851486	9.920956774	Japanese	42.48310444	0	61	221.9570694
123	42	21.98383647	0.2542978	Japanese	43.00845485	1	32	200.6812301
124	57	22.39747705	19.84830801	Italian	41.8023207	0	17	306.330244
125	32	25.89272462	13.4290987	American	22.94923302	0	72	130.277355
126	71	27.07155621	6.338449547	Mexican	40.60775335	0	32	289.8920296
127	97	41.99128088	17.5853732	Japanese	15.72645431	1	31	448.3815013
128	46	23.97547996	5.168632406	American	36.30864589	1	60	156.9067705
129	53	28.72927314	10.88033158	Mexican	35.66274267	0	53	389.0985156

130	95	34.99903948	12.72141527	Mexican	11.02279196	0	16	476.4561107
131	44	25.1090379	4.421267429	Mexican	41.51880534	1	18	160.3854626
132	74	43.46259253	19.56794558	American	46.1299242	1	90	490.2307266
133	56	33.49814417	14.96383812	American	20.69661114	0	69	288.7082461
134	87	21.76178717	11.15626451	Italian	42.49562897	0	98	278.6971706
135	12	38.56236799	3.611891964	American	13.29623887	1	33	58.90983174
136	10	31.10590921	1.07716414	American	33.99871433	1	83	76.56291267
137	14	31.37823565	6.40119665	American	29.42587461	0	24	157.5013266
138	99	29.24351794	15.06374672	Mexican	31.83694133	1	67	461.1858325
139	23	29.88047795	5.386625966	Japanese	47.04820808	0	73	266.4799919
140	36	40.61425124	10.79112441	Italian	44.85260989	0	73	101.7397794
141	18	14.11925917	14.55894353	Japanese	41.40334629	0	87	209.8282253
142	88	23.37451245	2.158874293	Japanese	20.94793831	1	84	333.8217984
143	24	13.02013455	13.70596032	American	46.47274515	0	10	181.7307571
144	99	40.12984883	1.837417754	Italian	13.9003077	0	90	341.0259733
145	51	20.89207419	19.30259484	Japanese	44.13316866	1	15	271.4703792
146	86	45.89721639	9.767882601	Japanese	23.12640289	1	89	489.9498646
147	60	31.06313867	3.013522154	Italian	12.15089247	0	60	282.188614
148	72	42.02999086	6.847672721	American	36.77427937	0	93	357.4846908
149	61	49.15725742	8.268328986	Japanese	19.25119713	1	79	371.7087958
150	13	43.59155438	15.16370022	Mexican	19.74284503	1	6	115.0898635
151	32	44.6797504	14.86847572	Japanese	22.24188665	1	15	151.8647318
152	24	26.31937981	12.23881011	Mexican	12.5473907	0	98	63.00779063
153	52	32.06890381	2.118614251	American	17.08508577	0	15	203.3759881
154	38	20.15554256	3.031387432	Japanese	10.2055851	0	39	224.6209835
155	45	17.84452399	1.875138413	Italian	27.81638084	1	11	174.6028056

156	22	30.22031516	11.54345567	Italian	13.3702151	0	34	160.4799298
157	41	33.80195919	17.54276883	Japanese	37.3061042	1	20	252.6859901
158	80	23.57084984	8.676010245	American	19.42491405	0	49	355.5429149
159	68	32.77774071	3.216963479	American	47.74296576	1	42	245.170225
160	95	45.4984193	11.1608465	Italian	44.8500651	0	99	472.4465972
161	37	32.26886339	16.23504434	Japanese	21.5776358	0	16	152.3316354
162	75	38.83252385	14.83424123	American	41.32861819	1	31	461.4226661
163	51	42.21259869	17.62901559	Italian	42.12979619	1	40	280.4429944
164	54	49.54915128	5.558041945	Mexican	49.46252456	0	5	264.632079
165	71	34.12789609	3.879077345	Japanese	33.91359947	1	87	299.6345076
166	66	42.27400596	17.59632972	Mexican	13.27862939	1	26	384.9893461
167	15	48.50597274	13.15120626	Mexican	35.87348916	0	19	208.8922763
168	37	47.77638902	6.2185786	American	34.99918994	1	24	247.3137544
169	37	15.64327395	14.84696101	American	35.5340231	0	65	114.5216964
170	53	26.25569233	16.30236769	American	22.28582592	0	8	366.7007793
171	93	22.95960244	12.29956481	Japanese	44.3871834	1	91	374.6946211
172	39	13.4770006	13.7576822	Mexican	48.37888732	1	52	282.1283025
173	71	35.329631	7.505198089	Japanese	18.70850052	0	60	284.2144476
174	84	39.43618453	10.77391335	American	13.52608633	0	37	347.2142781
175	98	43.92512382	10.48318341	Mexican	17.78140133	0	2	383.5234315
176	71	14.91107559	16.96095582	Japanese	42.82208557	0	83	399.7027714
177	10	45.05776304	16.79927961	Japanese	37.06578539	0	69	157.593608
178	36	35.71702254	16.65360939	Italian	35.44189369	0	96	281.1399417
179	71	38.15784541	0.220632285	Japanese	26.0717254	1	54	243.0342553
180	86	46.42496014	14.99090969	American	33.11311886	1	0	340.6267178
181	12	34.99045084	15.1735278	Japanese	22.31496501	1	69	162.6838914

182	79	23.43463429	13.56757646	Mexican	29.08800318	0	44	279.2886934
183	81	43.00429326	15.12529316	Japanese	41.72252113	1	97	385.0488273
184	36	24.52236849	16.83940743	Italian	49.2496639	0	46	238.0065897
185	18	11.36912443	9.406381082	Mexican	47.29955694	0	78	113.6556177
186	71	43.22618026	10.04100427	Mexican	47.35463771	1	25	359.898749
187	46	23.8076858	2.494281138	Italian	26.08894939	0	13	227.949084
188	60	40.95338093	7.707517661	American	28.75159522	0	53	231.6345442
189	53	24.51031579	12.8673036	Mexican	38.22541585	0	70	344.3766981
190	33	44.4427208	15.60799552	American	46.88767712	0	45	258.3128436
191	88	18.780421	4.344181954	Mexican	15.67977037	0	30	292.4033827
192	68	48.98189061	2.756457983	Mexican	22.16967455	0	87	217.3504514
193	41	41.19035099	13.26084009	American	15.56987378	1	58	269.644218
194	97	14.56951704	0.03759767	Mexican	34.4803412	0	94	300.3599903
195	61	32.6310648	3.567451686	American	22.25203988	0	28	275.4038574
196	71	49.41469249	0.581427607	Italian	23.0443141	0	95	270.5133607
197	67	28.84261811	3.337702357	American	12.83663838	0	16	366.2391636
198	61	17.28428555	3.626385599	Italian	23.99184247	1	13	205.3492661
199	21	29.39113469	7.755471895	Mexican	44.4998603	1	35	226.3917444
200	48	30.49830624	8.046542494	American	19.32291248	0	57	362.2557787
201	11	39.66747777	17.48524941	Mexican	37.80749865	0	23	163.5019343
202	12	37.95383413	10.20219425	American	35.1851338	0	62	140.2683836
203	65	26.10200174	14.46809531	Italian	44.53938433	1	90	258.3387278
204	90	18.7209243	2.082747158	Italian	15.29151791	1	24	284.7478321
205	68	35.80216671	4.508775109	American	49.10359841	1	58	261.2324669
206	11	26.86815738	14.05886882	Italian	33.30514584	1	31	136.8781096
207	11	15.29104845	1.056079209	Mexican	49.81896609	0	81	138.2696452

208	63	46.02289205	9.731953245	Mexican	46.46058903	1	17	363.4186198
209	96	37.6488743	11.98737931	Japanese	18.86069088	1	36	384.7706242
210	10	37.32735426	12.04291288	Italian	12.2464804	0	25	236.4496811
211	28	42.90335504	0.458656225	Mexican	20.22556375	0	8	164.6375348
212	11	31.18646715	6.442160085	American	31.10254596	1	98	57.31941977
213	62	42.60217714	12.72981302	Italian	16.86385468	1	63	325.5445588
214	53	29.92363014	2.979994723	Mexican	16.69838978	1	71	233.7958127
215	99	12.687016	13.90831297	Italian	28.50374188	1	19	436.7806564
216	41	26.20992352	14.91850418	American	11.33071885	0	53	215.0685303
217	79	29.89859576	19.63938381	Mexican	14.8755233	0	85	432.0267035
218	41	38.80810327	4.196839897	Japanese	26.63445358	1	82	278.8993453
219	77	14.26314921	16.50261335	Japanese	12.52519637	1	29	308.9682697
220	64	15.60074155	10.40593154	Italian	25.87020226	1	52	254.1584923
221	84	20.54735211	7.469251923	American	28.96025064	1	91	267.6304698
222	65	20.65993573	2.447535147	Mexican	28.05066456	1	98	199.9807309
223	26	39.28738314	14.71011213	American	44.7669943	0	62	224.9284042
224	47	20.02922642	4.717151983	Mexican	31.2024982	0	10	258.7153264
225	33	35.32392253	7.428440037	Italian	28.7192185	0	30	269.238539
226	78	29.77095401	1.250844538	Mexican	26.42836982	1	48	290.8349706
227	79	32.93177513	16.02733832	American	42.70493513	1	99	262.777638
228	95	43.52427853	15.33177048	American	45.80226416	0	93	409.3034161
229	20	26.17526006	14.6723166	Italian	44.12969877	0	86	193.0134803
230	25	40.83645508	5.98887021	American	36.94186313	1	18	180.661047
231	82	26.88144238	3.469601485	Mexican	15.49446984	0	11	285.4671803
232	68	24.15485506	12.75106529	Italian	21.07479878	1	14	328.6551331
233	79	48.34508709	9.923327682	Mexican	11.84252255	1	30	443.9866368

234	89	17.37483683	3.565049338	American	47.232963	0	42	301.1595741
235	12	13.61347842	13.71822542	Italian	36.43194119	0	36	96.37982086
236	29	40.83943702	19.82252864	Mexican	39.85451537	0	78	252.9834074
237	68	12.72998821	14.94439175	American	22.46286958	1	28	315.8337353
238	45	43.48315963	14.79972773	American	43.82614089	1	25	397.6446395
239	28	27.48773585	19.00625506	Japanese	20.38413135	0	1	237.5759044
240	99	46.62054709	4.058727351	American	10.06575028	1	83	404.5022097
241	76	38.86131444	11.3109993	Mexican	25.0565985	1	14	350.3093698
242	28	34.41618668	19.5929343	Italian	32.83045924	1	3	304.476143
243	29	47.97227188	3.305882219	Japanese	29.14819432	0	85	300.8587017
244	80	25.86597492	14.00503919	American	44.4830457	1	49	356.1079961
245	61	48.16347746	11.71620292	American	14.21621144	0	37	343.3628636
246	42	15.4000511	11.72958184	American	32.15293763	0	22	228.6471401
247	49	29.35019768	13.95341255	Mexican	30.51917399	1	58	166.4859381
248	48	20.7964489	18.59233846	Japanese	21.25571929	0	89	277.1865625
249	91	31.59371116	15.6516388	Mexican	42.8891245	1	74	424.0417604
250	10	16.48055512	1.122722068	Japanese	40.11394624	0	27	46.67513578
251	20	43.66118641	17.0545604	Italian	39.31436749	0	65	202.2323645
252	66	43.85631397	4.510826599	Mexican	45.04769913	1	7	268.0038012
253	98	48.20945251	7.538283295	Mexican	38.91138402	0	28	458.7283629
254	59	16.18688564	9.234656272	Italian	19.27365167	0	10	310.2055017
255	32	34.93330204	9.618120015	Japanese	38.22250847	0	17	219.0541329
256	40	28.96949235	8.703607985	American	22.30811254	1	54	388.9162198
257	51	23.77534928	18.74706588	Italian	19.68698601	1	49	291.3070946
258	16	24.06160645	14.12038015	American	12.92014372	0	13	116.011228
259	25	26.51174202	12.62341292	American	39.76114321	0	73	149.1359037

260	99	38.19702199	0.996840556	Italian	38.44003848	0	98	356.1414009
261	69	33.93144233	18.07826661	Japanese	19.37997911	1	8	308.5063722
262	11	28.38726051	18.18780384	American	39.97516067	1	98	241.8505908
263	10	13.03733641	15.60426127	Italian	46.71363532	0	10	142.6744769
264	57	13.11061376	0.140584076	Mexican	14.3211583	1	62	221.8492868
265	21	10.09357563	1.145605022	Mexican	19.30546554	1	75	158.559878
266	78	48.70797553	4.504474045	Mexican	40.01043176	1	66	307.9686104
267	46	10.21015274	2.626988163	Italian	20.98026694	1	52	120.2269498
268	41	14.12374984	7.976720594	Mexican	34.98655897	0	38	276.8719589
269	18	22.58386227	18.45246116	American	13.24656898	1	77	89.21713909
270	28	42.29721007	9.441208684	Mexican	28.08809916	1	55	153.3087716
271	57	48.48395143	0.268052529	Mexican	39.56365688	0	27	283.6463855
272	89	41.60054235	5.654913792	American	37.46055772	0	74	424.2944992
273	12	37.6224093	0.891898167	Mexican	42.60288403	1	38	73.048537
274	29	30.8520267	0.725752452	American	45.09808506	1	24	193.8420028
275	33	13.46820847	5.514397062	Mexican	40.92734005	0	20	181.3720251
276	63	48.32683172	5.512246774	American	36.86350972	0	9	295.9440034
277	42	40.33569656	15.74253205	Mexican	11.26889967	0	42	306.2799142
278	33	35.55797541	19.49669306	American	31.39866694	1	14	206.8633648
279	84	40.36143382	7.424141878	Italian	11.507909	1	82	413.004619
280	81	38.96413317	8.577737577	American	42.32271268	0	47	434.1430938
281	45	35.48926319	16.89578873	Mexican	22.82898098	1	85	213.3767241
282	47	49.22307201	5.362780402	Mexican	32.49717496	0	20	186.8682213
283	93	46.13455251	3.127636088	Japanese	21.17781229	1	59	308.1363814
284	98	35.86659676	17.06179247	American	41.30000578	0	95	464.1886581
285	34	37.72700846	0.930011729	Italian	45.46010107	0	49	221.8614914

286	27	12.06383227	12.3939888	American	25.94569977	1	22	158.0483244
287	91	36.79669858	11.23985695	American	44.51689574	1	12	382.7124179
288	75	11.76661751	6.91840895	Italian	20.6830003	1	68	296.3168077
289	63	33.53055028	6.303790422	American	31.38643692	1	3	318.5696854
290	44	49.97413998	15.43715452	Italian	44.35045719	1	91	330.7138193
291	89	32.23226326	1.728492215	Italian	23.0352537	1	77	330.3941751
292	70	28.92840472	4.735686186	American	48.1616906	0	71	349.2052774
293	50	22.51419426	14.64897464	American	14.87518007	1	8	307.8453531
294	42	14.82090315	10.96641167	Mexican	10.73100279	0	34	208.1580688
295	77	39.24094686	2.559764171	Mexican	46.1654141	0	20	398.5663825
296	42	17.70396111	17.64271229	Japanese	12.74594665	1	57	269.0819814
297	23	14.61128085	7.868956931	Japanese	49.67556784	0	17	131.166999
298	30	26.87050639	16.55736751	American	48.05587448	0	96	391.9728499
299	57	41.81367352	15.75426031	Italian	15.4808364	1	71	288.5322251
300	29	39.80258192	12.48850893	Japanese	31.50338503	1	80	165.6309604
301	17	12.19530459	9.889622626	American	17.3968119	1	42	99.8022674
302	16	28.2057474	19.59101681	Japanese	27.12620692	0	94	281.0757323
303	76	30.89738304	2.899301448	Italian	12.45700517	1	97	406.3648505
304	26	35.78186603	14.50311455	American	18.95698351	1	5	271.721235
305	42	36.00669402	3.823814029	Italian	24.53091186	1	85	249.136435
306	57	24.58788998	6.765262107	American	27.54590001	1	77	187.7977805
307	85	32.42708776	6.173974867	Japanese	49.09985033	0	66	346.2718972
308	68	29.24247535	18.94071975	Mexican	42.31462583	1	11	233.1435954
309	95	45.39993426	19.34712356	Japanese	39.49639777	0	99	522.7666363
310	31	31.21060531	3.660308885	Japanese	31.48165259	1	11	235.4213325
311	39	27.64128647	0.440703206	Mexican	43.59991042	0	65	172.9336307

312	47	26.17941799	14.40016079	Mexican	23.57291343	0	1	321.7673344
313	60	32.89711038	8.359798677	American	38.35596451	0	14	234.9335908
314	63	42.15907918	1.776804902	American	49.75135902	1	22	230.3129584
315	17	31.53605962	13.67508211	Italian	22.58515636	0	28	201.3184206
316	36	36.466207	10.90211057	Italian	36.9607014	1	80	246.5948799
317	36	39.51423068	19.61033644	Japanese	43.77259457	1	95	291.8101925
318	30	30.8067244	4.548189917	Mexican	46.63230333	1	6	27.15113597
319	39	27.07268227	8.079557527	Italian	38.71730764	1	93	244.6490118
320	37	45.06661511	10.85091607	Japanese	21.34431694	1	27	285.2456746
321	73	26.70024336	15.83399977	Mexican	23.05128606	0	13	308.8283974
322	78	28.47771402	10.97677313	American	10.83032128	1	43	374.4070008
323	70	49.61365253	12.25800322	Japanese	40.43543911	0	29	300.3489903
324	57	10.00950094	4.522941032	American	11.70958424	0	80	244.5486958
325	28	17.41891638	16.93837456	Italian	44.58609418	0	31	159.6923291
326	13	25.3585556	8.572610939	Italian	14.74642458	0	69	210.8077187
327	44	47.30359638	2.81919255	Mexican	16.87506049	0	87	252.1820121
328	73	12.80118823	13.28375829	Japanese	29.06105883	0	5	372.3558883
329	58	10.37941735	4.321809064	American	39.51521661	0	26	270.6884077
330	26	12.10735344	1.951250205	Mexican	15.56008821	1	81	184.7136561
331	53	13.54173642	17.23973817	Italian	27.602843	0	87	251.3293657
332	39	11.50387093	9.20324529	Italian	29.31405098	1	23	139.5220891
333	55	29.19125307	14.98567425	Japanese	18.49142567	0	33	286.278497
334	15	33.23028829	5.811751196	American	14.62233507	0	39	155.7636668
335	46	20.86747884	15.7394027	Japanese	43.45683692	0	46	286.1998055
336	33	25.93149536	2.23871854	American	30.09862599	0	78	143.3425137
337	55	13.6679734	7.119467476	American	10.57524944	1	74	140.1916724
338	62	23.45481855	19.56123636	Japanese	42.11925205	1	85	245.661809
339	69	30.90076767	15.87979433	Mexican	49.23175222	1	23	264.8550625
340	72	39.29087229	9.945118606	Japanese	34.23704782	0	64	402.7605942
341	94	10.13385448	6.795549431	Japanese	30.14833548	0	17	254.6415017
342	41	28.69243066	17.442714	Italian	45.13663452	0	43	267.4654382
343	96	21.88011506	10.89695518	Italian	48.10067159	1	75	460.829895
344	42	44.11287287	15.16850036	Japanese	40.18147664	0	56	314.1676277
345	76	38.62213261	15.08215361	Japanese	27.81811719	1	27	484.9508482
346	27	33.56434138	0.358913571	Italian	33.68885597	1	70	164.7211441
347	34	21.09062583	2.525075656	American	38.0374146	0	22	216.4565973
348	63	46.41356408	13.19258733	Japanese	17.77094284	0	84	281.4750959
349	67	11.80402709	3.232396226	Mexican	27.88804129	1	33	234.2347987
350	76	14.37870395	5.109866893	Mexican	43.19285059	1	10	174.0834241
351	55	25.66520562	18.93275908	Mexican	36.18901394	1	85	364.0336258
352	33	14.97750229	11.20364517	Japanese	26.44816269	0	34	300.0270934
353	41	48.2566279	15.01459014	American	36.53450136	0	92	263.8036196
354	56	41.91957289	11.15961478	American	14.72889648	0	41	318.3322992
355	95	20.35930606	11.0502723	Italian	26.05990442	0	27	294.5890405
356	32	33.54216916	1.883038203	Japanese	41.72514152	0	81	202.2351329
357	75	49.2932206	6.873263852	American	18.82035149	0	77	406.0549437
358	36	45.36795913	5.648617974	Japanese	17.4384603	1	81	144.5060822
359	11	34.03193786	7.814567312	Italian	48.0327464	1	91	163.929111
360	99	46.14895869	15.96688376	Italian	18.39590518	1	23	459.0697423
361	26	49.56739639	7.438653358	Italian	20.43674319	0	7	238.4688011
362	42	39.73411292	9.033936385	Mexican	48.78463756	1	52	256.6323989
363	18	12.59945638	7.94762781	Japanese	39.38834281	0	71	126.5993077
364	52	26.06320946	1.86815944	Italian	37.98386132	0	3	249.2636199
365	57	43.51302941	8.821443165	Japanese	44.11705535	1	90	249.1352861
366	48	19.22276783	3.912185461	Mexican	14.63292329	0	99	227.1060701
367	51	43.22570071	3.734465915	Mexican	38.23824946	1	4	256.9056662
368	35	14.81355048	1.880722089	Italian	30.14754437	1	96	122.2985143
369	59	11.87781429	7.133419483	Italian	43.16602561	0	48	158.6342068
370	34	25.36362446	15.1496377	American	30.32566875	1	97	240.5400057
371	33	11.4692775	5.528833812	Japanese	34.80965417	1	29	237.9437898
372	22	48.29081052	6.246982038	Italian	29.10968506	1	71	198.886609
373	69	43.04822783	12.85349315	American	29.12635694	1	62	380.6134146
374	16	42.03078749	3.916816252	Japanese	40.71879912	0	46	264.0589653
375	66	35.19111692	17.93915861	Italian	35.14265603	1	25	339.0892755
376	45	18.65296092	10.40381047	American	35.33052806	1	56	197.3275369
377	54	30.73041467	8.952058806	American	28.01129935	1	36	173.0227527
378	29	33.8909629	11.15640384	Japanese	31.24437985	0	97	50.91547366
379	74	31.00126003	3.320686268	American	39.82870665	1	6	269.1761795
380	17	20.40933472	12.26946277	Japanese	43.03763676	0	39	181.1958148
381	25	30.60991697	7.262014733	Japanese	37.86245609	1	91	109.2163583
382	23	29.64745363	17.94372682	Japanese	46.96291955	1	38	197.817202
383	85	49.85343103	3.377084134	Mexican	42.81296353	1	61	374.1600239
384	96	45.89744592	10.14102208	American	32.47455159	0	91	353.8519756
385	24	28.53522993	11.96225259	Japanese	36.85624291	0	77	185.7354753
386	75	34.91749532	0.667672036	Italian	47.05233108	0	52	326.8374472
387	41	39.91486416	7.273862495	Mexican	34.76672177	0	3	336.2227678
388	96	11.39460641	14.65270469	Mexican	42.99227792	0	81	324.0212621
389	72	45.79685765	16.40755015	American	37.55242014	0	24	376.373954

390	95	44.40507266	6.263239896	Japanese	32.07745444	1	6	399.2691103
391	60	28.29728391	15.97186545	American	19.67352782	0	27	293.0874415
392	34	25.64679064	19.35511754	Mexican	27.03990078	1	68	278.8705847
393	67	20.91837499	17.13747872	Japanese	48.70971299	0	66	300.9057773
394	72	29.06707313	2.352879763	Italian	36.31061533	0	94	302.6967588
395	71	11.77506654	14.79460664	Japanese	32.57581521	0	19	288.9048253
396	31	44.08774133	6.393336649	Mexican	47.31826502	1	82	234.3481646
397	67	11.41689474	19.76831979	Italian	33.46047124	0	10	263.463877
398	67	23.16205461	9.707179663	Japanese	31.04330074	0	66	300.5994886
399	95	47.26357953	2.448677967	Italian	40.37243421	1	86	480.7545138
400	58	39.40850618	14.50371096	Italian	38.50662476	1	6	255.1177213
401	61	21.25563842	6.460223968	Mexican	17.72640034	1	22	214.8227571
402	51	20.10596376	19.94156314	Italian	28.85233098	1	60	267.7270824
403	79	10.20328559	19.59564916	American	28.90061771	0	70	408.1658206
404	24	37.82857171	13.12268036	Mexican	25.12898061	1	33	231.9747141
405	63	11.95134327	2.596528979	American	30.69173929	0	18	83.0636909
406	69	31.30410078	13.73721995	Italian	26.52429591	0	57	412.9268001
407	17	48.4848789	1.860609729	American	17.97381958	1	8	211.389448
408	62	14.3787544	0.587283182	Japanese	26.49465949	0	47	141.5435209
409	69	45.77919724	16.45916892	Italian	12.8910779	1	81	325.0684297
410	14	49.65901341	4.195058954	Japanese	33.03143451	0	40	292.2978294
411	77	12.46068105	7.580542524	American	44.14618196	1	47	343.4945802
412	15	45.31625077	19.4691478	Japanese	39.01148337	1	44	299.3528926
413	56	30.64316082	6.736810037	Italian	24.4157616	1	24	270.0704893
414	64	46.26054878	5.929795626	Japanese	13.48284822	1	77	259.5662004
415	49	32.85275768	18.36415999	American	18.08597597	1	24	287.2146326

416	61	36.4844804	2.214968906	American	31.81961362	1	18	325.1546481
417	25	31.75115632	15.1916482	American	18.28228334	0	38	174.6629697
418	22	44.57971521	19.76131133	Italian	31.75892445	0	78	222.7186198
419	39	39.35639676	18.02592022	American	19.26396099	0	79	359.0860101
420	28	30.85041249	13.03335837	Italian	43.95695478	1	26	208.4240831
421	26	44.72643216	6.2812159	Japanese	25.89085891	0	73	278.4208386
422	72	19.81142129	5.086439624	American	19.82202504	1	62	246.1307927
423	28	16.29362639	16.3851021	Japanese	42.49304721	0	67	165.2538355
424	67	16.41600421	17.59643828	Mexican	23.71355859	0	18	276.6655938
425	64	23.36035515	10.50869825	Japanese	14.45950751	1	27	314.105031
426	99	19.54760125	4.886798816	Mexican	32.81637966	0	84	319.8655012
427	99	47.20361402	2.258864168	Japanese	33.26256818	0	77	394.4393476
428	71	14.23336375	1.743589357	Italian	38.58782701	0	32	218.4024908
429	32	37.95962952	14.38154895	Japanese	19.60877733	1	9	291.0891079
430	18	17.56886159	8.402374557	Italian	27.75085356	0	82	192.8954576
431	21	45.26548531	18.10719749	Italian	21.29471676	1	29	185.675556
432	10	48.45261153	14.79342601	American	13.23814944	0	21	260.8661242
433	67	28.61880035	3.825208358	Italian	45.06455308	0	2	424.0145018
434	10	45.16285921	12.27554838	Mexican	39.15479218	0	81	155.7844327
435	43	10.20580683	8.710109088	American	22.05852238	0	74	171.4008164
436	57	42.7978767	11.54499462	American	33.97491971	1	47	472.8480308
437	98	19.14500646	14.76138247	Japanese	18.05055153	0	1	371.681354
438	10	46.34805237	16.12391762	Italian	45.60946487	1	75	228.9437953
439	25	10.04480446	16.87474653	Italian	14.59625373	1	53	220.3678807
440	70	32.2472694	5.079497491	Mexican	10.15348582	0	5	364.1679974
441	73	40.26543215	0.622311998	Japanese	30.08032791	0	39	310.2675784

442	72	20.34753323	14.7858947	Japanese	33.32511294	1	92	297.4458824
443	78	17.72531458	2.126874525	American	33.15861281	1	93	240.5142952
444	31	13.36392898	6.298391772	American	12.1804771	0	1	79.86631534
445	76	23.05734306	0.076342682	American	22.56175228	0	72	264.1027406
446	85	32.39377249	13.63338965	Mexican	29.96795787	0	1	391.66196
447	35	14.63375427	3.342640229	Mexican	18.66316743	0	92	110.3208532
448	25	25.06781448	13.10564819	American	44.90503542	0	96	40.07049395
449	60	10.36289619	19.83485234	Japanese	13.48437158	1	73	361.5529383
450	95	42.74717682	1.453417432	Mexican	29.49537767	0	97	346.9235192
451	66	15.13794872	1.541560239	Japanese	37.6566423	0	54	145.6654655
452	38	21.33371793	6.807601853	Japanese	20.47004401	1	71	162.101083
453	87	10.73710875	17.32773029	Italian	34.96962948	0	59	371.7323479
454	78	27.28963136	4.310597376	American	43.17540601	0	96	205.1618195
455	56	43.60486011	7.207556399	Italian	17.613479	1	5	324.9862276
456	71	25.68604846	1.370108133	American	46.68882367	0	18	196.4012515
457	78	17.17989294	15.26310583	American	32.14234131	1	95	434.8650083
458	85	40.14795336	8.166126855	Japanese	21.96928017	0	26	437.7259277
459	25	41.55227927	15.18497466	American	27.23433102	0	0	212.9365565
460	99	16.95013901	18.07215689	Japanese	22.48767535	0	32	394.6819877
461	99	40.83416434	4.655262942	Japanese	39.0044464	1	39	405.4757996
462	57	47.89191553	16.14792061	American	41.85202654	1	31	378.4028782
463	94	16.65510348	1.181877802	Japanese	43.31749569	1	68	317.9346753
464	48	40.8393717	17.97694776	Japanese	49.43072998	1	16	287.5643152
465	42	38.34288018	13.01537095	Japanese	41.6513717	0	7	275.2114984
466	32	28.94295554	17.48453448	Mexican	34.05536283	1	56	202.6176117
467	19	12.31681842	0.003767991	Japanese	23.56881194	1	41	110.8748855

468	78	12.98777671	3.381621407	Mexican	31.78381795	1	97	196.9283363
469	43	44.09658637	7.795187749	Mexican	11.64272691	0	8	261.2715461
470	61	49.60593491	7.590058067	Japanese	39.79300302	0	86	293.1873092
471	19	37.04209029	4.534607658	Japanese	27.8382218	1	34	111.1706306
472	28	27.83833228	7.632238797	Mexican	10.80092919	1	70	260.9154433
473	67	35.90294574	2.57697424	Mexican	12.50027766	0	78	284.5457691
474	10	33.77583437	8.255222585	Japanese	25.49865855	1	52	106.5554469
475	78	25.34449042	16.04708658	Japanese	33.33686076	1	53	417.2333618
476	13	12.62504578	9.581401262	American	28.56815796	1	83	141.2283856
477	25	42.8605179	0.360001779	Mexican	34.76817269	1	24	225.6358839
478	33	24.87334117	8.428880832	Mexican	29.21177035	0	66	127.9365596
479	89	42.75142573	0.543500322	American	36.46815382	0	61	380.8633043
480	11	20.10034999	19.32486821	Mexican	18.76504259	0	54	139.3655501
481	41	10.04117919	2.549079188	Italian	14.51372749	1	45	226.6879005
482	93	16.35753133	14.15376653	Mexican	33.85735819	1	23	220.9561244
483	33	28.74157667	19.45552646	American	29.38758068	0	97	167.1026649
484	21	47.7721454	16.52078103	Italian	11.07470616	0	45	336.3950397
485	59	14.08774479	13.9863075	Japanese	25.21927542	0	21	270.5425511
486	44	44.16839098	16.78921819	American	14.24502342	1	10	297.2885877
487	42	31.08096385	19.63597387	American	31.25427327	0	47	239.9143063
488	42	43.03683063	12.42688995	Japanese	25.83110719	1	7	258.6604559
489	70	21.93497724	11.18369614	American	34.52607644	1	38	162.4303339
490	60	43.59788659	12.56476221	Mexican	42.22408875	0	34	357.71479
491	52	40.9129112	10.65351774	Japanese	21.98674735	0	48	264.9095348
492	21	19.20214732	12.70178212	American	48.48408417	1	45	151.7332266
493	76	15.25340506	8.996844159	Japanese	31.42538002	0	37	315.4251889

494	74	44.7221729	4.368863658	Japanese	34.73163195	0	7	291.4544727
495	42	48.52946614	2.40494453	Japanese	37.22689493	1	53	330.2964304
496	49	11.25209255	1.366818711	Italian	36.66926119	1	36	156.0776544
497	83	39.25080268	5.266315675	American	30.43882972	1	90	329.2473389
498	52	46.38098044	0.922471287	Italian	27.15057164	1	91	274.1840667
499	53	44.29440368	11.3434156	American	12.15127857	1	39	309.479463
500	38	48.58081889	5.67656894	Mexican	10.88343381	0	33	341.970237
501	22	40.40376314	16.49069616	Mexican	24.31844778	1	95	247.0698273
502	21	27.6919505	1.48481746	Mexican	23.34643563	1	78	111.0873077
503	55	22.47807962	4.527029887	American	25.50780619	1	16	290.5908109
504	11	22.82970962	14.60492294	Japanese	15.53472804	0	32	49.23777578
505	44	38.9769069	11.97616593	Japanese	39.52673082	1	46	282.5435137
506	96	39.83348048	1.289825425	Italian	44.27857659	1	51	263.9744397
507	90	18.01436574	17.54478914	Italian	47.54858541	1	79	333.5772185
508	99	44.72491254	17.54523526	Mexican	29.69449214	0	92	518.3270032
509	17	38.59696161	6.608160078	Italian	38.03462797	1	8	204.3623864
510	35	32.51001145	6.808116867	American	44.22204494	1	32	256.8131171
511	83	28.8612984	19.78056717	American	26.73164444	1	59	399.7575479
512	99	31.72931223	12.52352343	Italian	22.37651179	0	92	341.8038146
513	43	16.36982953	10.62633041	Italian	47.46898223	0	76	229.5792084
514	16	18.14804997	16.76304367	Italian	30.62706204	0	20	104.6343107
515	77	13.64657962	5.705806767	Italian	29.89015546	0	57	402.9054623
516	67	16.14632153	8.583651868	Mexican	17.13664386	1	84	143.4209598
517	84	28.14774538	5.35553351	Mexican	14.32620058	1	68	390.9255975
518	38	31.21417575	15.78321693	Italian	43.6842745	0	9	219.190575
519	45	10.7144757	11.46698691	American	34.56532705	0	5	217.894581

520	98	41.87191311	13.1765014	Mexican	23.03298336	1	12	449.0744415
521	30	21.76671893	5.046144997	American	17.6694487	1	64	171.7534695
522	45	46.4734414	4.404506028	Japanese	32.21989121	0	19	296.0598969
523	19	48.70045054	9.76749258	American	33.35677157	0	56	151.762282
524	82	17.41835499	14.75131498	Italian	18.27974405	0	45	371.6100961
525	33	28.48389163	10.50479515	Japanese	44.05622992	0	7	190.6073138
526	73	10.04927341	0.652913551	Mexican	13.37057465	1	9	187.0222094
527	58	33.9330906	17.37103013	Mexican	24.24372341	0	72	329.386519
528	45	34.29225793	1.29373787	Japanese	37.21537655	0	66	233.3603624
529	91	19.90159822	19.11648967	American	20.96481455	1	10	461.8865537
530	33	32.3365069	0.502839166	Mexican	43.10700779	1	11	186.6819617
531	32	19.0216057	13.88900912	Italian	39.59471306	1	9	201.4229832
532	71	16.43301876	1.644141274	Japanese	41.56091378	1	24	157.6961069
533	46	36.06077867	18.21353159	Italian	14.76426144	1	48	344.6122706
534	21	23.03701856	9.288664725	Japanese	15.62188512	0	48	139.5235088
535	64	33.61055756	2.750475496	Mexican	21.45894951	0	88	335.9146027
536	22	27.32986996	6.816313996	American	30.72308661	0	10	106.8139275
537	32	49.3740563	3.296493507	Japanese	39.92064307	1	61	239.9972406
538	98	15.98322571	3.411638231	Japanese	43.46123563	0	35	311.2183851
539	39	28.58394387	18.9621664	Italian	38.17341264	1	38	302.7671014
540	26	37.21826505	8.904922246	Mexican	47.14387526	0	52	193.27788
541	71	25.82180336	15.77604528	Italian	39.54974038	0	89	304.2486844
542	93	48.98799407	12.79211122	Mexican	22.25071548	1	77	466.1583023
543	98	38.87550512	13.52852725	Italian	20.60325137	1	55	429.5654261
544	95	36.2793543	2.633868769	Japanese	37.84769801	0	0	339.6173814
545	22	40.2823707	4.683688936	Italian	17.71482275	0	55	249.7515456

546	68	48.10594552	15.7350067	American	13.70859243	0	3	462.1045294
547	28	46.31274612	19.41532413	Mexican	31.28891499	0	29	273.8164197
548	58	38.97141233	5.253512852	Mexican	42.86566279	0	11	296.2868417
549	21	39.96646556	5.431375295	Japanese	14.0551937	1	34	182.5007747
550	70	15.83688296	9.457942656	Mexican	41.67565295	1	76	412.926019
551	28	14.11169618	19.67680626	American	42.46288017	1	51	141.3203232
552	85	32.83618007	5.379178917	Italian	29.29096	1	30	383.5548713
553	18	10.73470548	8.139438193	Japanese	12.2924124	0	86	34.49421058
554	80	29.29425526	17.82960031	American	16.98359405	0	67	256.6920825
555	37	18.4139832	0.058420652	Mexican	38.29705101	1	51	110.2473821
556	87	23.84066271	0.7953193	Mexican	38.4711178	0	79	338.7863398
557	61	41.03218943	12.08143354	American	45.93421101	1	6	316.3952225
558	92	47.64950017	18.14760329	Italian	33.30744586	0	21	454.6898702
559	25	12.89665089	4.777133208	Italian	14.96429471	0	92	273.8694174
560	78	31.36968824	9.392835494	Japanese	14.85659622	0	74	329.6884102
561	21	45.30133943	9.009306033	Mexican	38.84529429	1	55	217.0683431
562	34	13.30034138	9.459426922	American	17.3513866	1	96	127.5885612
563	61	42.98056188	12.45039391	Italian	23.13541998	1	57	353.1623885
564	94	24.31205404	9.39487526	Japanese	29.48848028	1	92	380.2115742
565	62	22.88378809	8.949362826	Mexican	10.9905951	0	63	232.8714287
566	32	49.09865545	9.485768409	Mexican	19.95276108	0	11	320.5490125
567	25	18.03436359	11.81734883	Japanese	26.31558434	0	6	222.8760669
568	66	37.72775763	10.74014977	Japanese	21.18545274	0	54	334.1379957
569	48	42.85762912	17.08271995	Italian	15.4548131	1	61	196.0726971
570	62	11.64733981	0.98993707	American	39.12049031	0	89	215.9855398
571	51	36.81541833	16.66634143	Mexican	42.52590621	0	72	395.5429116
572	67	48.06199897	16.87849225	American	20.29915905	0	97	325.7322599
573	48	14.96879867	5.698160431	Mexican	40.97677978	1	76	263.3963601
574	23	45.84885211	9.2570391	Mexican	25.42214258	1	54	260.9994444
575	14	33.7910655	8.770769771	American	30.26743163	1	35	190.9114448
576	44	34.66418327	4.370196687	Japanese	44.240043	0	88	216.182297
577	96	34.51776699	7.935936942	Italian	39.99814727	1	23	281.9620174
578	84	21.95343545	0.887755443	Mexican	42.20467405	0	43	302.3945611
579	27	47.40807261	5.556441456	Japanese	15.47380666	0	16	194.4983578
580	85	47.89401851	16.33364351	Mexican	12.78877703	0	1	447.230857
581	18	35.01218641	5.691246035	Italian	45.73618916	1	12	45.38150684
582	83	41.33690904	16.36413337	Mexican	25.51337078	1	30	358.2770185
583	67	34.49031944	10.63198929	Italian	21.51518712	0	92	370.3349418
584	26	29.3759379	17.59725355	Italian	38.98785017	0	27	278.7019
585	16	36.43280473	11.49559109	Mexican	43.45662386	1	59	47.08816126
586	55	32.1200581	0.593364461	Mexican	44.49958488	1	26	251.3187591
587	22	33.89903451	13.50840415	Italian	22.63988565	1	14	205.7210154
588	49	41.35330214	0.89707136	Japanese	44.08849565	0	94	149.9137104
589	51	29.23615326	9.01194038	American	11.63735777	0	78	272.136316
590	18	11.63016424	18.58853389	Italian	11.98645966	0	32	170.063181
591	59	16.55997924	3.304078787	Mexican	34.94076781	1	19	254.3045988
592	36	27.54428866	7.797447059	Mexican	12.52425476	1	41	270.3574392
593	75	38.99306239	14.59720618	Japanese	22.98436674	1	11	351.0214661
594	14	33.73114593	17.64517112	American	36.91023945	0	72	223.4694812
595	38	36.01176743	13.41886009	Italian	23.49067584	0	22	204.0202548
596	46	29.17115714	5.163465084	Italian	17.18399126	1	71	282.7589763
597	47	34.62968088	13.49215706	American	21.25402487	0	95	291.659038
598	92	23.16069751	5.934552822	Italian	28.30301329	1	13	316.402066
599	17	28.80730371	14.3685846	American	29.01543421	0	90	133.4754751
600	74	39.77116262	19.08066255	Mexican	41.32000925	1	24	479.8152954
601	95	44.87331116	14.42786902	American	40.00612918	0	76	374.8505099
602	26	27.34621336	17.25761006	Italian	28.27271218	0	84	227.7664606
603	80	11.74646631	0.042076864	American	43.82669396	1	30	241.9958293
604	98	16.46690177	14.35144249	Japanese	14.56948554	1	53	405.9516291
605	54	10.83444557	16.59175114	Mexican	20.21413401	1	4	186.5776652
606	13	36.13052077	0.811174751	Mexican	43.11905854	1	12	-28.92089006
607	45	23.29258501	17.43195455	American	25.11976155	1	27	361.9730309
608	79	31.69914543	5.133884566	Italian	49.24726235	0	42	284.9133731
609	40	29.73107387	13.30584204	Japanese	28.65013126	1	88	399.6521997
610	28	13.26105958	16.00709758	Mexican	41.67742492	1	98	162.370576
611	70	26.13045444	13.25329026	Mexican	26.09228166	1	46	317.9799849
612	63	18.822445	1.845695841	Italian	26.99814076	0	70	216.5969576
613	48	26.73599155	11.92643038	Italian	25.31532589	1	37	249.8414055
614	83	18.14313121	10.99167313	Japanese	38.76302482	1	95	323.3042369
615	99	43.61809625	18.00642536	American	31.90071536	1	66	492.3054636
616	28	16.96773346	8.084794593	Italian	47.75112445	0	61	116.9003707
617	48	48.43122206	6.421621942	American	20.03561266	0	38	252.7354543
618	76	26.63629465	14.99701529	Japanese	14.02250928	1	36	289.0246845
619	54	43.84981168	8.253506522	American	25.7304594	0	85	185.9878407
620	22	35.15059273	12.9219235	Mexican	34.66850831	1	7	197.162077
621	67	39.41401195	12.9290018	American	44.43195748	0	80	353.7264368
622	29	40.70895552	17.76078129	Japanese	23.84787547	1	51	193.313743
623	81	30.19379528	11.97384805	Japanese	36.15147004	0	90	426.678802

624	70	31.73202093	10.0430722	Japanese	24.00794724	1	32	461.9567431
625	48	30.55983471	2.777426528	American	34.68878332	1	80	238.6512822
626	10	33.52233512	2.314752134	Mexican	45.39231788	0	41	3.819308481
627	12	11.17577783	3.086431306	Italian	19.25430569	1	75	27.3522199
628	86	25.82060679	3.436969627	Mexican	26.16966568	1	26	277.7392203
629	71	34.24754854	12.75754185	American	49.73060786	1	7	338.4156567
630	72	33.90894261	14.81962501	American	44.74228113	1	22	371.2592927
631	34	41.71824378	10.59983046	Italian	32.58559432	0	63	223.2275064
632	65	36.20037355	16.58035465	American	30.66510367	1	92	335.6903101
633	42	49.68533265	19.82526453	Japanese	21.38401529	0	19	279.3952723
634	47	21.35418114	16.40025258	Japanese	30.68837711	0	15	286.006945
635	15	24.36604119	13.89764747	Japanese	43.92725993	0	16	156.6351039
636	67	25.35388607	10.64985399	Japanese	37.40698053	0	79	201.3518213
637	53	28.63251228	8.977061433	Japanese	21.60176113	1	18	373.8083687
638	54	43.43992255	1.687847611	American	18.05855684	1	7	178.141588
639	41	19.43168376	7.987374968	Italian	20.60834148	0	37	111.1652405
640	54	41.04000828	18.20269274	Italian	16.79036806	1	37	373.8231073
641	70	26.21240732	16.1604526	Japanese	13.82562784	0	52	345.390677
642	56	34.07421535	4.3210273	American	26.76544698	1	92	281.5320603
643	30	35.30162133	9.332155694	Mexican	14.53198418	1	99	51.26926144
644	89	24.91394549	16.33446405	Mexican	16.87650231	1	89	459.0826148
645	94	30.50460821	1.347691096	American	31.89298737	0	30	363.0077801
646	84	26.52124699	7.747851713	Japanese	16.58397095	0	43	267.6115943
647	45	12.76531145	17.03682474	Italian	12.5551907	1	59	208.7354245
648	28	27.48661555	8.077957283	American	22.695554	0	75	186.8546548
649	29	13.8957357	1.552292895	Mexican	18.81276394	0	16	73.3422458

650	66	26.20083196	10.85068737	Mexican	30.61220752	1	78	248.4243693
651	27	33.71673599	17.2348499	Mexican	14.09468338	0	72	83.60227642
652	56	33.17659597	11.80571086	Japanese	40.07083151	1	38	293.4075583
653	58	36.50485116	4.525582164	Mexican	46.24934078	1	41	215.8023562
654	23	13.81046032	7.684792403	American	39.73740696	1	47	131.8227091
655	24	36.08492982	9.002467224	American	31.3130815	0	11	144.7603508
656	40	22.48592227	11.58532708	Mexican	44.42671287	1	95	191.9012814
657	10	25.96183785	15.18702052	American	37.79673519	1	88	178.1240069
658	63	44.31945016	1.977472353	Mexican	10.4220449	0	17	301.3408821
659	12	10.36121617	17.79304066	Mexican	25.76360611	0	78	45.96339532
660	25	33.53650533	0.24619406	Italian	44.03807598	0	95	227.6197437
661	96	33.9485592	7.401046693	Japanese	23.83435432	0	5	398.2725912
662	66	25.86639105	0.548406199	Japanese	11.40860694	1	42	304.0339168
663	84	19.48111428	7.859484507	Italian	39.84718554	0	34	336.1401243
664	21	19.38696153	10.51625198	Italian	42.08153081	0	97	27.11148877
665	83	45.34838679	10.23362773	Japanese	16.4237478	0	53	429.637912
666	25	27.68097495	3.312566941	Mexican	31.49361987	0	61	177.4576776
667	81	23.23557452	12.9148102	Mexican	17.67821756	1	68	283.4391256
668	85	27.95110295	0.89556901	Japanese	38.76937676	0	81	254.478197
669	33	36.77561606	15.2381364	Mexican	20.86201821	1	15	223.7356099
670	37	15.83026273	18.93288868	Italian	22.6486049	0	7	59.59001063
671	17	27.97735341	6.795376543	Mexican	43.79883415	1	88	231.0412336
672	45	33.51537484	10.34478763	Mexican	29.05289409	1	95	209.3143248
673	99	47.94333419	7.578205561	American	36.79017805	0	89	423.6922028
674	17	20.10111128	0.134207806	American	26.09492203	1	85	105.2123856
675	67	24.37002992	7.717490058	American	21.92027429	0	30	396.712849

676	69	25.1743436	15.83440519	Mexican	35.01912242	1	9	367.5586697
677	59	34.22152631	11.91977684	Japanese	16.19430296	1	55	216.9907778
678	37	24.26928212	2.025198292	Italian	24.91999342	1	75	184.6027302
679	50	31.34047701	16.94278979	Mexican	18.3614815	0	91	359.6951421
680	73	41.27161319	10.6217069	Japanese	37.83179868	1	69	319.6722433
681	36	38.21542376	16.83446723	Mexican	47.76835353	0	19	333.6194123
682	72	27.85474586	11.60443038	Italian	49.82043872	1	4	313.4948686
683	26	44.56864711	11.1937689	Mexican	28.01107242	1	22	291.8783286
684	82	32.32560998	2.105183704	Italian	20.10121559	1	8	321.3732353
685	42	31.59623358	3.347194663	Mexican	27.71317482	1	57	249.3390645
686	93	11.36751064	19.44050513	Italian	43.8907764	1	58	381.7734352
687	86	49.46771921	4.569734935	Mexican	18.24022714	1	92	305.1816839
688	38	14.91883234	19.54904236	Italian	48.08382607	0	79	119.7428306
689	22	19.2452365	10.63825413	American	23.67463075	0	53	192.5891779
690	55	12.06787349	2.715780941	Mexican	49.43214638	0	45	251.3877262
691	44	35.60011867	1.986044055	Mexican	43.5443555	1	21	275.0092103
692	15	18.98804708	4.177281414	Japanese	42.34282137	0	63	175.8507115
693	91	17.2004175	18.5127466	Japanese	22.71618695	1	7	343.7171118
694	78	13.54222337	8.283004487	Italian	10.35108848	0	19	292.7757897
695	56	14.13235828	7.90868536	Mexican	25.28531635	0	50	214.6059107
696	34	33.38682802	6.264125221	Italian	22.07237479	1	77	113.2507774
697	75	37.52831463	2.941237011	American	11.49587962	1	38	405.4127559
698	19	25.89515954	9.757023944	Japanese	30.21461026	0	36	173.0702607
699	65	29.19588326	2.60095565	Italian	26.12125245	0	49	250.5318112
700	39	37.43870944	16.70960473	Italian	35.38225871	0	33	188.7639422
701	14	10.69519161	7.407939726	Japanese	34.84745289	0	74	169.0619304

702	42	22.92522628	7.496071356	Mexican	14.82986544	1	32	345.767038
703	74	48.92390992	9.48012857	Italian	49.07862185	0	30	348.0330389
704	27	33.20448919	8.882554263	Japanese	17.47725718	1	53	291.4428205
705	58	20.1689832	19.89715946	Mexican	27.81096099	1	43	351.2672812
706	20	30.86920187	10.90762242	American	29.52470174	0	33	82.19211368
707	94	23.70613952	1.764034076	Japanese	42.52251189	1	70	353.7325253
708	35	31.28979955	12.13754683	Italian	13.96918384	0	61	292.1149576
709	72	15.02966047	3.031685857	American	27.14577117	1	22	246.7498564
710	98	15.30858615	0.618752895	Mexican	35.14008681	1	45	434.8571416
711	95	15.74169448	19.41484307	Mexican	18.60622486	0	55	404.6011166
712	68	47.59279168	15.47077783	Japanese	35.09871595	1	6	369.817071
713	36	39.31595673	19.60217985	Japanese	46.9465935	1	88	336.2908285
714	58	16.49527579	9.55756896	Japanese	49.90072525	0	60	271.8847397
715	86	17.53396335	10.64515062	Mexican	11.29804307	1	2	350.9447573
716	42	39.23317899	3.366739118	American	19.42352913	1	37	311.6230058
717	10	16.24655379	4.938148533	Mexican	22.41228463	0	83	109.309554
718	30	19.38245173	16.7709843	Italian	25.96504402	0	33	239.5469493
719	64	48.53387331	7.123712942	Mexican	19.6902863	1	43	295.5068596
720	15	20.21226099	16.53629948	Italian	18.13951978	1	29	171.9434908
721	90	36.46529401	14.18027449	American	48.1821537	1	16	350.0895856
722	78	49.1452515	3.543418505	Japanese	46.74086411	0	58	460.1955813
723	14	24.97187069	10.14892732	Japanese	11.87970004	1	9	72.06850544
724	12	10.70063963	13.54774643	Mexican	23.55252086	1	98	206.8992174
725	62	19.76589162	4.700744296	Japanese	26.20594215	1	83	271.8175688
726	32	43.23508765	16.61377021	American	19.56058139	1	17	202.2117539
727	62	44.58435344	10.89655315	Japanese	35.15491588	1	44	410.9441884
728	46	40.74077458	3.304841009	Mexican	35.61522987	1	90	283.2552865
729	83	14.52439276	11.65968852	American	44.02016913	0	58	404.8393303
730	83	41.82041487	19.99427608	Italian	39.63477074	0	61	539.3716035
731	92	46.74711188	3.010429833	Mexican	31.70293308	1	58	403.8434455
732	26	34.0243817	16.3351386	Italian	36.1434708	1	48	246.4661319
733	94	47.63501425	4.146180064	American	20.95955126	0	99	457.5787006
734	87	43.07355535	15.64776659	Mexican	33.7222476	0	62	430.4849068
735	82	36.67886068	14.69155846	Mexican	29.56909019	0	86	470.9541908
736	10	33.08740947	6.009787712	American	15.33377869	1	31	-7.627381568
737	60	23.94294669	6.516462351	Japanese	19.70177106	1	9	343.525308
738	54	16.37149781	15.85042784	Mexican	21.66773803	1	10	233.602843
739	86	34.19815691	9.228910238	Mexican	46.97238268	1	69	418.6631077
740	13	22.72446214	14.61636057	Italian	19.21546558	0	61	208.2901531
741	71	44.6575467	2.959682947	Japanese	21.56975611	0	46	317.7867501
742	74	49.73347749	17.05115564	Italian	24.7741652	0	33	491.8124456
743	41	40.0971105	14.21769433	Italian	36.54970955	0	99	211.3736255
744	43	13.61926781	18.13885618	Japanese	14.7295357	1	16	337.5291186
745	81	21.54742382	2.647447581	Japanese	48.1578085	1	53	356.132807
746	48	14.36438527	11.86029001	American	44.6228488	0	75	276.0633235
747	35	36.02046591	11.9824117	Japanese	33.08864292	0	84	293.1514665
748	43	40.60763458	7.069214811	Mexican	36.9615147	0	39	237.903674
749	63	47.02699324	9.169328491	Japanese	10.66641885	0	18	276.1945934
750	12	27.52689386	0.759631266	American	23.55779101	0	66	35.49014088
751	59	46.05502598	14.91720077	American	41.82640923	1	81	369.3788105
752	21	45.11564636	10.98644959	American	13.09931895	1	52	191.0570621
753	74	22.68782867	1.154821037	Japanese	21.29255337	0	63	326.4703193
754	63	44.75602502	13.61511786	Japanese	33.41187313	0	82	408.5871306
755	14	25.02689008	9.335641545	Italian	20.25652111	0	46	146.1889314
756	66	21.87477098	0.689225189	Italian	47.35911897	1	5	297.0599452
757	26	35.96298635	19.39215454	Mexican	15.24234523	1	15	298.4217333
758	56	23.13524912	18.98224286	Italian	41.89577432	1	71	274.9351707
759	32	35.20834993	10.25804832	Mexican	30.31794768	0	17	281.3999748
760	88	34.39327222	1.873448414	American	40.01323103	1	51	287.7318019
761	94	32.53755709	15.09242443	Italian	22.42861313	0	47	439.9755497
762	23	25.21678817	9.838622002	Japanese	30.57958409	1	42	129.1016805
763	75	31.93998887	6.048949566	Italian	44.27341399	1	38	297.077829
764	84	33.859419	9.828363551	Mexican	28.67512396	1	48	312.7619961
765	60	31.82747554	10.44888291	Italian	24.38879448	0	11	223.7768258
766	47	27.77677811	11.25423398	American	31.44621545	0	9	221.2461748
767	73	10.58668228	8.824474273	Japanese	38.69451012	1	70	255.1313447
768	47	33.48246831	12.11510606	Mexican	27.74253382	0	54	261.1605952
769	59	16.73777541	18.37469421	Japanese	47.55507901	0	0	256.3932878
770	91	35.70303876	7.245361742	Japanese	21.79783296	0	48	264.3819665
771	39	40.37377312	14.51432926	Mexican	29.01854425	0	35	363.5761001
772	88	30.01781237	17.88905114	Japanese	40.27710637	0	20	422.7111839
773	60	31.6945466	9.119585515	Japanese	37.56484362	0	36	253.1879957
774	72	48.05914239	8.869116728	Italian	44.5456421	0	23	395.1728992
775	61	43.88669445	12.04233971	Japanese	49.87236723	1	54	367.1711011
776	47	46.56946064	11.82813519	Japanese	27.07433315	0	24	303.220511
777	97	42.55065153	16.13178685	Japanese	14.44603519	0	6	485.5397629
778	88	14.35673367	9.529667676	Japanese	11.21663523	1	80	373.3018117
779	39	15.95967424	19.92245899	Mexican	39.07507625	1	92	377.7741581

780	60	31.06495189	1.189431377	American	44.96054685	1	93	250.4347176
781	90	19.84278635	0.063356972	Italian	18.82850344	1	70	285.8297505
782	14	29.067819	12.80396623	Japanese	16.33792149	0	96	151.6778928
783	38	25.77461295	5.968698247	Japanese	12.48820366	1	70	221.9794741
784	13	31.97656954	4.807647774	Mexican	42.90811687	1	23	205.3872912
785	19	39.87502865	6.2379816	American	22.4552776	0	10	268.1428526
786	65	38.61963628	5.120951302	Japanese	26.06787571	0	72	226.0204173
787	26	30.8493653	0.615800993	Mexican	30.11000766	0	11	106.9629654
788	83	42.89573675	19.43066085	Italian	43.57812937	1	34	464.2216957
789	26	27.23448366	4.414909769	Italian	36.03332886	0	62	195.2643128
790	93	45.86857159	14.68059525	Japanese	34.60258519	1	67	344.1993751
791	97	12.8512108	15.25192789	Japanese	17.026884	0	99	295.258914
792	78	48.1864012	18.58514796	American	30.52759567	0	91	400.8338011
793	43	36.18772326	0.610989142	American	38.61524069	1	56	81.08254005
794	15	25.20298036	17.44096677	Japanese	48.93043532	0	12	154.8507163
795	62	17.88616506	10.54452362	Mexican	18.05625237	0	22	284.7019046
796	75	29.16732918	12.6508822	American	10.38173412	0	82	282.8244957
797	86	17.78233293	12.88940554	Japanese	13.01205734	0	38	256.1592646
798	52	16.65326431	16.3959561	American	25.03409861	0	87	302.9909028
799	84	34.6940703	6.668146443	Japanese	42.82178591	1	79	378.5244392
800	32	30.92338898	14.6610238	Italian	10.28568327	1	96	86.55486645
801	64	36.80384138	10.57156219	Japanese	28.26922372	0	11	367.4060483
802	89	47.1842351	15.86788029	Italian	23.16922461	1	33	503.329166
803	84	31.29040853	19.81092066	American	12.0992684	1	70	437.4591935
804	25	33.23728213	4.013020348	Japanese	28.00945147	1	74	205.6737961
805	17	13.58505357	19.89297283	American	20.18994829	0	30	268.2640559

806	13	38.70543144	1.796243086	Italian	24.07539337	0	92	97.18787503
807	13	25.99620827	0.046350958	Japanese	38.22461219	0	61	129.9082558
808	65	49.06900955	0.883238724	American	30.86203936	1	12	273.3105159
809	34	42.92696062	6.835648095	Italian	14.98589716	1	30	247.3188103
810	76	30.89924162	12.65812661	American	48.45658162	0	1	400.6678351
811	76	14.92890617	1.380590818	Mexican	35.43643063	0	14	185.9993296
812	36	44.37663173	0.113316291	Italian	40.63473703	1	56	182.8877758
813	41	41.99124382	18.74182057	Italian	13.168526	1	74	345.3824754
814	59	17.38340518	16.43853272	Italian	11.88233518	0	33	375.3253931
815	70	21.81113157	16.18366342	Japanese	10.77792201	1	67	297.5324752
816	60	41.73629033	17.68848551	Japanese	17.55176111	1	74	384.9852328
817	28	17.40913786	14.00068546	Japanese	26.57633591	1	28	255.9873727
818	30	46.71009957	2.088082595	Mexican	47.43304534	1	65	255.7997732
819	14	26.98109701	7.982003761	Mexican	28.28238637	1	15	143.9580666
820	91	13.4815585	14.46298973	Italian	21.61020421	1	38	442.052764
821	51	42.62994574	5.606689216	Mexican	32.47741014	1	76	405.1891926
822	70	24.94038732	13.74812728	American	14.47738313	0	36	366.1779693
823	31	31.05996777	0.489968965	Italian	34.38023564	0	80	158.3399963
824	30	14.49217378	6.122746818	Italian	32.73727817	1	20	146.6532546
825	79	10.10336059	10.98260673	American	41.67972134	1	43	304.3043918
826	10	35.51940823	12.64179659	Italian	49.46750446	1	18	261.672131
827	14	30.60700681	11.54081329	Mexican	20.03493887	0	70	193.1946485
828	21	33.74154276	13.95512552	Japanese	30.38625803	0	7	227.7738898
829	99	41.74975402	0.121774965	American	46.35461625	1	76	285.1572148
830	55	35.86423654	2.018824	Mexican	41.40759538	0	65	342.8978375
831	43	16.99093771	17.37888505	American	29.49625212	1	40	280.9039375

832	58	24.43090076	13.63497288	American	47.81486598	1	74	262.2404707
833	87	46.23564419	2.258632854	Italian	12.28946774	1	87	366.2153963
834	99	40.79778801	16.84148875	Italian	47.23994479	1	69	393.0292532
835	54	32.26548232	4.844624356	Mexican	12.0202957	1	60	262.0070617
836	36	20.22579991	9.565197684	American	31.89218976	1	25	251.2230906
837	82	22.08242007	2.903930258	Italian	45.61763792	1	70	367.8091836
838	35	39.2783227	14.20220113	Italian	20.50883774	0	82	265.850783
839	56	44.04697043	13.33708842	American	28.88993996	1	69	278.8031741
840	95	36.85698678	4.222249423	Mexican	22.97315517	0	28	337.7828919
841	65	42.38119903	17.26801534	Mexican	48.08575668	1	65	293.8505188
842	72	40.66624541	8.026120484	American	42.94582379	1	18	395.0988777
843	57	15.98815872	5.850301821	Mexican	38.36006045	0	67	302.4194781
844	70	26.95779252	3.046591789	Italian	33.50507946	0	57	300.2114316
845	90	16.11935203	5.32336025	American	42.82367157	1	69	383.6502538
846	35	19.40942339	0.440538424	Italian	25.69068166	0	56	133.5035742
847	45	44.77038499	11.66514985	Japanese	27.9020742	1	36	283.3447157
848	10	28.91567863	16.84461493	Japanese	39.20006884	1	60	130.3139074
849	17	31.41432407	10.97960899	Mexican	24.31019053	0	91	162.8811506
850	61	17.13783861	4.385479238	Japanese	12.55603768	0	29	155.1209845
851	88	22.49087144	0.673383564	Italian	12.97338436	0	70	346.4633262
852	56	19.20984674	4.292561465	Japanese	14.61451205	0	72	255.5380316
853	65	18.24744658	8.322428561	Italian	44.78725293	1	24	340.035489
854	95	35.51843173	15.19222857	Mexican	28.88905703	0	96	446.2173328
855	23	20.27535391	3.756857051	Mexican	38.51529477	0	64	133.2447446
856	99	35.28061944	14.39582907	Italian	25.71637342	0	5	386.2982019
857	37	12.33818456	17.33866937	Japanese	20.3484379	0	17	246.3065685

858	96	15.67097574	9.078936962	Mexican	23.8876867	0	9	425.3635387
859	87	20.98161016	16.42863983	American	41.76729381	1	91	280.9195953
860	97	33.88187668	11.09314696	Mexican	35.008645	1	10	429.9535917
861	11	37.0336498	12.74181674	Japanese	47.5055912	0	96	223.4724251
862	35	27.70867867	10.52556921	American	37.18819114	0	64	166.4532554
863	23	27.83163658	9.263454335	Japanese	32.80726527	1	76	124.3513203
864	68	38.98067406	4.693068836	Italian	34.40784036	0	83	304.3613309
865	65	10.63284367	6.063881189	American	47.69485758	1	64	144.7677626
866	16	23.64360098	5.799402379	Japanese	41.52803719	1	51	145.5599182
867	12	42.74828228	11.26577862	American	10.96164949	0	66	133.9567022
868	32	34.12877641	2.39907386	Japanese	31.36558785	1	81	235.3064118
869	27	23.48908531	18.33583381	American	37.53353646	0	77	349.0280256
870	47	46.22099169	14.41310291	Japanese	39.94793914	0	30	238.5298558
871	24	40.9049091	8.028912618	Italian	18.52568477	1	90	228.8607504
872	73	20.59581789	3.172482891	American	44.68709418	1	16	253.0129148
873	98	35.8115935	17.63984339	American	48.23372589	0	54	491.2183607
874	37	33.83002913	16.4755618	American	19.46861223	1	92	210.3784868
875	83	26.04751656	14.43443105	American	45.3046183	0	35	399.7177251
876	48	24.67824317	5.430940948	Italian	13.8791505	1	25	157.9260496
877	66	23.74241899	8.048932749	American	12.22827344	1	52	283.8639332
878	26	34.68623134	2.054965049	Mexican	15.168952	0	16	97.37470186
879	95	27.69639746	3.780790495	Italian	12.30964318	0	85	421.3262292
880	99	43.27452453	15.66033275	Mexican	43.80048949	0	24	412.8267037
881	53	36.78410465	14.30077438	American	12.36200578	1	1	292.9773267
882	34	12.79726423	11.86971526	Japanese	20.86656765	0	85	177.7478129
883	26	35.60672895	17.91453452	Mexican	14.15025374	0	4	239.9446093
884	22	20.84589746	11.66533036	American	21.81688145	1	3	140.9721287
885	93	33.71247967	9.339359999	Mexican	13.6223787	1	35	377.5466092
886	34	37.20527724	6.807414926	American	36.72013003	0	5	241.9126099
887	77	19.23316757	18.80897011	American	15.33307265	0	20	413.9285081
888	19	28.2743458	1.65430923	American	44.07383703	0	76	63.12670942
889	76	17.61299451	18.61774744	Japanese	42.68764203	0	88	410.645005
890	27	12.09355503	16.5342932	Mexican	10.54646742	1	25	239.9978938
891	95	25.13905694	2.598598491	Italian	45.82905422	1	54	326.6752724
892	43	24.51777087	11.29968715	Italian	11.15777267	0	38	166.2504057
893	17	49.81921963	17.33095963	Italian	19.5858314	0	83	219.5845681
894	49	27.85409405	11.43618327	Italian	11.29556551	0	32	207.8446009
895	92	23.34278198	18.14399643	American	23.43373948	1	13	542.4672824
896	51	33.21900027	18.85967634	Mexican	30.55466103	0	55	361.5436306
897	50	13.55069576	8.447690444	Mexican	41.86579476	0	99	195.3917426
898	15	48.46662944	7.546273228	Japanese	23.51926202	1	80	115.9683918
899	61	35.39840182	18.72642308	Mexican	38.21960099	1	20	388.7252796
900	35	40.1935128	13.50280122	Mexican	17.50017828	0	45	240.8328156
901	73	16.81911753	2.320453627	Mexican	40.02393827	0	15	268.7095592
902	68	26.37911412	0.845834984	Japanese	19.43986424	1	15	260.475397
903	65	26.90756454	11.29258102	Italian	15.74179024	0	83	312.0211561
904	68	42.87225776	18.60341931	Mexican	17.41635401	0	28	420.8392686
905	79	10.04946719	16.88497405	Japanese	17.28252516	1	91	362.8129601
906	42	22.03412433	13.51695493	Japanese	30.51451518	1	7	239.0709131
907	62	49.24945497	2.782316978	American	43.79190325	1	87	308.8429181
908	31	28.99546174	5.084627675	Japanese	13.18420308	0	2	109.6767795
909	30	25.30105121	17.51022453	Italian	38.27825274	1	88	279.2171329
910	79	14.79316008	3.656144733	Mexican	36.17385839	1	18	255.5759364
911	79	48.96658515	5.568788165	Japanese	44.90079807	0	96	358.8951704
912	13	10.80105424	2.592013024	Italian	19.76590242	0	65	145.0991174
913	84	39.487209	13.52216463	Japanese	28.00439435	1	78	466.0920367
914	71	40.74399219	18.87964611	Japanese	47.03309498	1	85	323.4522257
915	71	19.74626864	5.379360632	Japanese	36.39973801	0	73	254.7235652
916	33	36.86233774	9.508132157	Japanese	25.55073651	0	81	124.0926846
917	64	13.71716578	12.68679479	Japanese	15.25649531	1	21	305.9501209
918	18	20.28189066	19.98932496	Italian	20.49940032	1	45	200.7899088
919	12	41.51607953	2.0961613	Mexican	33.2486191	0	90	177.1210592
920	40	44.58744768	9.97141599	Mexican	38.7267179	1	12	217.9615588
921	49	34.33945651	7.702402398	Mexican	14.97687547	0	32	250.2667553
922	45	30.27353727	17.3164033	Mexican	33.9838904	1	4	184.9534427
923	33	16.37216082	9.51935462	Italian	32.01405115	0	89	213.333911
924	15	12.52436132	2.219550492	American	30.55821787	0	76	109.8213697
925	75	14.70596114	12.68848291	American	37.60492471	1	26	253.7858907
926	93	19.094292	15.41561237	Italian	43.4979838	1	77	298.4670263
927	84	23.66698008	0.057625033	Italian	22.91958733	0	23	333.0678659
928	13	25.23638102	17.17913371	Japanese	45.5604876	1	66	114.8993948
929	88	23.85287074	3.72046	Japanese	16.4253445	1	62	372.0462826
930	15	19.56221329	11.04138729	Japanese	16.92095479	1	93	123.7338349
931	60	30.38896987	8.315950014	Mexican	20.67088254	0	98	267.6772312
932	71	43.01144433	14.39307633	Japanese	14.78123478	1	47	390.2413143
933	66	21.18814154	7.423232296	American	19.21276899	1	62	218.6318728
934	75	47.69548076	17.39526353	Mexican	14.25647052	0	77	454.4659268
935	88	44.93346647	14.60992722	Japanese	42.87009668	1	11	436.2816219

936	84	16.47056845	12.95426342	Mexican	15.55875968	1	27	279.6345038
937	17	41.45103827	15.92481191	American	39.8919023	0	43	238.0279882
938	35	32.98349529	15.0804607	Japanese	11.91655472	0	92	222.0412565
939	60	32.66301449	0.718773498	American	47.9408131	1	42	155.7640963
940	54	25.64171052	7.894687077	American	38.4464035	1	16	192.8635996
941	53	30.04611997	18.61614595	American	10.27186194	0	39	400.9114189
942	14	25.8871511	2.51448538	Japanese	40.55426001	0	48	101.8025857
943	79	31.58930475	14.74193101	Italian	30.56327683	1	84	437.0783908
944	35	35.11800211	19.49183623	Japanese	41.17460681	1	56	249.0056499
945	77	36.97215226	1.043972581	American	30.89987236	1	86	233.8430346
946	28	31.2992477	13.64387761	Japanese	35.64099431	0	83	304.9517932
947	93	38.85688269	9.363082826	American	41.45326622	1	63	438.0029383
948	29	15.65112197	11.58350511	Italian	26.20432218	0	74	145.5666443
949	21	18.47929964	1.70682673	Mexican	10.10832599	0	56	159.4053054
950	56	42.74940016	0.860139037	Italian	43.99785984	0	97	348.6973192
951	10	22.90111911	13.33733321	Italian	25.2356145	1	73	107.7497047
952	99	10.915753	16.4661886	Mexican	29.84170904	1	73	442.9207013
953	23	16.15164599	15.6403067	Japanese	10.31066569	1	69	247.2031429
954	73	18.94457738	10.87611456	Mexican	29.68368041	0	99	246.4845497
955	47	48.29291552	1.997336423	Japanese	49.45949606	1	87	206.7248048
956	46	32.92165716	13.76088888	Japanese	21.13070643	0	75	359.1215615
957	20	47.30922962	4.837777671	American	25.81673081	0	42	33.86572806
958	86	33.00653402	18.22137406	American	37.48759986	0	22	378.5792732
959	12	29.17417761	16.76530719	Mexican	42.5917467	0	26	81.61933104
960	42	22.54910765	14.18765934	Italian	26.69566496	0	57	283.679517
961	15	34.25483778	18.97691912	Mexican	22.38675482	0	36	188.1066823
962	59	15.56792098	0.62843643	Japanese	47.513032	1	5	152.4383726
963	19	45.93046509	14.7240085	American	37.08579867	0	64	141.7841522
964	14	43.17329621	6.604855587	Mexican	14.76085218	1	5	87.77458858
965	32	22.70125792	18.12131771	Mexican	12.75711513	0	49	340.7090042
966	19	48.1119011	5.333503933	American	29.34199403	1	21	301.7396333
967	53	43.0080307	14.49712999	American	18.80360264	1	87	284.9687753
968	11	36.65227543	17.55294039	American	20.15939014	1	24	272.9846978
969	22	14.87047645	0.833136368	Japanese	45.10768715	1	78	66.30830262
970	49	29.12936092	7.952951592	American	13.58264097	0	89	261.6709091
971	11	30.24430219	15.24618064	Italian	38.61456461	0	71	114.0934273
972	93	19.30205433	9.09837779	Mexican	20.84569014	0	50	389.5409924
973	74	13.19861577	5.345052596	American	40.41312886	0	87	343.9382796
974	72	41.06894522	4.396106976	American	38.89645538	1	47	352.4224584
975	82	14.88140208	2.630365338	Japanese	10.29567606	1	69	232.8624582
976	26	33.77163002	17.1425912	Japanese	30.36541652	1	98	204.5062031
977	18	47.45796696	11.50710807	Mexican	15.5727793	0	64	147.9712087
978	84	45.0442077	1.5276989	Italian	44.64908122	1	59	239.0811604
979	24	21.11010495	7.651606471	Mexican	44.86343531	0	45	151.9142486
980	33	44.0579455	7.354673016	American	26.7338323	1	35	162.1793239
981	47	10.01330024	3.634783814	Italian	27.77582474	1	57	133.5663818
982	44	10.70751238	8.710152509	Italian	22.13131758	1	75	238.6227594
983	58	34.10212655	6.641987308	Italian	38.66425196	1	37	306.182215
984	78	24.59192877	6.83607411	Italian	24.34645452	1	10	286.8389313
985	71	25.45843365	14.78053116	Mexican	20.71169898	0	11	321.8223496
986	69	32.57764363	4.46526771	Japanese	27.28924238	1	41	346.8057992
987	59	17.67129822	12.83719918	Japanese	13.65950074	0	38	281.8251634
988	87	18.47021403	11.41299248	American	39.86412819	0	70	369.9691039
989	84	29.38406348	11.36185025	Japanese	10.50385605	0	10	320.9757559
990	18	40.17014177	13.26014692	Japanese	35.77940246	0	35	188.8383018
991	43	34.77991006	10.96764072	Japanese	17.31458107	0	51	323.6103599
992	85	21.89420193	17.5851023	Japanese	19.29616185	0	30	296.138161
993	44	27.2134871	13.51160184	American	35.42527995	0	32	341.4194508
994	10	16.82659034	10.93926503	Mexican	49.71773863	0	12	139.1802368
995	49	12.98519257	16.59526241	Italian	41.10869956	1	17	229.9531199
996	73	41.30784163	12.1229313	Japanese	19.03358529	1	40	249.3120337
997	31	20.61549637	5.822884956	Mexican	17.0409902	0	57	110.2287675
998	69	17.11065551	4.141898364	Japanese	44.64931481	0	55	312.2125517
999	73	37.66472222	3.046555713	Japanese	27.76735792	0	23	272.4822043
1000	81	34.72206697	17.98910425	Italian	15.48211211	1	72	379.9730719

2. Multiple Linear Regression

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: Monthly_Revenue

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	6604293	2201431	522.38	<.0001
Error	996	4197348	4214.20514		
Corrected Total	999	10801641			

Root MSE	64.91691	R-Square	0.6114
Dependent Mean	268.72417	Adj R-Sq	0.6102
Coeff Var	24.15745		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	56.37349	8.80441	6.40	<.0001	.	0
Average_Customer_Spending	1	-0.27487	0.17910	-1.53	0.1252	0.99935	1.00065
Menu_Price	1	2.17157	0.18224	11.92	<.0001	0.99843	1.00158
Number_of_Customers	1	2.90646	0.07795	37.28	<.0001	0.99867	1.00134

Correlation of Estimates

Variable	Intercept	Average_Customer_Spending	Menu_Price	Number_of_Customers
Intercept	1.0000	-0.5932	-0.5968	-0.4593
Average_Customer_Spending	-0.5932	1.0000	-0.0213	0.0146
Menu_Price	-0.5968	-0.0213	1.0000	-0.0338
Number_of_Customers	-0.4593	0.0146	-0.0338	1.0000

Pearson Correlation Coefficients, N = 1000
Prob > |r| under H0: Rho=0

	Number_of_Customers	Menu_Price	Average_Customer_Spending	Monthly_Revenue
Number_of_Customers	1.00000	0.03347 0.2903	-0.01390 0.6607	0.74524 <.0001
Menu_Price	0.03347 0.2903	1.00000	0.02081 0.5110	0.25958 <.0001
Average_Customer_Spending	-0.01390 0.6607	0.02081 0.5110	1.00000	-0.03566 0.2598
Monthly_Revenue	0.74524 <.0001	0.25958 <.0001	-0.03566 0.2598	1.00000

Linear Regression Results

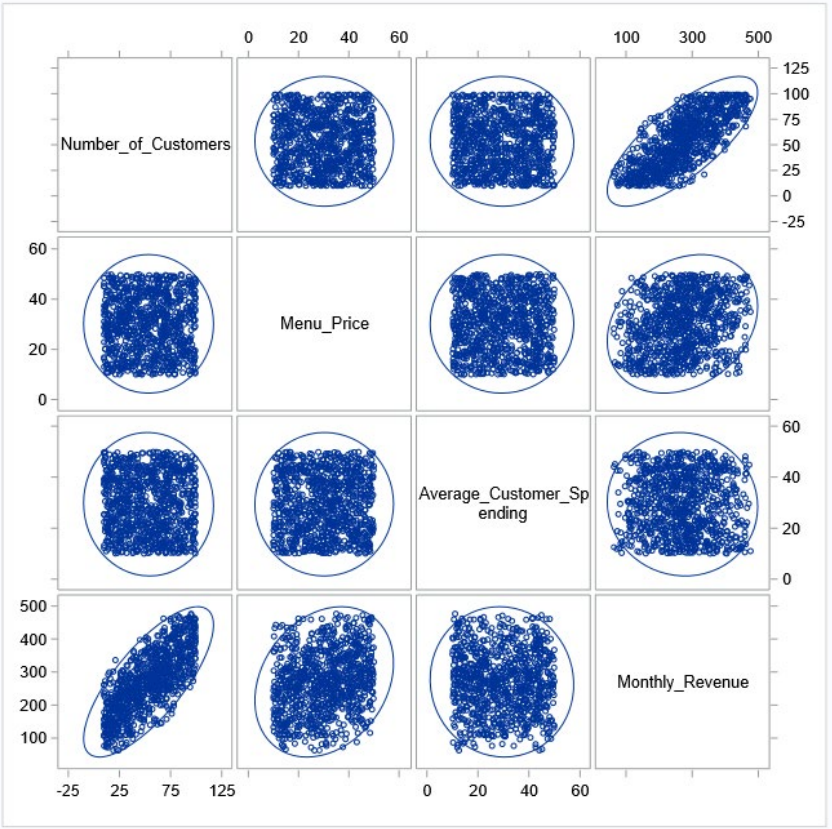
The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: Monthly_Revenue

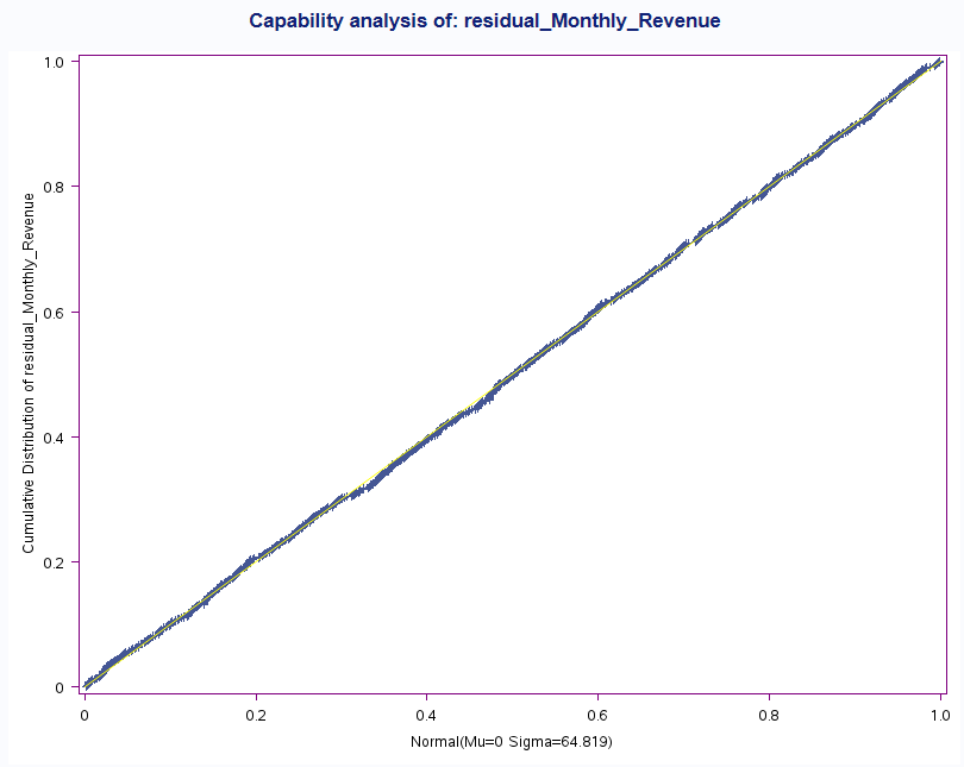
Durbin-Watson D	2.141
Number of Observations	1000
1st Order Autocorrelation	-0.071

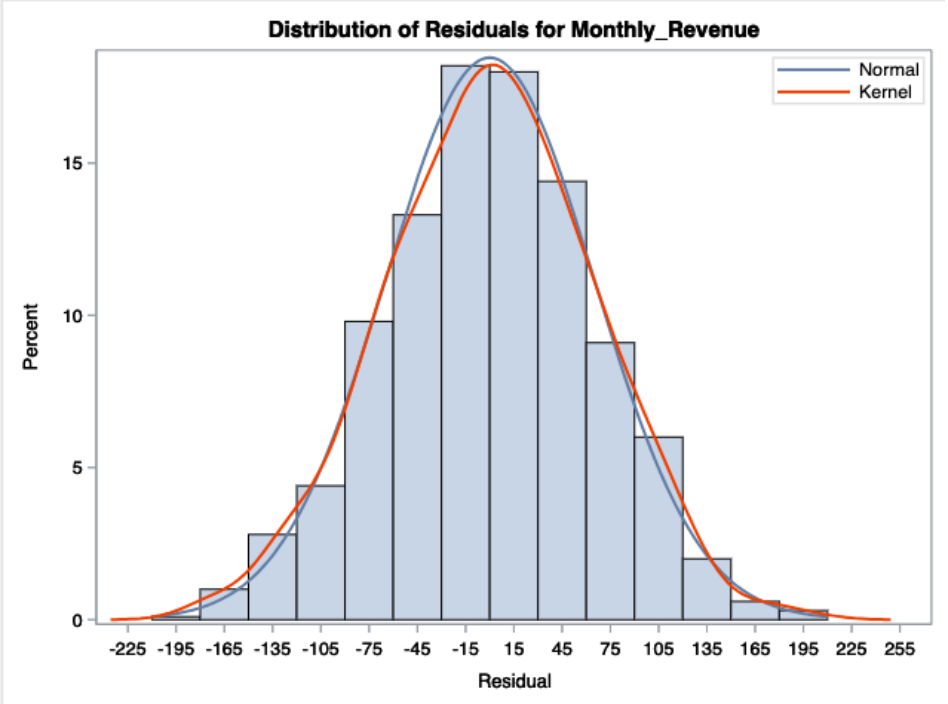
Goodness-of-Fit Tests for Normal Distribution				
Test	**Statistic**		**p Value**	
Kolmogorov-Smirnov	**D**	0.01458460	**Pr > D**	>0.150
Cramer-von Mises	**W-Sq**	0.01871405	**Pr > W-Sq**	>0.250
Anderson-Darling	**A-Sq**	0.15744849	**Pr > A-Sq**	>0.250

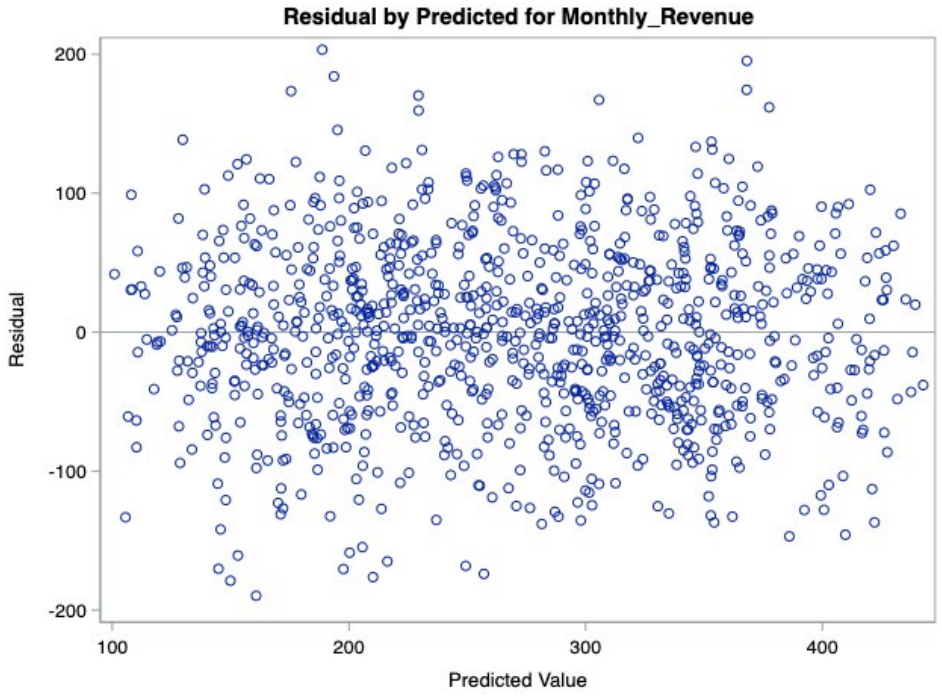
Number in Model	R-Square	Adjusted R-Square	C(p)	MSE	Variables in Model
1	0.5554	0.5549	143.6252	4812.23884	Number_of_Customers
1	0.0674	0.0664	1394.442	10094	Menu_Price
1	0.0013	0.0003	1563.890	10810	Average_Customer_Spending
2	0.6105	0.6097	4.3554	4219.93440	Number_of_Customers Menu_Price
2	0.5560	0.5551	143.9832	4810.12521	Number_of_Customers Average_Customer_Spending
2	0.0691	0.0672	1392.118	10086	Menu_Price Average_Customer_Spending
3	0.6114	0.6102	4.0000	4214.20514	Number_of_Customers Menu_Price Average_Customer_Spending

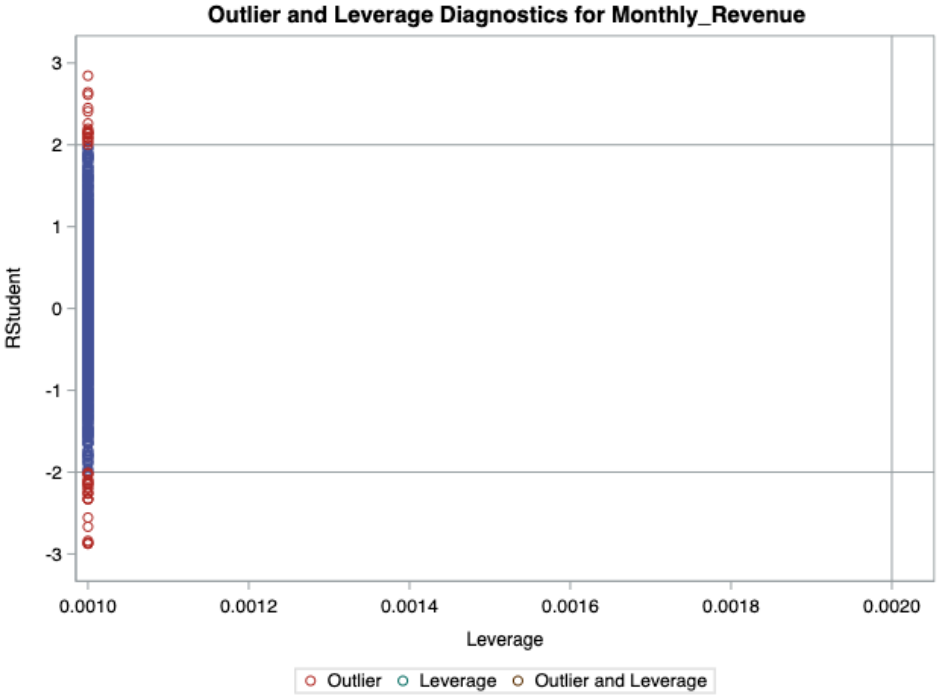
Scatter Plot Matrix

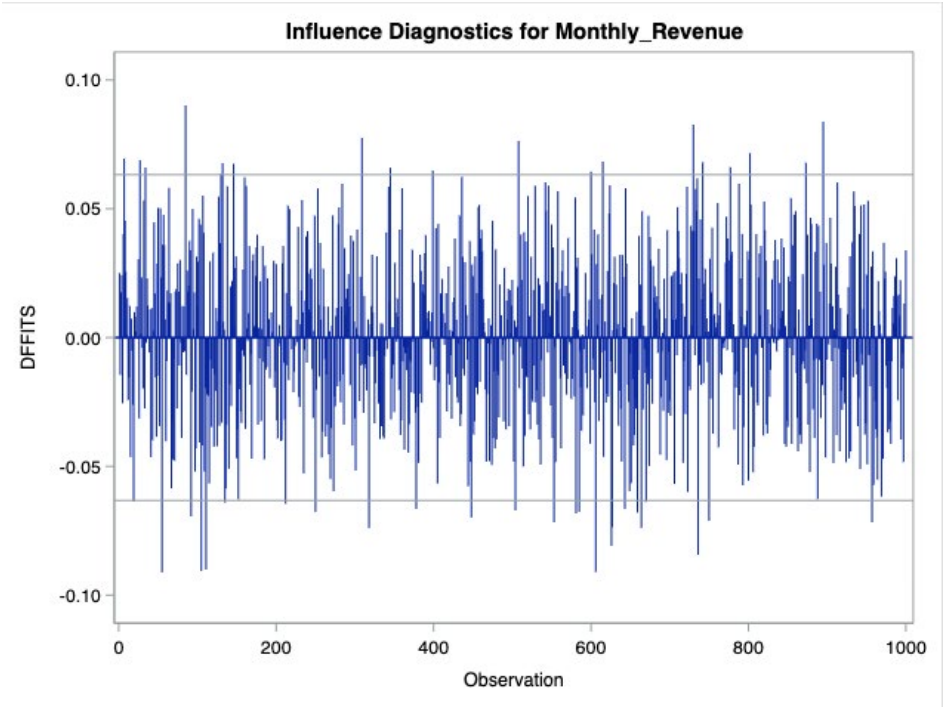


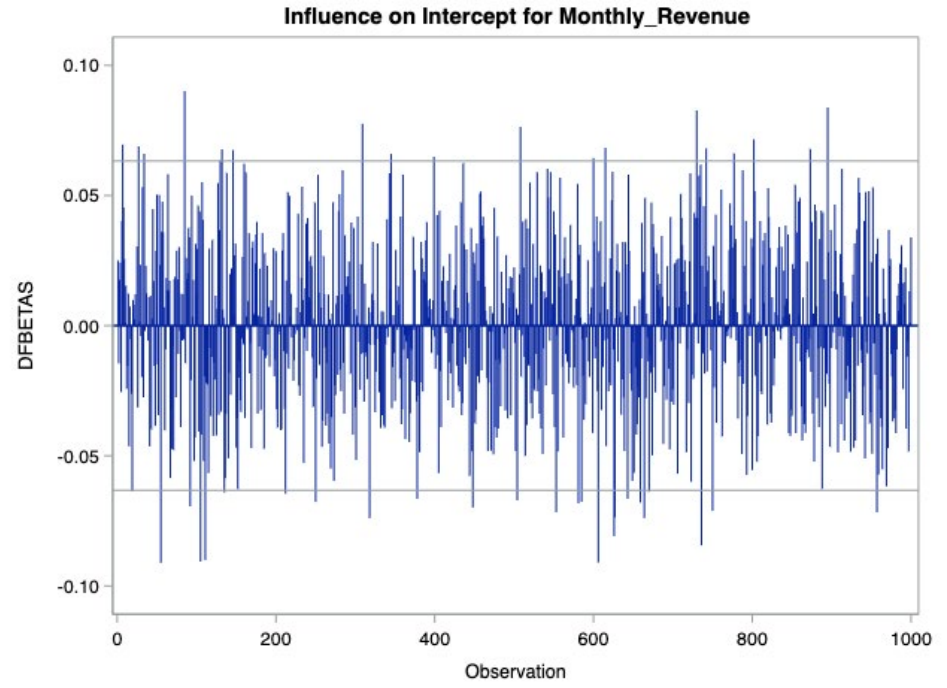

Capability analysis of: residual_Monthly_Revenue
Cumulative Distribution of residual_Monthly_Revenue
1.0
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1.0
Normal(Mu=0 Sigma=64.819)

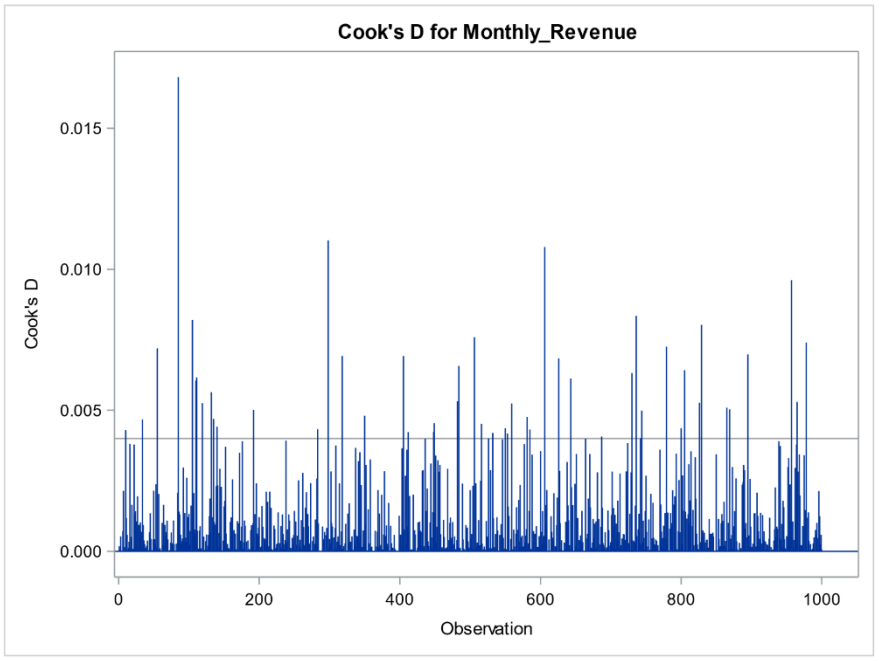

Distribution of Residuals for Monthly_Revenue
Normal
Kernel
Percent
15
10
5
0
-225 -195 -165 -135 -105 -75 -45 -15 15 45 75 105 135 165 195 225 255
Residual

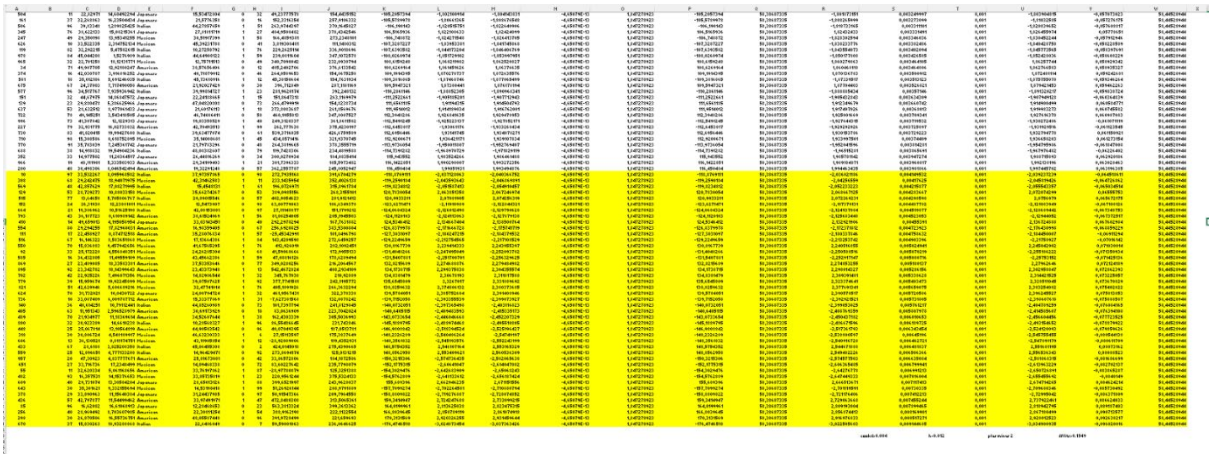

Residual by Predicted for Monthly_Revenue
Residual
200
100
0
-100
-200
100
200
300
400
Predicted Value


Outlier and Leverage Diagnostics for Monthly_Revenue
RStudent
3
2
1
0
-1
-2
-3
0.0010
0.0012
0.0014
0.0016
0.0018
0.0020
Leverage
Outlier
Leverage
Outlier and Leverage


Influence Diagnostics for Monthly_Revenue
DFFITS
0.10
0.05
0.00
-0.05
-0.10
0
200
400
600
800
1000
Observation


Influence on Intercept for Monthly_Revenue
DFBETAS
0.10
0.05
0.00
-0.05
-0.10
0
200
400
600
800
1000
Observation


Cook's D for Monthly_Revenue
Cook's D
0.015
0.010
0.005
0.000
0
200
400
600
800
1000
Observation



AFTER REMOVE OUTLIERS

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: Monthly_Revenue

Number of Observations Read	932
Number of Observations Used	932

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	5093241	1697747	539.93	<.0001
Error	928	2918010	3144.40685		
Corrected Total	931	8011251			

Root MSE	56.07501	R-Square	0.6358
Dependent Mean	270.38107	Adj R-Sq	0.6346
Coeff Var	20.73925		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	68.61265	8.05015	8.52	<.0001	.	0
Average_Customer_Spending	1	-0.19328	0.16039	-1.21	0.2285	0.99973	1.00027
Menu_Price	1	2.07019	0.16342	12.67	<.0001	0.99998	1.00002
Number_of_Customers	1	2.71379	0.07108	38.18	<.0001	0.99974	1.00026

Correlation of Estimates

Variable	Intercept	Average_Customer_Spending	Menu_Price	Number_of_Customers
Intercept	1.0000	-0.5907	-0.6115	-0.4820
Average_Customer_Spending	-0.5907	1.0000	-0.0039	0.0160
Menu_Price	-0.6115	-0.0039	1.0000	0.0017
Number_of_Customers	-0.4820	0.0160	0.0017	1.0000

Pearson Correlation Coefficients, N = 932
Prob > |r| under H0: Rho=0

	Number_of_Customers	Menu_Price	Average_Customer_Spending	Monthly_Revenue
Number_of_Customers	1.00000	-0.00181 0.9561	-0.01596 0.6265	0.75647 <.0001
Menu_Price	-0.00181 0.9561	1.00000	0.00388 0.9058	0.24952 <.0001
Average_Customer_Spending	-0.01596 0.6265	0.00388 0.9058	1.00000	-0.03498 0.2861
Monthly_Revenue	0.75647 <.0001	0.24952 <.0001	-0.03498 0.2861	1.00000

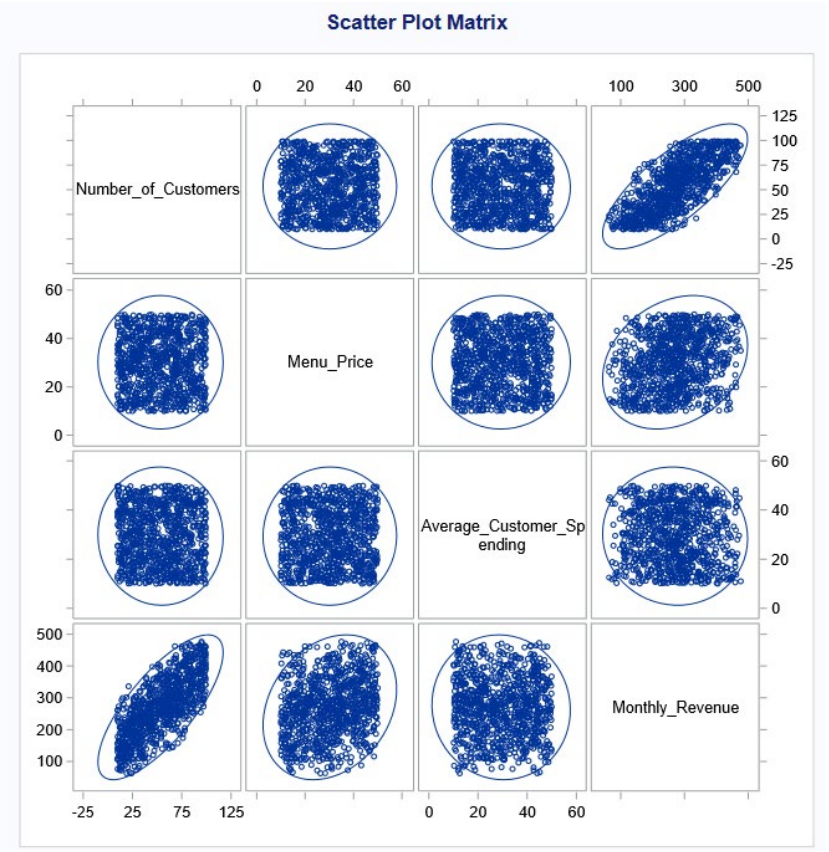
Linear Regression Results

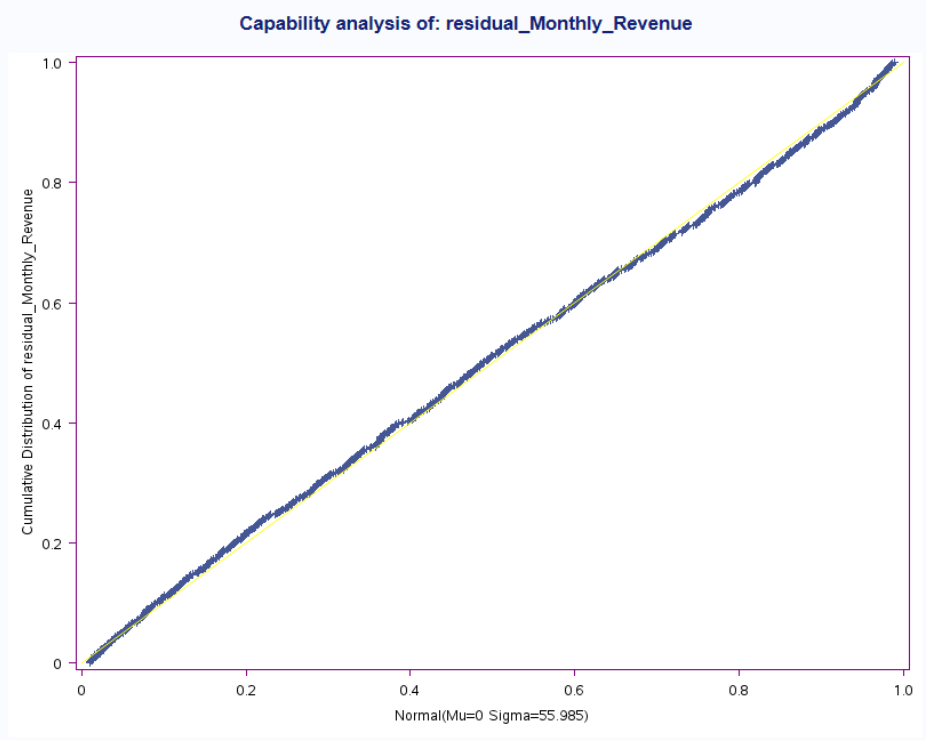
The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: Monthly_Revenue

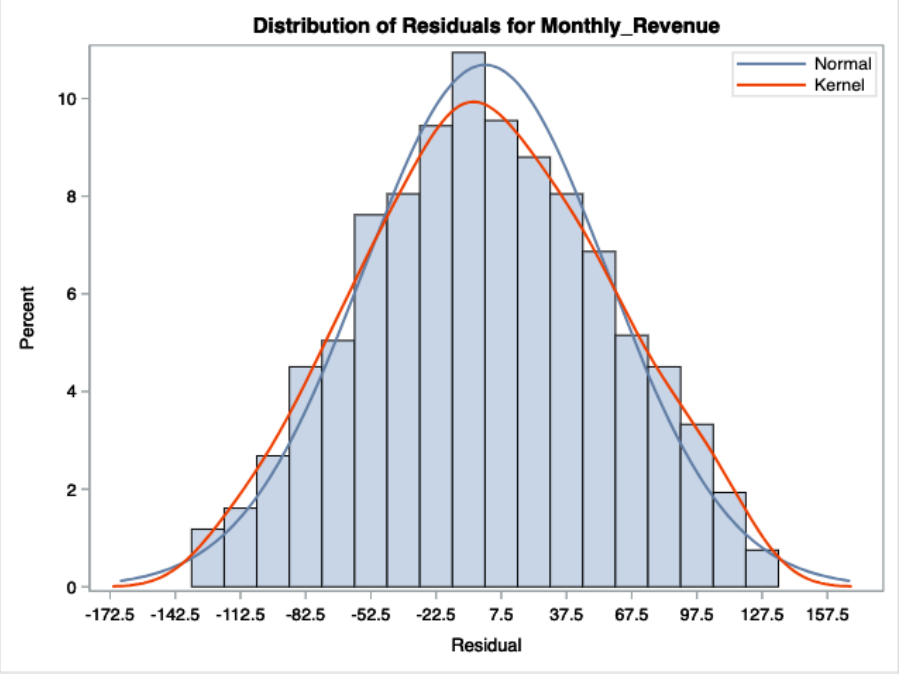
Durbin-Watson D	0.017
Number of Observations	932
1st Order Autocorrelation	0.986

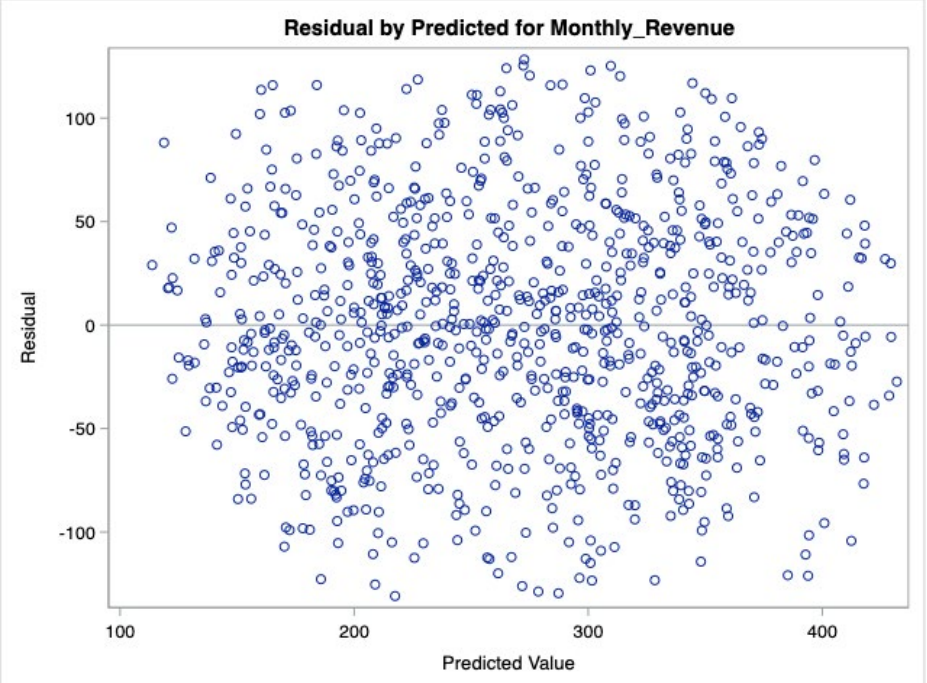
Goodness-of-Fit Tests for Normal Distribution				
Test	Statistic		p Value	
Kolmogorov-Smirnov	D	0.02181855	Pr > D	>0.150
Cramer-von Mises	W-Sq	0.09455623	Pr > W-Sq	0.136
Anderson-Darling	A-Sq	0.85357345	Pr > A-Sq	0.029

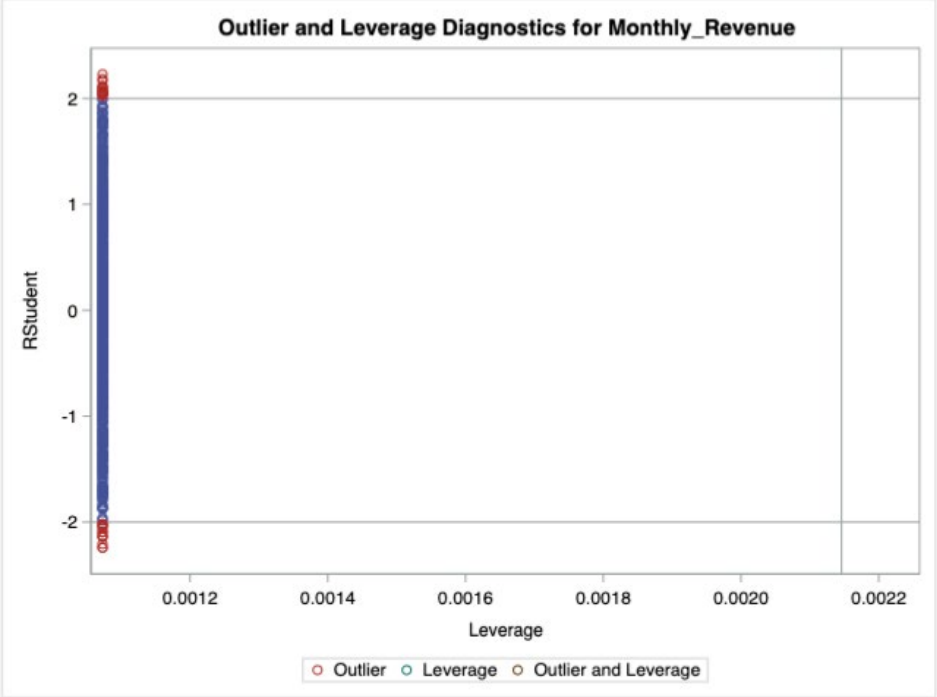
Number in Model	R-Square	Adjusted R-Square	C(p)	MSE	Variables in Model
1	0.5722	0.5718	161.8204	3684.77292	Number_of_Customers
1	0.0623	0.0613	1461.153	8077.92344	Menu_Price
1	0.0012	0.0001	1616.661	8603.70834	Average_Customer_Spending
2	0.6352	0.6344	3.4521	3145.93719	Number_of_Customers Menu_Price
2	0.5728	0.5719	162.4835	3684.21433	Number_of_Customers Average_Customer_Spending
2	0.0636	0.0615	1459.861	8075.47494	Menu_Price Average_Customer_Spending
3	0.6358	0.6346	4.0000	3144.40685	Number_of_Customers Menu_Price Average_Customer_Spending

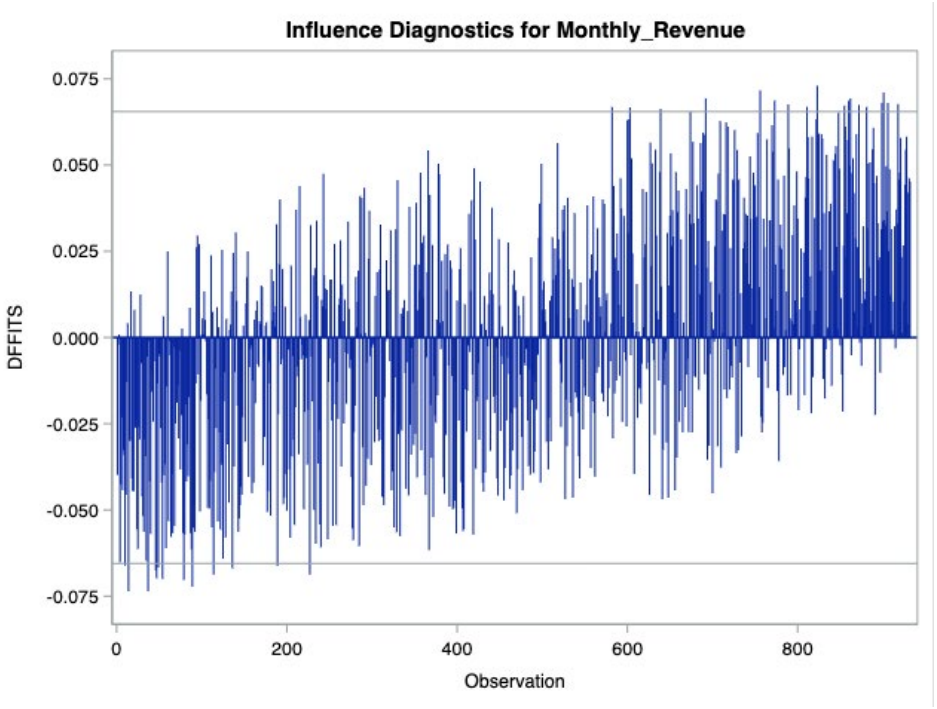


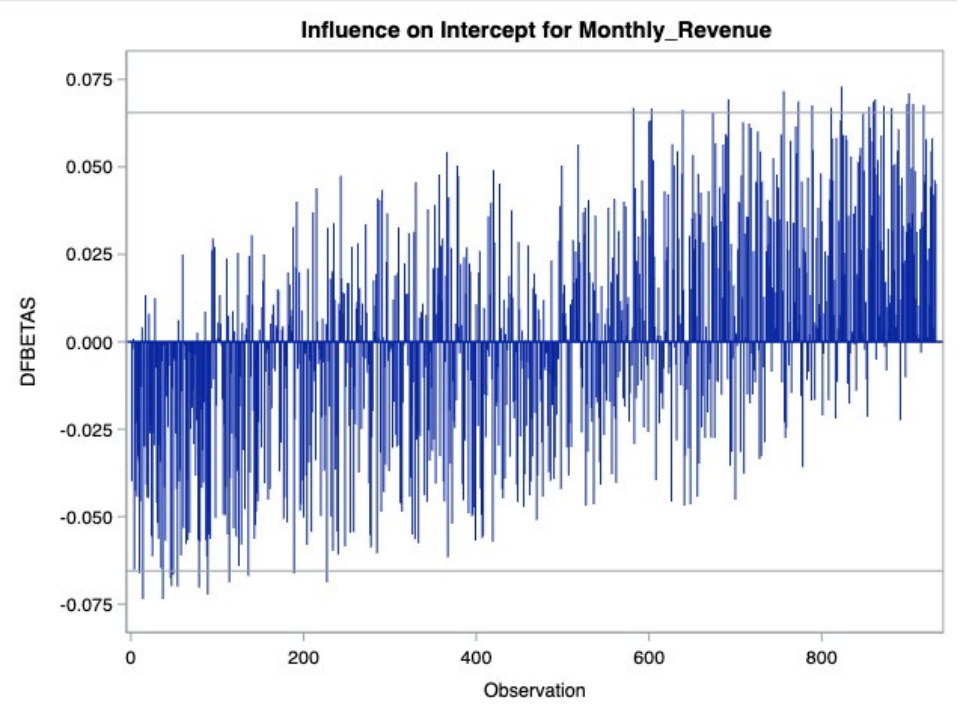

Capability analysis of: residual_Monthly_Revenue
Cumulative Distribution of residual_Monthly_Revenue
0
0.2
0.4
0.6
0.8
1.0
Normal(Mu=0 Sigma=55.985)

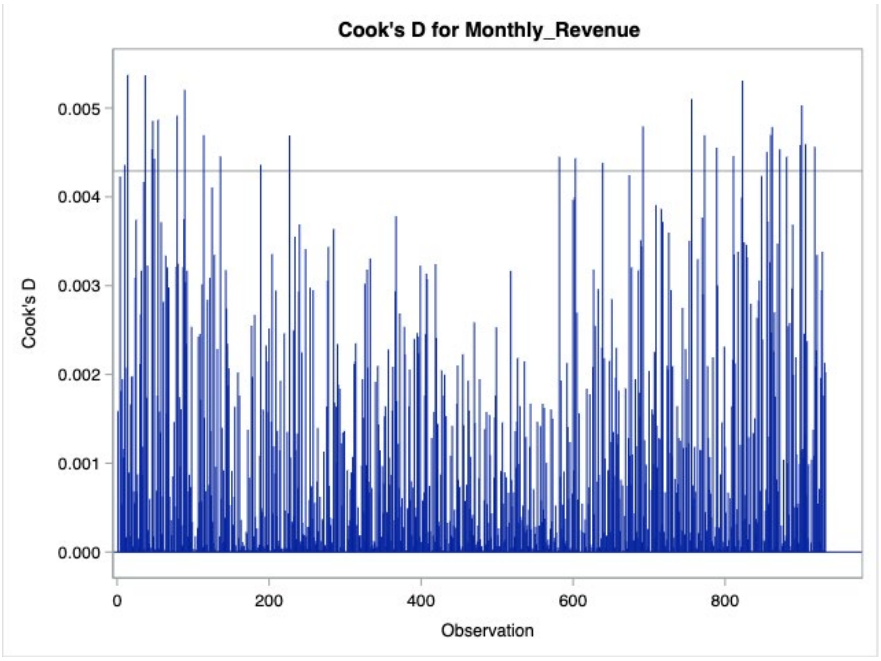

Distribution of Residuals for Monthly_Revenue
Normal
Kernel
Percent
0
2
4
6
8
10
-172.5
-142.5
-112.5
-82.5
-52.5
-22.5
7.5
37.5
67.5
97.5
127.5
157.5
Residual

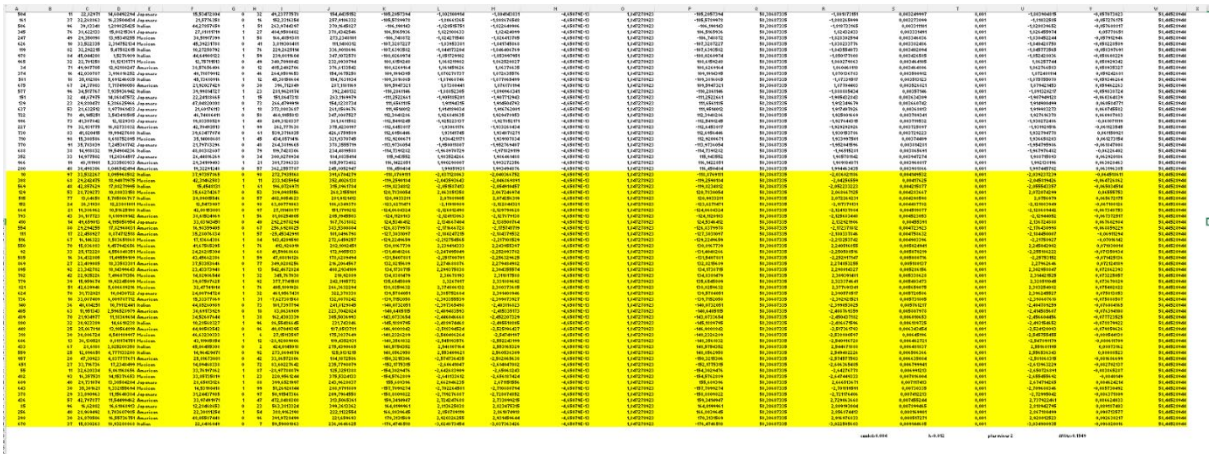

Residual by Predicted for Monthly_Revenue
Residual
100
50
0
-50
-100
100
200
300
400
Predicted Value


Outlier and Leverage Diagnostics for Monthly_Revenue
RStudent
2
1
0
-1
-2
0.0012
0.0014
0.0016
0.0018
0.0020
0.0022
Leverage
Outlier
Leverage
Outlier and Leverage


Influence Diagnostics for Monthly_Revenue
DFFITS
0.075
0.050
0.025
0.000
-0.025
-0.050
-0.075
0
200
400
600
800
Observation


Influence on Intercept for Monthly_Revenue
DFBETAS
0.075
0.050
0.025
0.000
-0.025
-0.050
-0.075
0
200
400
600
800
Observation





	AD	AE	AF	AG	AH	AI	AJ	AK
1	lual_Monthly_Revenue3	stdr_Monthly_Revenue	student_Monthly_Revenue3	cookd_Monthly_Revenue	h_Monthly_Revenue	rstudent_Monthly_Revenue3	dffits_Monthly_Revenue	stdi_Monthly_Revenue
2	-298,8306253	98,7933758	-3,024804274	0,009570545	0,001044932	-3,037793467	-0,098249262	98,89666216
3	-267,4010408	98,7933758	-2,706669741	0,007663244	0,001044932	-2,715679215	-0,087831343	98,89666216
4	-243,8681294	98,7933758	-2,468466407	0,006373772	0,001044932	-2,475075423	-0,08004966	98,89666216
5	-236,7261387	98,7933758	-2,396174205	0,006005911	0,001044932	-2,402145046	-0,077690923	98,89666216
6	-235,7302084	98,7933758	-2,386093263	0,005955482	0,001044932	-2,391978309	-0,077362107	98,89666216
7	-225,8388425	98,7933758	-2,28597151	0,005466178	0,001044932	-2,291045818	-0,074097717	98,89666216
8	-225,256954	98,7933758	-2,280081556	0,005438046	0,001044932	-2,28511047	-0,073905754	98,89666216
9	-224,5452135	98,7933758	-2,272877222	0,005403735	0,001044932	-2,277850956	-0,073670965	98,89666216
10	-221,9825735	98,7933758	-2,246937831	0,005281098	0,001044932	-2,251715984	-0,072825699	98,89666216
11	-213,9009296	98,7933758	-2,16513433	0,004903563	0,001044932	-2,169326894	-0,070161046	98,89666216
12	-212,3105176	98,7933758	-2,149035964	0,004830916	0,001044932	-2,153118761	-0,069636837	98,89666216
13	-210,8471286	98,7933758	-2,134223341	0,004764549	0,001044932	-2,138206701	-0,069154547	98,89666216
14	-208,0936399	98,7933758	-2,106352154	0,0046409 2	0,001044932	-2,110152435	-0,068247207	98,89666216
15	-204,9120467	98,7933758	-2,074147634	0,004500092	0,001044932	-2,077742834	-0,067199006	98,89666216
16	-199,1518439	98,7933758	-2,015842077	0,004250648	0,001044932	-2,019083266	-0,06530182	98,89666216
17	-198,1718123	98,7933758	-2,005922064	0,004208916	0,001044932	-2,009105203	-0,064979106	98,89666216
18	-197,8781035	98,7933758	-2,002949104	0,004196449	0,001044932	-2,006114967	-0,064882395	98,89666216
19	-194,6574366	98,7933758	-1,970349075	0,004060958	0,001044932	-1,973329168	-0,063822026	98,89666216
20	-194,5668473	98,7933758	-1,969432117	0,004057179	0,001044932	-1,972407081	-0,063792204	98,89666216
21	-191,354034	98,7933758	-1,936911584	0,003924295	0,001044932	-1,939708025	-0,062734641	98,89666216
22	-189,6010183	98,7933758	-1,91916732	0,003852723	0,001044932	-1,921869096	-0,062157689	98,89666216
23	-189,0282356	98,7933758	-1,913369536	0,00382948	0,001044932	-1,916040797	-0,061969188	98,89666216
24	-187,661385	98,7933758	-1,899534088	0,003774299	0,001044932	-1,902133351	-0,061519389	98,89666216
25	-183,4457607	98,7933758	-1,856862965	0,003606632	0,001044932	-1,859247388	-0,060132358	98,89666216
26	-182,0032102	98,7933758	-1,842261273	0,003550132	0,001044932	-1,844574648	-0,059657808	98,89666216
27	-174,8405285	98,7933758	-1,769759633	0,003276202	0,001044932	-1,771738451	-0,057302117	98,89666216
28	-174,0324743	98,7933758	-1,761580398	0,003245989	0,001044932	-1,763523344	-0,057036421	98,89666216

933	98,7933758	1,925253745	0,003877199	0,001044932	1,927987776	0,062355581	98,89666216
934	98,7933758	1,929949279	0,003896134	0,001044932	1,932708347	0,062508255	98,89666216
935	98,7933758	1,932155658	0,003905048	0,001044932	1,934926538	0,062579997	98,89666216
936	98,7933758	1,953251493	0,003990786	0,001044932	1,956136815	0,063265986	98,89666216
937	98,7933758	1,953585904	0,003992153	0,001044932	1,956473062	0,063276861	98,89666216
938	98,7933758	1,97251775	0,004069902	0,001044932	1,975509993	0,063892559	98,89666216
939	98,7933758	1,973188499	0,00407267	0,001044932	1,976184507	0,063914375	98,89666216
940	98,7933758	2,021733136	0,004275528	0,001044932	2,025009096	0,065493475	98,89666216
941	98,7933758	2,036839477	0,00433966	0,001044932	2,040205619	0,065984965	98,89666216
942	98,7933758	2,077424318	0,004514322	0,001044932	2,081040071	0,067305646	98,89666216
943	98,7933758	2,111426443	0,004663307	0,001044932	2,115259668	0,068412387	98,89666216
944	98,7933758	2,12093334	0,004705396	0,001044932	2,12482875	0,068721873	98,89666216
945	98,7933758	2,163409208	0,004895753	0,001044932	2,167589924	0,070104868	98,89666216
946	98,7933758	2,166338781	0,004909021	0,001044932	2,170539628	0,070200268	98,89666216
947	98,7933758	2,169370283	0,004922769	0,001044932	2,173592024	0,07029899	98,89666216
948	98,7933758	2,214009933	0,005127448	0,001044932	2,218546725	0,071752929	98,89666216
949	98,7933758	2,216852856	0,005140624	0,001044932	2,221410185	0,07184554	98,89666216
950	98,7933758	2,226849823	0,005187092	0,001044932	2,231479818	0,072171215	98,89666216
951	98,7933758	2,232863231	0,005215145	0,001044932	2,237537277	0,072367127	98,89666216
952	98,7933758	2,237853627	0,005238482	0,001044932	2,242564424	0,072529717	98,89666216
953	98,7933758	2,257696325	0,005331792	0,001044932	2,26255501	0,073176259	98,89666216
954	98,7933758	2,281866933	0,005446566	0,001044932	2,28690958	0,073963942	98,89666216
955	98,7933758	2,349437043	0,005773906	0,001044932	2,355016601	0,07616668	98,89666216
956	98,7933758	2,501247193	0,006544182	0,001044932	2,508159062	0,081119661	98,89666216
957	98,7933758	2,546185763	0,006781446	0,001044932	2,553526758	0,082586957	98,89666216
958	98,7933758	2,714263502	0,007706304	0,001044932	2,723357341	0,088079671	98,89666216
959			COOKD>0.0042		plus minus 2	DFFITS>0.1584	

3. Process of Autocorrelation

```
/* Step 1: Fit the initial model and output residuals (last data) */
proc autoreg data=orion.restaurantNEW;
   model monthly_revenue =
         number_of_customers
         menu_price
         average_customer_spending;
   output out=residuals r=resid;
run;

/* Step 2: Fit the autoregressive model on residuals (autocorrelation)*/
proc autoreg data=residuals;
   model resid = / nlag=2;
   output out=co r=re;
run;

/* Step 3: Use PROC REG to fit the model on the residuals (analysis)*/
proc reg data=co;
   model re =
         number_of_customers
         menu_price
         average_customer_spending / spec;
run;
```

4. Selection Prediction Variable

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: Monthly_Revenue

R-Square Selection Method

Number of Observations Read	933
Number of Observations Used	932
Number of Observations with Missing Values	1

Number in Model	R-Square	Adjusted R-Square	C(p)	MSE	Variables in Model
1	0.5722	0.5718	161.8204	3684.77292	Number_of_Customers
1	0.0623	0.0613	1461.153	8077.92344	Menu_Price
1	0.0012	0.0001	1616.661	8603.70834	Average_Customer_Spending
2	0.6352	0.6344	3.4521	3145.93719	Number_of_Customers Menu_Price
2	0.5728	0.5719	162.4835	3684.21433	Number_of_Customers Average_Customer_Spending
2	0.0636	0.0615	1459.861	8075.47494	Menu_Price Average_Customer_Spending
3	0.6358	0.6346	4.0000	3144.40685	Number_of_Customers Menu_Price Average_Customer_Spending

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Rubric for Written Report on Multiple Linear Regression Analysis

Criteria	Excellent (4 points)	Competent (3 points)	Adequate (2 points)	Needs Improvement (1 point)	Points	Marks
Introduction & Background	Clearly defines the research question, objectives, and relevance with comprehensive background information.	Defines research question and objectives with relevant background, but lacks some detail.	Basic introduction with limited background information or relevance.	Unclear or missing research question and objectives, with minimal background.		
Data Description & Preparation	Provides a thorough description of the dataset, including source, variables, and data cleaning steps. Demonstrates clear understanding.	Adequate description of dataset and preparation steps, with minor gaps.	Basic description of dataset with limited explanation of preparation steps.	Incomplete or unclear data description and preparation.		
Model Development	Accurately develops the multiple linear regression model with clear explanation of variable selection and thorough checks of assumptions.	Develops model with most variables appropriately chosen and some assumption checks.	Basic model development with limited explanation of variable selection and assumptions.	Poorly developed model with minimal explanation or assumption checks.		
Analysis & Interpretation	Provides in-depth and accurate analysis, interpreting coefficients, significance, and model fit. Explains implications clearly.	Mostly accurate analysis and interpretation, with minor errors or omissions.	Basic analysis with limited interpretation and missing some key implications.	Misinterpretation of analysis or inadequate explanation of findings.		
Conclusion & Recommendations	Offers well-supported conclusions, discusses limitations, and provides insightful recommendations for future work.	Conclusions and recommendations are reasonable but lack depth in discussing limitations.	Basic conclusions with limited recommendations or discussion of limitations.	Conclusions are vague, with little to no recommendations or discussion of limitations.		
Organization & Clarity	Report is well-organized, with logical flow, clear headings, and concise language. Free of errors.	Report is organized and mostly clear, with minor issues in flow or language. Few errors.	Report has some organizational or clarity issues, with noticeable errors.	Disorganized and unclear report with multiple errors.		
Technical Accuracy & Use of Tools	Demonstrates excellent use of statistical software (e.g., R, Python) with accurate code and outputs.	Good use of statistical software with minor inaccuracies in code or outputs.	Basic use of software with some inaccuracies in code or outputs.	Poor use of software with significant errors.		

Rubric for Oral Presentation on Multiple Linear Regression Analysis

Criteria	Exemplary (4 points)	Proficient (3 points)	Adequate (2 points)	Needs Improvement (1 point)	Points	Marks
Introduction & Objective	Clearly introduces the topic, objectives, and relevance. Captures audience interest effectively.	Introduces topic and objectives clearly but lacks some engagement.	Basic introduction of topic and objectives with minimal engagement.	Introduction is unclear or fails to capture audience interest.		
Clarity of Content	Presents content in a clear, logical sequence, making it easy for the audience to follow.	Content is presented clearly, with minor lapses in sequence or clarity.	Content is somewhat clear, but with noticeable gaps in logic or flow.	Content is disorganized, making it difficult for the audience to follow.		
Explanation of Analysis	Thoroughly explains the multiple linear regression model, including assumptions, interpretation, and significance.	Adequately explains the model and its interpretation, with minor omissions.	Basic explanation of the model with limited interpretation or missed key points.	Inadequate explanation of the model and interpretation.		
Visual Aids & Data Presentation	Uses visual aids (charts, graphs) effectively to enhance understanding. Data is presented clearly and accurately.	Uses visual aids well, but with minor issues in clarity or accuracy.	Visual aids are used but may lack effectiveness or clarity.	Poor use of visual aids, with unclear or inaccurate data presentation.		
Engagement & Delivery	Engages the audience with confident delivery, clear voice, and eye contact. Handles questions effectively.	Engages audience adequately, with clear voice and reasonable handling of questions.	Limited audience engagement, with some issues in delivery or response to questions.	Lacks engagement, with poor delivery and difficulty responding to questions.		
Time Management	Manages time effectively, covering all key points within the allocated time.	Manages time well, with minor deviations from the allocated time.	Somewhat poor time management, missing or rushing key points.	Poor time management, significantly over or under time, with incomplete coverage.		
Team Collaboration	All team members contribute actively, with smooth transitions and shared responsibility.	Most team members contribute, with generally smooth transitions.	Some team members contribute minimally, with noticeable gaps in collaboration.	Lack of collaboration, with one or few members dominating or gaps in participation.		

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